BT404 - FINALTERM MERGED PPT
(LECTURE 92 TO 185)
Food Biotechnology

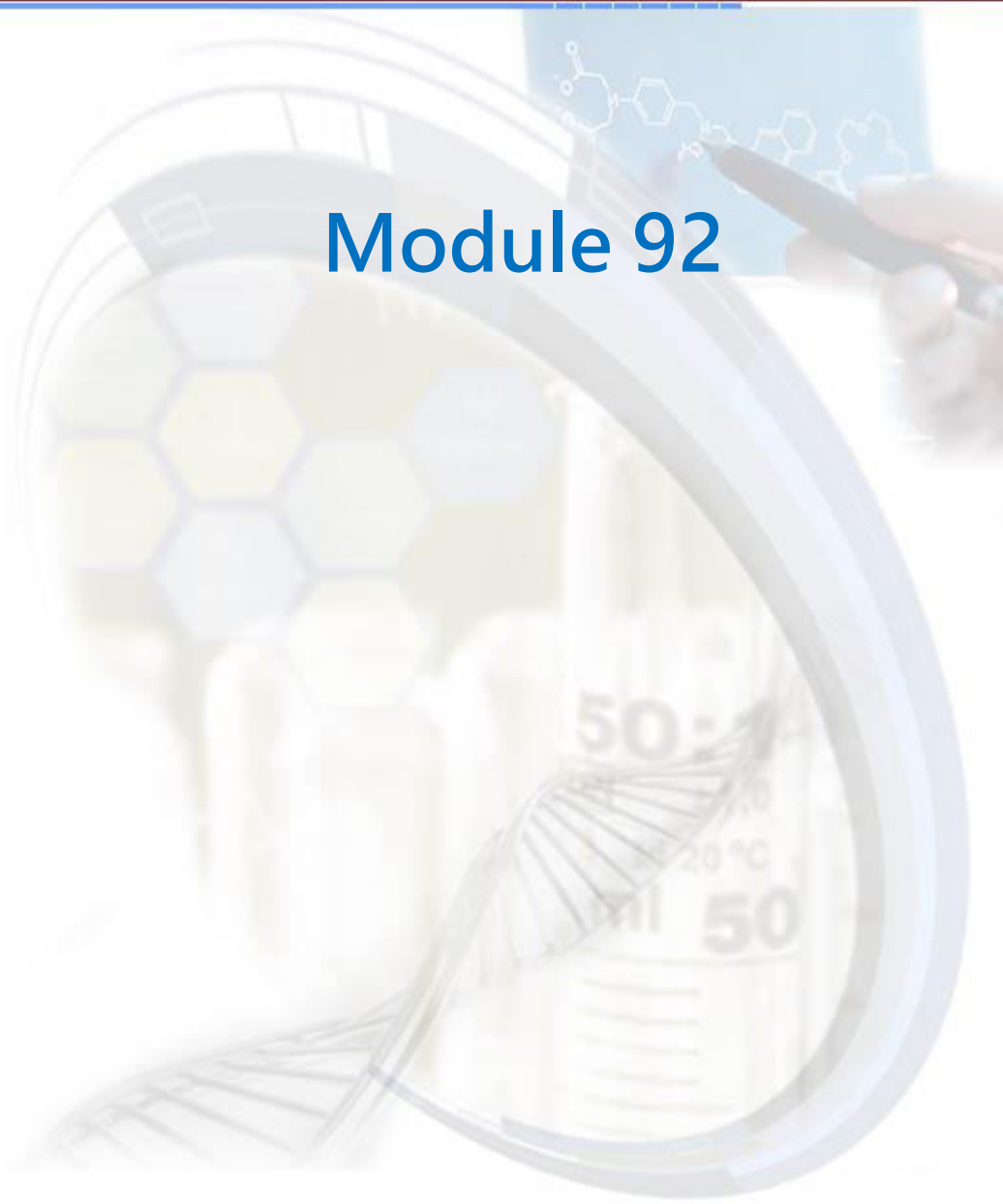
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Food Biotechnology

Module 92



Food Biotechnology

Ingredients used for Cheese Making

Ingredients used for Cheese Making

→Good cheese requires high-quality milk and carefully selected starter cultures.

→Additional ingredients are often utilized to enhance visual appeal (annatto), coagulation properties (calcium chloride or enzymes), and flavor development (adjunct cultures or enzymes) to make great cheeses.

Ingredients used for Cheese Making

Ingredients for cheese production :-

1) Milk
2) Calcium Chloride
3) Starter Culture
4) Adjunct Culture
5)Color
6) Enzyme
7) Salt
8) Optional ingredients

Ingredients used for Cheese Making

1) Milk :-

→must start with high-quality milk.

→Cheddar and most Cheddar-like cheeses can be made from raw, heat-treated, heat-shocked, or pasteurized milk, nonfat milk or cream, alone or in combination

2) Calcium Chloride :-

→ Use as coagulant (0.02 %)

Ingredients used for Cheese Making

3) Starter Culture :-

→ Isolates that produce sufficient acid to reduce the pH of milk to 5.3 in 6 h at 30–37.8°C, and aid in curd digestion and flavor development

→ For cheddar cheese → *Lactococcus lactis Sub spp + lactococcus cremoris* is used.

Ingredients used for Cheese Making

4) Adjunct Culture :-

→An adjunct culture is a one that is added, along with starter culture, for the desirable characteristics it may impart upon the cheese other than acid.

→Adjunct cultures are select nonstarter lactic acid bacteria (NSLAB)

Ingredients used for Cheese Making

5) Color :-

→The recognizable yellow to orange color of Cheddar and Colby cheeses is derived from annatto, an extract from seeds of the “Lipstick tree,

6) Enzyme :-

1)Rennet

2) Chymosin

Other clotting enzymes

Ingredients used for Cheese Making

7) Salt :-

→Cheddar cheese typically contains 1.6–1.8% salt in the form of NaCl.

→Salt enhances flavor, encourages syneresis, and slows or stops growth of salt-sensitive bacteria

Ingredients used for Cheese Making

END

8) Optional ingredients :-

→ sodium benzoate ,
→sodium propionate,
→sorbic acid →Hydrogen peroxide is allowed, if followed by a quantity of catalase preparation sufficient to eliminate the hydrogen peroxide

Food Biotechnology

Module 93 :-

Food Biotechnology

Selection of Cheese Culture

Selection of Cheese Culture

Introduction :-

→Culture quality is of the utmost

→Culture manufacturers commonly work closely with processing facility operators to effectively meet specific needs.

→Culture manufacturers continuously develop unique culture combinations for a given product

Selection of Cheese Culture

Types of cultures :-

1. Liquid (for propagation of mother culture; rarely used today)
2. Deep-frozen concentrated cultures (for propagation of bulk starter)
3. Freeze-dried concentrated cultures in powder form (for propagation of bulk starter or DRI-VAC, for preparation of mother culture)

Selection of Cheese Culture

4) **Deep-frozen or freeze-dried,** super-concentrated cultures in readily soluble form for direct inoculation of the product (direct vat set, DVS).

Selection of Cheese Culture

→The availability of frozen or freeze-dried cultures eliminates the need for to make cultures

→The culture room is a separate room in the dairy plant because it limits opportunities for contamination by airborne yeast, mold, and bacteriophage

Selection of Cheese Culture

→ Larger plants are typically supplied with frozen or freeze-dried cultures for the manufacture of bulk starters in aseptic bulk culture rooms

→Cooling and storage conditions and shelf-lives of cultures vary.

Selection of Cheese Culture

END

Generally, deep-frozen and freeze-dried cultures can be stored for 9–12 months at -18 C and -45 C.

Maintenance of consistency across cheese lots requires constancy of culture handling

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Food Biotechnology

Pre-treatment of Milk for Cheese Making

Pre-treatment of Milk for Cheese Making

Introduction :-

→ Pre-treatment of milk is essential for good cheese making

→ It ensures Quality and Safety of cheese

Pre-treatment of Milk for Cheese Making

Pre-Treatment Methods :-

1) Centrifugation :-

Centrifugal clarifier-separators are used to separate the cream and skim fractions, as well as remove solid impurities from milk

Pre-treatment of Milk for Cheese Making

Standardization :-

→Essential to maintain consistency in the quality of cheese

→ Fat addition with cream

→ Protein addition with Casein

Pre-treatment of Milk for Cheese Making

3) Homogenization :-

→Most efficient when fat globules are in the liquid state ,so milk is preheated in the plate heat exchanger in the regeneration section of the HTST pasteurizer, where the temperature is raised to at least 60C prior to homogenization

Pre-treatment of Milk for Cheese Making

→ In a two-stage homogenizer, pressures typically range from 10 to 25 MPa in the first stage and 5 MPa in the second stage

Pre-treatment of Milk for Cheese Making

END

4) Pasteurization :-

→The only step in the dairy processing system that guarantees the killing of pathogenic microorganisms is pasteurization.

→Thus, pasteurization may be considered the most critical segment of the cheese processing line

→ 72 C / few minutes

Food Biotechnology

Module 95 :-

Food Biotechnology

Cheese Making Process

Cheese Making Process

Ingredients used for cheese making :-

1) Milk
2) Calcium Chloride
3) Starter Culture
4) Adjunct Culture
5)Color
6) Enzyme
7) Salt
8) Optional ingredients

Cheese Making Process

Preparation for cheese making :-

1) Culture Selection / Propagation:-

→ Most important

1)Liquid (for propagation of mother culture; rarely used today);

2) Deep-frozen concentrated cultures (for propagation of bulk starter);

Cheese Making Process

3.)Freeze-dried concentrated cultures in powder form (for propagation of bulk starter

4. Deep-frozen or freeze-dried, super-concentrated cultures

Cheese Making Process

2) Cheese Milk Pretreatments

→ Centrifugation prior to standardization

→ Homogenization

→ Pasteurization

Cheese Making Process

Steps for cheese making :-

1) Addition of Culture , Calcium Chloride and color in pre-treated milk

2) Ripening for 15-30 minutes

3) Add coagulant and check after regular interval

Cheese Making Process

4) After coagulation cut into proper sizes Cutting of the curd is an extremely important step in the cheese making process because it influences whey drainage and cheese yield

Cheese Making Process

HEALING :-

Healing. Freshly cut curd is fragile and shatters easily, so curds are allowed to “heal” for 5–10 min prior to agitating and cooking

5) Cook it for 40-50 minutes at 102 Fahrenheit

6) Drain whey via proper filtration aid

7) Trenching

8) Milling

Cheese Making Process

9) Salting :-

1. Encourages further syneresis, 2. Inhibits further growth and metabolism of most microorganisms (thus arresting lactic acid production), and 3. Provides flavor.

10) Hooping

11) Pressing (HARD CHEESE)

12) Packing

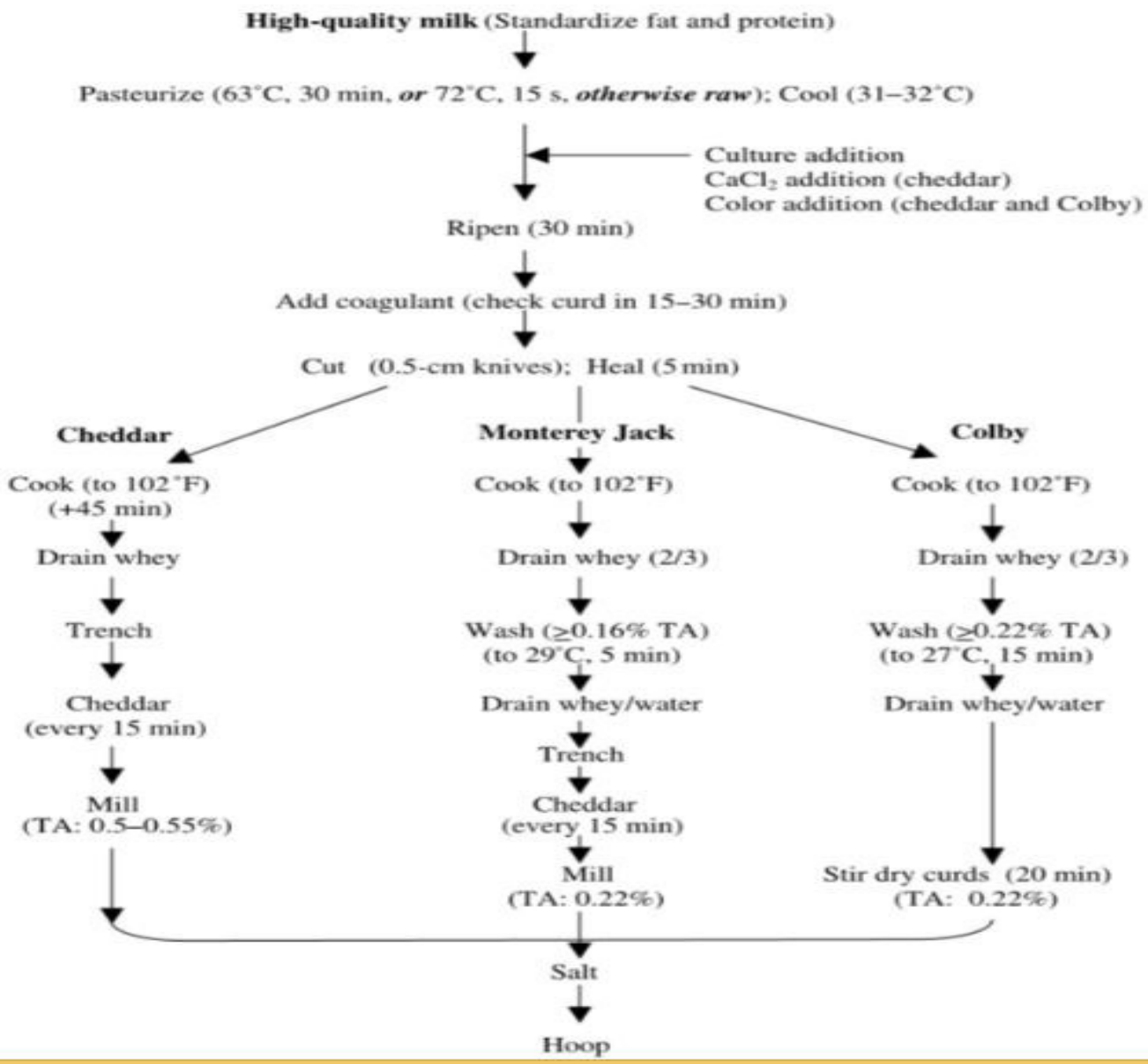
Cheese Making Process

Ageing :-

Aging enables flavor and texture development of hard cheeses

END

CHEESE MAKING PROCESS



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Module #96



Food Biotechnology

Quality Control of Cheese Making

Quality Control of Cheese Making

- The shelf-life of cheeses is limited by quality, not safety.
- The quality of good cheese improves with storage.
- ripend cheese may be removed from shelves due to flavor, body, or appearance defects.

Quality Control of Cheese Making

- Common flavor defects are high acid, bitter, unclean, and fermented /fruity.
- Common body defects are weak or crumbly body, gas holes, surface discoloration, and appearance of crystals on surfaces.

Quality Control of Cheese Making

Evaluation:

- High-quality Cheddar cheese has a full, balanced nutty, sharp, but not bitter flavor.
- The ideal texture is no gas holes and it should be firm, smooth and waxy.

Quality Control of Cheese Making

- There are as many as 200 different compounds that may contribute to cheese flavor however as few as 20 volatile compounds are pertinent to determinant of the eventual flavor of Cheddar cheese.
-

Quality Control of Cheese Making

- Cow diet, milk handling and sanitation practices, milk composition, and cheese manufacturing conditions all affect cheese chemistry.
- What appears to be critical is the relative proportions of the key chemical flavor compounds in providing peculiar flavor.

Quality Control of Cheese Making

- Ammonia-like and sulfur-like odors and bitter taste typically occur in aged cheeses, a consequence of amino acid breakdown.
- Although some attributes, such as sulfide, may be considered desirable in an aged cheese, mild cheeses are discredited for pronounced attributes.

Quality Control of Cheese Making

- Similarly certain functional properties in cheese (melting, grinding, slicing).
- Cheeses continually change during ripening, not only in flavor, but also in body/texture.
- Some of texture defects may be corky, crumbly, crystals, meaty, gassy, open, pasty, short and weak.

Quality Control of Cheese Making

Safety

- Cheddar cheese and other semi hard cheeses are generally considered safe and have rarely been associated with foodborne illness outbreaks.
- However, some pathogens can survive the cheese making process and during ripening.

Quality Control of Cheese Making

END

- Raw-milk Cheddar cheese was linked to major salmonellosis outbreaks in Canada in 1982, 1984, and 1998.
- Salmonella can survive in ripening Cheddar cheese for 7–10 months.
- *L. monocytogenes* can survive as long as 434 days in Cheddar cheese ripened at temperatures above 1.66 degree C.

Food Biotechnology

Module #97



Food Biotechnology

Food Enzymes

FOOD ENZYMES

- Enzymes, catalyze chemical reactions produced by living things and are critical for life.
- Enzyme reactions two steps:

In step 1:

E + S --- ES

In step 2,

ES + R --- P + E

FOOD ENZYMES

- Presence of necessary reactants plus energy input (activation energy) is required to start a particular reaction.

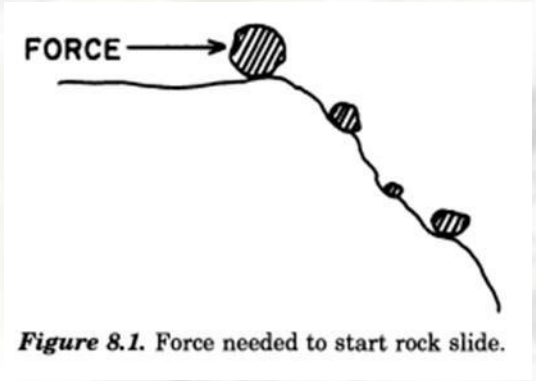


Figure 8.1. Force needed to start rock slide.

FOOD ENZYMES

- Enzyme reactions are dependent on temperature.
- reactions can be very quick at higher temperatures but living organisms have particular body temperatures at reaction have to take place so enzyme are needed.

FOOD ENZYMES

- Enzymes are needed by all levels of life, simple or complex, plant or animal kingdoms, including humans.
- All life depends on enzymes to convert foods or nutrients to a form in which they can be utilized, and to carry out cellular functions.

END

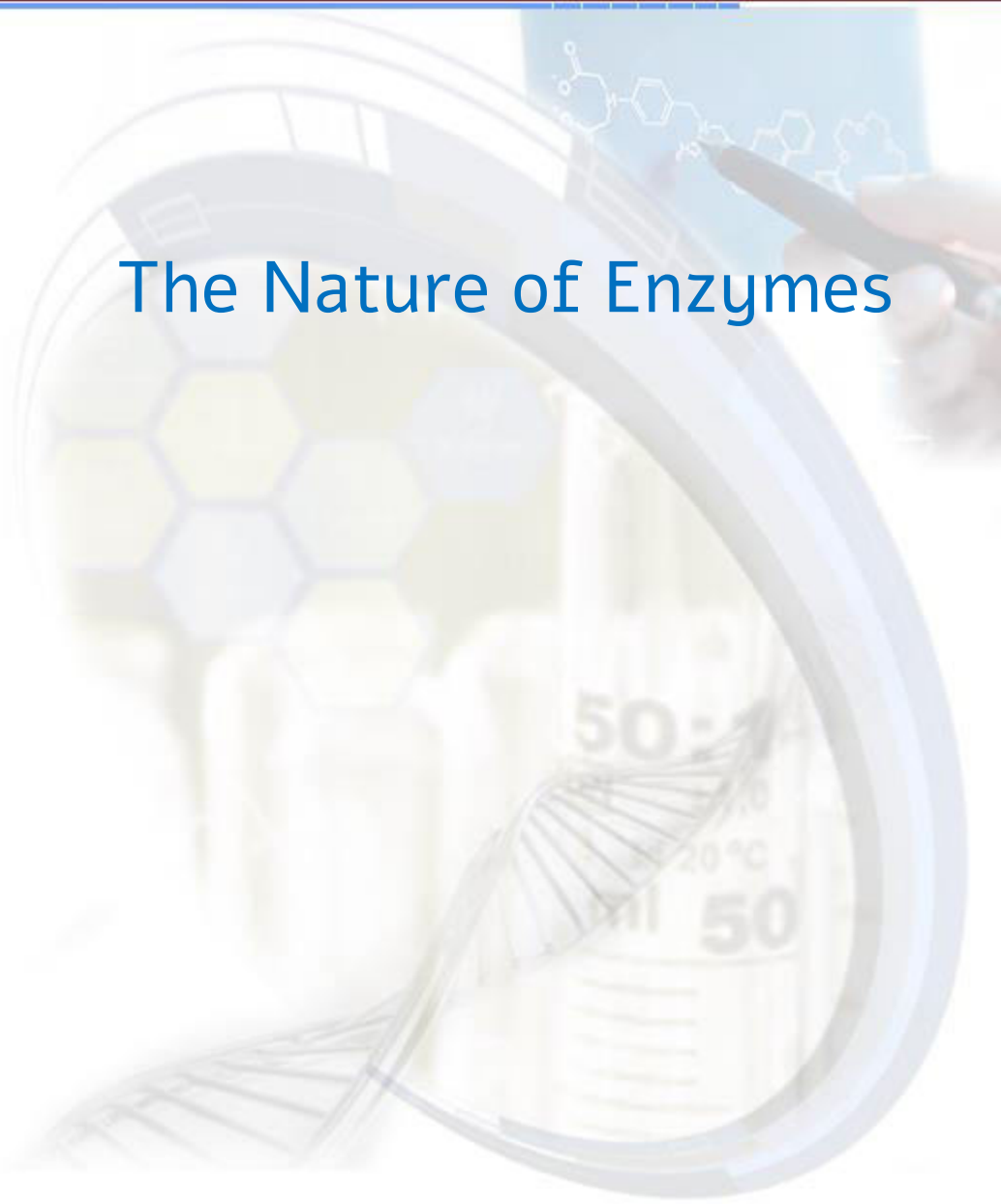
Food Biotechnology

Module #98



Food Biotechnology

The Nature of Enzymes



THE NATURE OF ENZYMES

- In composition, enzymes always contain a protein.
- They may also contain or require complex chemical compounds in order to become functional.
- These compounds are known as prosthetic groups (also called coenzymes) and are usually vitamins, especially those belonging to the B group

THE NATURE OF ENZYMES

- Some enzymes also require trace amounts of a metal, such as copper, to function.
- Enzymes therefore consist of either pure proteins, proteins with a prosthetic group, or proteins with a prosthetic group plus a metal cation.

THE NATURE OF ENZYMES

- Through their action, enzymes convert foods into less complex chemical substances that can be utilized for energy and for the building of cellular protoplasm.
- Proteins, fats, and carbohydrates are thus broken down to less complex compounds by enzymes in order that they may be utilized.

THE NATURE OF ENZYMES

- More complex carbohydrates are broken down to glucose, a source of ready energy eventually converted to carbon dioxide and water or built up into fats.
- When reserve energy is needed, fats are first hydrolyzed (water is inserted to split the molecule), breaking down into glycerine and fatty acids.

THE NATURE OF ENZYMES

- The fatty acids are then converted to acetates that can be utilized for energy, remaining may be reformed into fats and deposited as an energy reserve.
- Similarly, proteins break down to amino acids and incorporated into cellular protoplasm used for cell repair or growth.
- In some instances, oxidized to provide energy.

THE NATURE OF ENZYMES

- Enzymes always contain a protein, the protein itself having a particular shape or configuration.
- 20 to 22 known amino acids in a particular sequence produce proteins.
- Also, chains of amino acids are cross linked one to another and the different proteins form special configurations and shapes.

THE NATURE OF ENZYMES

- Proteins are also sometimes combined with carbohydrates, lipids, or phospholipids called as conjugated proteins.
- sequence of amino acids shape or relationship of these chains and conjugation with all affect the functional properties and determine the manner win which a protein will react to physical and chemical energy.

THE NATURE OF ENZYMES

- Ideal temperature in the range of 15.6 to 65.5°C,.
- However, some enzymes are able to react below freezing point or temperatures above 71°C.
- At temperatures between 71.1 and 93.3°C enzymes get inactivated due to structural changes.

THE NATURE OF ENZYMES

END

- An enzyme, which splits off fatty acids from fish phospholipids, is known to remain active even after steaming at 212°F (l00°C) for 20 min.
- Enzymes have an optimum pH at which they cause reactions to occur at the fastest rate.

Food Biotechnology

Module #99



Food Biotechnology

PROTEOLYTIC ENZYMES

PROTEOLYTIC ENZYMES

- Breakdown or splitting of proteins are called proteolytic enzymes or, proteases.
- Proteinases split the protein peptones, then into polypeptides and peptides.
- Peptidases split polypeptides and peptides into amino acids thus liquefying proteins.

PROTEOLYTIC ENZYMES

- Fish enzymes are much more active than in meats even when fish is held in the eviscerated state in ice or under refrigeration.
- In fish held uneviscerated, a concentrated source of enzymes present in blind tubules (the pyloric ceca) attached to the intestines accelerate enzyme action leading to spoilage.

PROTEOLYTIC ENZYMES

- Members of the herring and mackerel families handled in the uneviscerated state are quite subject to this type of enzyme deterioration, especially if they have been feeding when caught.

PROTEOLYTIC ENZYMES

- Similarly, lobster especially at high temperatures, proteolysis is accelerated resulting in liquefied products and poor texture.
- While cooking the flesh will be soft and crumbly (short-meated) and part of the tail portion gets dissolved, leaving only apart of this section intact hence shall be used in fresh state.

PROTEOLYTIC ENZYMES

- Shrimp, proteolysis is not extensive if heads are removed shortly after the shrimp are caught.
- The relatively high activity of enzymes in marine species is attributed to the low-temperature conditions in the marine environment.
- That is, in order to make reactions proceed at low temperatures, the activation energy system must be more efficient.

PROTEOLYTIC ENZYMES

END

- Plants proteolytic enzymes usually contribute little to deterioration, especially as long as the tissues are intact.
- Bromelin is found in unpasteurized pineapple can result in bleeding or skin eroding in handlers.
- Papain from green papaya fruit.
- Ficin from fig trees.
- Meat tenderizers

Food Biotechnology

Module #100



Food Biotechnology

Oxidizing Enzymes

Oxidizing Enzymes

- Loss of electrons in an atom or simply a chemical combination with oxygen.
- Such as wick of a candle burning.
- Oxidizing enzymes that bring about changes in foods that result in deterioration due to oxidation.

Oxidizing Enzymes

- In plants, peroxidases, ascorbic acid oxidase, tyrosinase, and polyphenolases may cause undesirable chemical reactions to occur.
- Peroxidases may oxidize certain phenol like compounds in root vegetables, such as horseradish, causing the prepared product to become darker in color when damaged.
-

Oxidizing Enzymes

- Ascorbic acid oxidase, in vegetables, oxidizes ascorbic acid cause loss of the vitamin C content of foods.
- Peroxidases also, indirectly, cause a loss of vitamin C by its by products.
- Phenolases are present in some fruits and vegetables cause brown or dark-colored compounds to be formed when the tissues are cut.

Oxidizing Enzymes

- Tyrosinase oxidizes tyrosine to form dark-colored compounds. The molecule rearranges and further oxidizes to form a red compound. Polymerization (combination of these compounds) results in the formation of dark-colored melanin compounds.

Oxidizing Enzymes

- The enzyme tyrosinase, cause discoloration of the cut tissue and will also oxidize compounds related to tyrosine.
- This enzyme is also present in shrimp and some spiny lobsters and may cause a discoloration called black spot as a black stripe on the flesh along the edges of the segments of the tail or as a pronounced band where the shell segments overlap.

Oxidizing Enzymes

END

- It is not generally recognized, but tyrosinases are also present in clams.
- Hence, shucked (deshelled) clams will darken at the surface if oxygen is present and if the enzymes have not been inactivated by heat.
- These reactions occur only after the shrimp or clams die.

Food Biotechnology

Module #101



Food Biotechnology

Fat Splitting Enzymes

Fat-splitting Enzymes

- Fats are composed of glycerine (glycerol) and fatty acids.
- Glycerine is a polyhydric alcohol (three alcohol groups), and fatty acids are short or long chains of carbon atoms to which hydrogen is attached, either to the fullest possible extent (saturated) or to a lesser extent (unsaturated), the latter resulting in reactive groups in the chain.

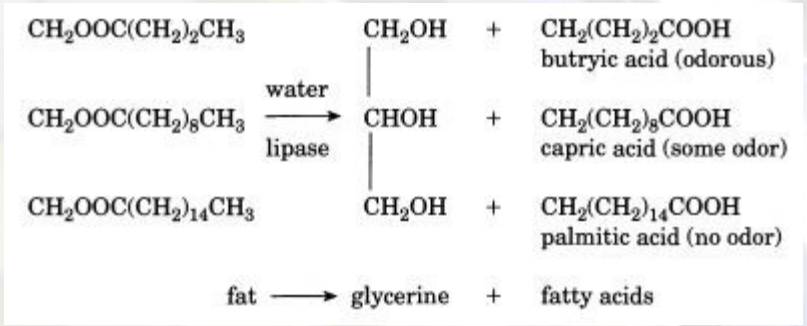
Fat-splitting Enzymes

- At one end of the fatty acid chain there is an acid group.
- In the formation of fats, each one of three fatty acids combines with one of the three alcohol groups of glycerine, splitting of water in each case.
- On the other hand, in the breakdown of fats, water and the enzyme lipase are present, and fats are split into their original component parts,

Fat-splitting Enzymes

- The fatty acids mostly consist of a chain of more than10 carbon atoms, and these fatty acids have no particular flavor or odor.
- Hence, when lipase acts on most natural fats, no bad odors are generated.
- However, when used for deep-fat frying, the oil may smoke during heating, which is undesirable.

Fat-splitting Enzymes



Fat-splitting Enzymes

- Short Chain fatty acids include butyric (four carbons), caproic (six carbons), caprylic (eight carbons), and capric (ten carbons) acids. All these fatty acids have an odor and flavor. Butyric acid, especially, is pungent and considered to be distasteful.

Fat-splitting Enzymes

- Thus butter releases butyric acid, which gives the butter a strong, undesirable rancid taste.
- Butter becomes rancid by the action of lipase produced by bacteria that grow in the water droplets.

Fat-splitting Enzymes

- Phospholipases, enzymes similar to lipase, are split phospholipids. Most phospholipids are similar to fats in that they contain glycerine, two alcohol groups of which are combined with fatty acids.
- The third alcohol group is combined with a molecule containing phosphoric acid.

Fat-splitting Enzymes

END

- Lecithin is a typical phospholipid. Phospholipase splits off fatty acids from phospholipids. Such action may cause deterioration in foods in resulting in destabilization of proteins causing a toughening of the tissues and a loss of succulence.

Food Biotechnology

Module #102



Food Biotechnology

Enzyme that decompose carbohydrates

Enzymes that Decompose Carbohydrates

- Pectin in fruits supports structure.
- In processed fruit juices broken pectin tend to settle to the bottom, leaving a clear serum on top.
- Pectin consists of a long chain of galacturonic acid molecules with the carboxyl groups partially esterified with methyl alcohol. It has a high water-holding capacity.

Enzymes that Decompose Carbohydrates

- There are pectic enzymes that will either break down the pectin molecule to smaller units or completely decompose the molecule to its primary unit, galacturonic acid.
- The emulsifying properties of pectin may be lost, causing settling in fruit juices and softening in fruit.

Enzymes that Decompose Carbohydrates

END

- Pectin in whole fruit breaks down resulting in deterioration of the fruit as a result of the action enzymes/microbes.
- In the sugar cane, invertase, breaks down sucrose, hence a part of sugarcane must be removed at harvesting.

Food Biotechnology

Module #103



Food Biotechnology

Glucose Oxidase

Glucose Oxidase

- Glucose oxidase is an enzyme that specifically catalyzes the oxidation of glucose to gluconic acid.
- Non enzymatic browning, is a undesirable browning reaction.
- The most important application of this enzyme is in the treatment of egg products, especially egg whites, prior to drying.

Glucose Oxidase

- In some cases, the enzyme is added to remove traces of oxygen to prevent oxidative degradation of quality. Examples of this type of application are mayonnaise and bottled and canned beverages (especially beer and citrus drinks).

END

Food Biotechnology

Module #104



Food Biotechnology

Catalases

Catalase

- Catalases are used to break down hydrogen peroxide (H_2O_2) to water and oxygen.
- Therefore, catalases are used when the presence of hydrogen peroxide is undesirable or when hydrogen peroxide is used for specific purposes, such as in bleaching, but then must be removed from the system.

Catalase

- Examples of the latter case are the uses of hydrogen peroxide for preserving milk in areas where heat pasteurization and refrigeration are unavailable and in the manufacture of cheese from unpasteurized milk.
- Hydrogen peroxide is produced during the spray-drying procss.
- Catalase is used to convert the unwanted H_2O_2 to water and

Catalase

- Catalases have health promoting activities in human by reducing the oxidative damage/inflammation etc.
- Damage proteins, cell membrane and even DNA, reduces with age and must come from food sources.

END

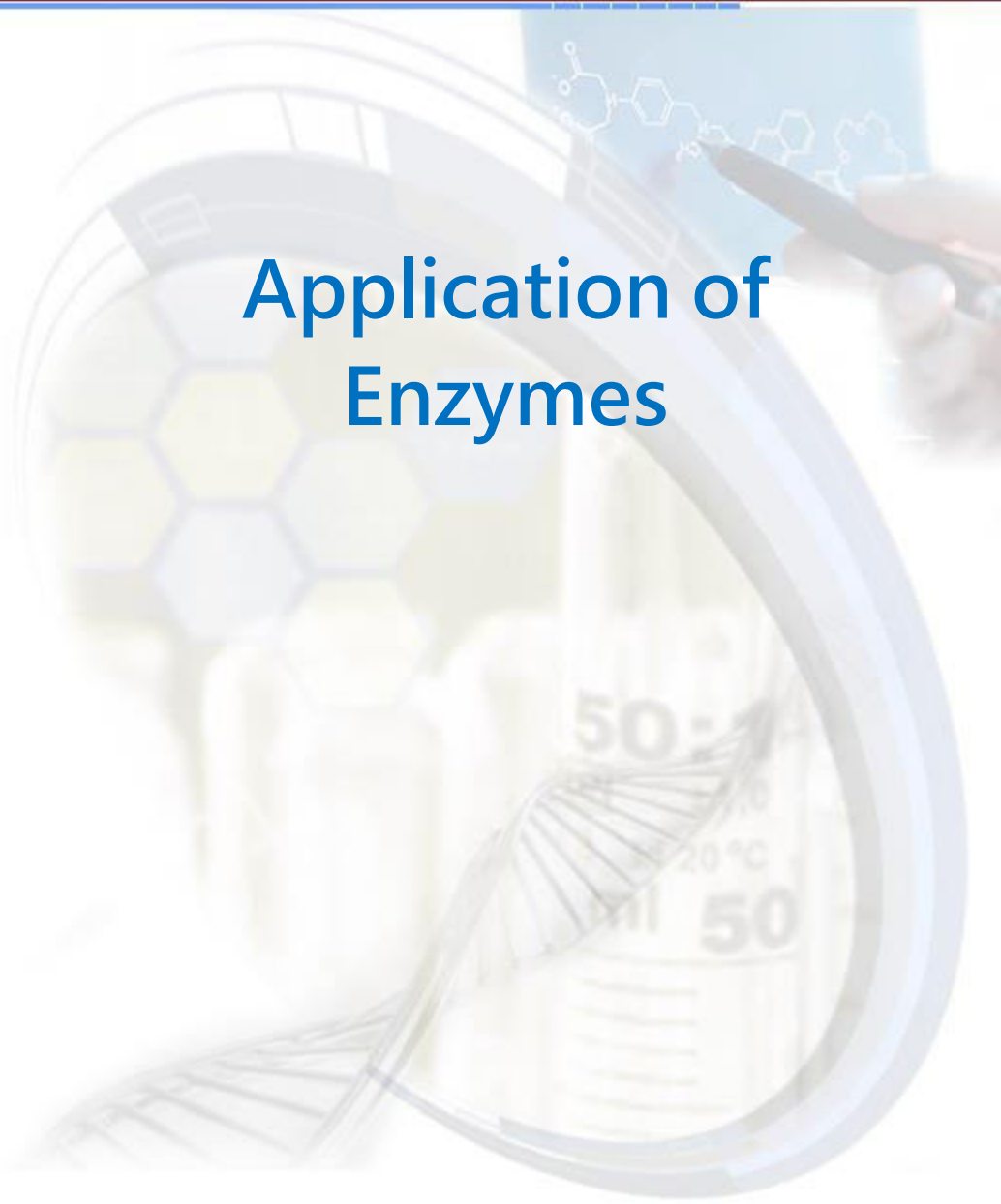
Food Biotechnology

Module #105



Food Biotechnology

Application of Enzymes



Application of Enzymes

- Whereas enzymes may cause a deterioration of foods, they may also be used in the processing of foods.
- Proteolytic enzymes obtained from plants may be used for tenderizing meat either by injecting animals with a solution of the enzyme prior to slaughter or by sprinkling the powdered enzyme on meat surfaces and allowing it to react, prior to cooking.

Application of Enzymes

- In the manufacture of certain kinds of milk powder (e.g., to be used in chocolate), the lipases may be allowed to act on the milk fat prior to drying to obtain a particular flavor in the finished product.
- Proteases may also be used to chill-proof beers, a technique used to remove proteins that would cause clouding during cool storage.

Application of Enzymes

- Protein hydrolysates can be produced from the use of proteases from both plant and animal sourees. They have many uses, for example, as flavorings, nutrients, and stabilizers.
- The characteristic flavor of certain cheeses is due to the action of lipases.
- In order to obtain the particular flavors of Roquefort or blue cheese, milk fat must first be broken.

Application of Enzymes

- fatty acids are produced by the molds allowed to grow in these cheeses. Enzymes produced by molds then oxidize the specific fatty acids that ultimately produce the unique characteristic flavors.
- Lipases also can be added to dried egg whites to improve the whipping quality.

Application of Enzymes

- Carbohydrate-splitting enzymes.
- In making malt, barley is germinated to obtain an enzyme that will convert starch to a sugar (maltose.

Application of Enzymes

- Carbohydrate-splitting enzymes are also used to modify starches used in foods and to modify starches used in sizing and laundering clothes.
- Corn syrup from corn starch and then isomerize glucose to fructose in the formation of high-fructose corn syrup which is used extensively in the manufacture of soft drinks.

Application of Enzymes

- Invertase (sucrase) is used in the production of chocolate covered cherries.
- During a short holding period, the enzyme splits the sucrose into a mixture of glucose and fructose, which has a higher water solubility and results in the sweet taste and creamy texture that is characteristic of the product.

Application of Enzymes

- The presence of pectins in fruit juices may cause clouding.
- Pectinases can be added to remove the pectin and clarify the juice.

Application of Enzymes

- One of the factors that will serve to widen the use of enzyme technology is the development of immobilized enzymes.
- It has been found that enzymes can be fixed chemically to the surface of inert substances, such as glass beads. In this form, they can be packed into a column through which a solution or suspension of the material to be acted on (called the substrate) is allowed to pass

Application of Enzymes

END

- Restoration and enrichment of flavor in some processed foods by addition of enzymes.
- e.g. conversion of all in in garlic into garlic oil by allinase and the addition of an enzyme preparation from mustard seeds to dehydrated cabbage which restores flavor by converting flavor precursors into the volatile sulfur-containing compounds responsible for the familiar flavors.

Food Biotechnology

Module #106



Food Biotechnology

Food Additives:
Definition and Functions

Food Additives

- Food additives have been defined as chemical substances deliberately added to foods, directly or indirectly, in known and regulated quantities, for purposes of assisting in the processing of foods; preservation of foods; or in improving the flavor, texture, or appearance of foods.

Food Additives

- An additive may be reactive or inactive; it may be nutritive or nonnutritive; it should be neither toxic nor hazardous.
- Some substances, such as pesticides, are added to foods unintentionally, and these are, of course, undesirable, and may be hazardous to health.

Food Additives

- Additives are classified as GRAS when they have been used without apparent harm for long periods of time, long before regulations had been put into effect.
- These include substances such as baking powder chemicals (e.g., sodium bicarbonate), fruit acids such as citric and malic, and gums such as agar-agar.

Food Additives

- Ionization radiation is considered an additive because the treatment may induce changes in food.
- In this case, the FDA tests foods that have been irradiated and approves irradiation sources and maximum dosages.

Food Additives

- Besides the requirements mentioned, food additives must also meet five general criteria:
  1. Intentional additives must perform their intended function.
  2. Additives must not deceive the consumer or conceal faulty ingredients or defects in manufacturing practices.

Food Additives

3. Additives cannot considerably reduce the food's nutritional value.
4. An additive cannot be used to achieve an effect that could be gained by good manufacturing practices.
5. A method of analysis must exist to monitor the use of the additive in foods or its incidental occurrence in foods (such as migration from a packaging material).

END

Food Biotechnology

Module #107



Food Biotechnology

Philosophy of Food Additives

Philosophy of Food Additives

- Foods are made entirely of substances that, in the pure form, can be described as chemicals or chemical compounds.
- It is important to note that our knowledge of the composition of foods, because of its complexity, is by no means complete.
- For instance, it is reported that one of the most important of our natural foods, human milk, contains several hundred chemical compounds.

Philosophy of Food Additives

- Unfortunately, the interpretation of the word *chemical* is too often inaccurate.
- Refined sugar, vinegar, spices, and other substances that are routinely added to foods are also chemicals or mixtures of chemicals, and the use of these is not questioned.

Philosophy of Food Additives

- A large number of chemical additives are unfamiliar, and there is a need for regulatory agencies to question their use from the standpoint of safety.
- For example, indiscreet use of certain additives for coloring candy and popcorn was reported to have caused diarrhea in children, resulting in the removal of these dyes from the FDA approved list of additives.

Philosophy of Food Additives

- Additives can come from natural food sources.
- Any additive or chemical compound can be injurious to health on improper use.
- It is necessary to evaluate each additive for usefulness and toxicity in a sensible, scientific way.

Philosophy of Food Additives

- Given present capabilities, our most reasonable evaluation of an additive for safety can be made through conventional animal feeding studies.
- chemical materials added to foods must be approved for use unless their use.

Philosophy of Food Additives

- Most food additives are components of natural foods and that without these additives the quality of many foods would be greatly inferior to that to which we have become accustomed.
- The shelf-life or availability of many foods would also be greatly limited were all additives to be eliminated from foods.

Philosophy of Food Additives

END

- Food additives may be used for a number of reasons.
- At present, over 3000 intentional additives are allowed and they can be divided into several major groups.

Food Biotechnology

Module #108



Food Biotechnology

Antioxidants in Foods

Antioxidants

- Antioxidants are food additives used to stabilize foods against oxidation.
- Oxidative quality changes in foods include:

•the development of rancidity from the oxidation of unsaturated fats resulting in off-odors and off-flavors and

•discoloration from oxidation of pigments or other components of the food.

Antioxidants

- Antioxidants function in different ways, the purpose of each is to prevent, delay, or minimize the oxidation of the food to which they are added.
- antioxidants make their combination with oxygen.
- or prevent oxygen from reacting with components of the food, as in a hermetically sealed container.

Antioxidants

- Some antioxidants lose their effectiveness when they combine with oxygen;
- Precautions are also necessary to minimize oxidation (heat, light, and metals are pro-oxidants)
- Many of the antioxidants used commercially occur naturally in foods (e.g., vitamin C, vitamin E, citric acid, amines, and certain phenolic compounds).

Antioxidants

- amines and the phenolic compounds can be toxic to , potency of the naturally occurring antioxidants is less than synthetic antioxidants.
- butylated hydroxyanisole (BHA), butylated hydroxytoluene (BHT), and propylgallate mostly used.
- These are used in combinations of two or all three of them, and often in combination with Citric acid.

Antioxidants

END

- Fats and shortenings, for bakery/fried foods, are subject to oxidative spoilage so chemical antioxidants in concentrations up to 0.02% of the weight of the fat component may be added.
- Their use is limited so that the maximum amount that can be added is generally 0.02% of the fat content of the food.

Food Biotechnology

Module #109



Food Biotechnology

Nutrient Additives

Nutrient Additives

- The need for a balanced and ample nutrient intake by the human body is well known.
- Although nutrients are available in foods, losses of fractional amounts of some of them through processing and other factors minimum daily requirements or sizable fractions of minimum daily requirements of a number of nutrients to popular foods.

Nutrient Additives

- Vitamin D is an example.
- The major source of vitamin D for humans is a precursor compound called 7-dehydrocholesterol which is produced in the liver.
- It circulates to an area just under the skin and is converted to pre vitamin D3 by the ultraviolet rays of sunlight.
- Previtamin D3 to active vitamin D.

Nutrient Additives

- However, in many cases, exposure to the sun is sporadic and insufficient, especially in areas where there is normally insufficient sunshine or in cases where sunlight exposure is of insufficient duration.

Nutrient Additives

- The addition of protein concentrate (produced from fish or soybeans) to components of the diet of inhabitants of underdeveloped countries has been used successfully to remedy the high incidence of protein malnutrition.
- It should be noted that soybean protein is incomplete and requires the addition of some amino acids in which it is deficient.

Nutrient Additives

- Children, especially, succumb in large numbers to the disease, kwashiorkor, that results from insufficient protein intake.
- Among minerals, iron has received major attention as a food additive, mainly because of its role in preventing certain anemias.

end

Food Biotechnology

Module #110



Food Biotechnology

Flavorings and Flavor Enhancers

Flavorings & Flavor Enhancer

- Flavorings are compounds, natural or synthetic, that are added to foods to produce flavors or to modify existing flavors.
- In the early days of human existence, salt, sugar, vinegar, herbs, spices, smoke, honey, and berries were added to foods to improve their taste or to produce a special, desirable taste.
- The variety of natural and synthetic flavorings available to the modern food technologist is very large.
- Essential oils provide a major source of flavorings.
- Essential oils are odorous components of plants and plant materials that give the characteristic odors of the materials from which they are extracted.

Flavorings & Flavor Enhancer

- Because of the large production of orange juice, quantities of essential oil of orange are produced as byproducts.
- For this reason, there is little need for the production of synthetic orange flavoring.
- Fruit extracts have been used as flavorings, but these are relatively weak when compared to essential oils and oleoresins.
- An oleoresin is a solvent extract of spices from which the solvent, usually a hydrocarbon, has been removed by distillation.
- Because of their weak effects, fruit extracts may be intensified by combining them with other flavorings.

Flavorings & Flavor Enhancer

- Synthetic flavorings are usually less expensive and more plentiful than natural flavorings.
- On the other hand, natural flavorings are often more acceptable, but they are quite complex and difficult to reproduce synthetically.
- natural flavorings are variable due to season and other uncontrollable factors unlike synthetic flavorings.

Flavorings & Flavor Enhancer

- Many artificial flavors, such as amyl acetate (artificial banana flavor), benzaldehyde (artificial cherry flavor), and ethyl caproate (artificial pineapple flavor), are added to confectioneries, baked products, soft drinks, and ice cream.
- These flavorings are added in very small amounts, often 0.03% or less.

Flavorings & Flavor Enhancer

- Flavorings either impart a particular flavor to food or modify flavors already present.
- Flavor enhancers, on the other hand, intensify flavors already present, especially when the desirable flavors are relatively weak.
- Monosodium glutamate (MSG) is one of the best known and most widely used flavor enhancers.

Flavorings & Flavor Enhancer

- While it is effective at relatively low levels (parts per thousand), there are other compounds called flavor potentiators that also enhance flavors but are extremely powerful, effective in parts per million and even per billion.

Flavorings & Flavor Enhancer

- These compounds have been identified as nucleotides, and their effect is attributed to their synergistic properties (properties that intensify the effect of natural flavor components).
- Two in this group are disodium inosinate and disodium GMP.

Flavorings & Flavor Enhancer

- One theory is that they increase the sensitivity of the taste buds, thus increasing flavor.
- A second suggests that an increase in salivation as a result of the flavor enhancers.
- A third theory of intensified flavor perception is flavor enhancers produce certain physical sensations in the mouth such as coolness and heat.

end

Food Additives

Module # 111



Food Biotechnology

Acidulants in Food

Acidulants in Food

- Acidulants are a class of compounds that tends to lower the pH of any food
- Enhance desirable flavor
- In some foods like pickle, they are a major taste component

Acidulants in Food

- Vinegar or acetic acid
- Added to relishes, chili sauce, ketchup and other condiments
- Acts as flavor component
- Also aids in preservation

Acidulants in Food

- By adding acidulants, it effectively decreases pH
- Lower pH or acidic conditions inhibit the growth of microbes

Acidulants in Food

- Many acidulants occur naturally in food

- Citric Acid - Citrus family
- Malic Acid - Apples
- Acetic Acid – Vinegar

- Figs contain all three acidulants

Acidulants in Food

- Tartaric acid is used commonly in food to provide tartness and enhance flavor
- Citric acid is used in carbonated drinks
- Phosphoric acid is an inorganic acidulant used in food

Acidulants in Food

- Can also be used as cleaners for dairy equipment
- Can also be used to improve gelling properties and texture

Acidulants in Food

- Used in processed cheese and cheese products: cheese spread
- Acidulants are used as emulsifying agents
- Also provide desirable tartness as to enhance flavor

Acidulants in Food

- Acid salts may be added to soft drinks

- Act as buffer to avoid excessive tartness
- Acid salts are also used to inhibit mold growth
- Calcium propionate added to bread

end

Food Biotechnology

Module # 112



Food Biotechnology

Alkaline Compounds in Food

Alkaline Compounds in Food

- Alkaline compounds tend to raise the pH of foods

- Sodium Hydroxide
- Potassium Hydroxide

- Can be used to neutralize excess acid that can develop in natural or culture fermented foods

Alkaline Compounds in Food

- Manufacture of butter
- Acid in the cream may be partially neutralized
- Must be done prior to churning
- Failure to do so can result in undesirable flavor

Alkaline Compounds in Food

- Rendered fats can be refined by:
- Sodium carbonate
- Sodium bicarbonate

- Alkaline compounds added to chlorinated drinking water

- Adjusts pH high enough to control corrosion of pipes

Alkaline Compounds in Food

- Sodium carbonate along with other compounds used to reduce the amount of hardness in drinking water

- Sodium bicarbonate used as an ingredient of baking soda

Alkaline Compounds in Food

- Sodium Hydroxide used in production of caramel
- Also used to modify starches

- Alkaline compounds also used in production of chocolate
- Also used to adjust acidity levels in grape and other juices that are to be fermented for wine production

Alkaline Compounds in Food

- Sodium hydroxide
- Potassium hydroxide

- Powerful reagents
- Should be handled with care

- Sodium bicarbonate is relatively mild and safe to use

Alkaline Compounds in Food

END

- Sodium hydroxide
- Potassium hydroxide

- Powerful reagents
- Should be handled with care

- Sodium bicarbonate is relatively mild and safe to use

Food Biotechnology

Module # 113



Food Biotechnology

Sweeteners

Sweeteners

- Sweetening agents used largely in food and beverage industry
- Natural and synthetic agents
- Varying strengths and caloric value

Sweeteners

- Most commonly used sweeteners:
- Table sugar (sucrose)
- Corn Syrup
- Many sweeteners are classified as nonnutritive

Sweeteners

- Fructose
- Molasses
- Honey
- Maple sugar
- Lactose
- Maltose
- Polyhydric alcohols
- Aspartame
- Saccharin

Sweeteners

- Fructose

- Natural
- Monosaccharide
- Twice as sweet as sucrose
- Most water soluble
- Hygroscopic

Sweeteners

- Greater flexibility of use over a range of temperatures
- Solution of fructose have low viscosity
- Does not crystallize out of solution

- Sometimes called fruit sugar due to its presence in fruits

Sweeteners

- Fructose
- Major component in honey, corn syrup, cane sugar, beet sugar
- Advised to diabetic individuals
- Accelerates the metabolism of alcohol

Sweeteners

- Fructose
- Rapid source of energy for athletes
- In combination with gluconate and saccharin; it is an effective, economic, safe and low-calorie sweetener for beverages

Sweeteners

- Molasses
- By-product of sugar production
- Use as sweetener mostly in baked goods
- Breads, cookies, cakes
- Adds flavor and acts as humectant

Sweeteners

- Molasses
- Greatest use of molasses is in animal feed
- Used in production of baked beans
- Also used in production of rum

Sweeteners

- Molasses
- Comprise of about 60% sucrose
- Can very depending upon raw material and grade of molasses

Sweeteners

- Honey

- Natural
- Viscous syrup

- Comprise mainly of invert sugar
- Relatively expsensive

Sweeteners

- Honey

- Produced from nectar of flowers

- Nectar is mainly sucrose
- Produced by action of invertase enzyme
- Enzyme secreted by honey bee

Sweeteners

- Honey
- Used as a direct sweetener
- Also used as an additive in many food products
- Included in some baked goods

Sweeteners

- Maple Sugar
- Sap of the sugar from maple tree
- Comprised mainly of sucrose and small amounts of other sugars like invert sugar

Sweeteners

- Maple Sugar
- Used in manufacture of candies, fudge, baked goods and toppings
- Among the most expensive sweeteners

Sweeteners

- Lactose
- Sugar component of mammalian milk
- Less sweet and lesser-water soluble than sucrose

Sweeteners

- Lactose
- Major source of lactose is whey, a cheese byproduct
- Because it is not as sweet as sucrose, it can be used in large amounts in those foods in which the texture benefits from a high solid content

Sweeteners

- Maltose

- Malt sugar
- Produced during enzymatic conversion of starch in the brewing process
- Converted to alcohol by action of yeasts
- Intermediate conversion to dextrose

Sweeteners

- Maltose
- Much less sweeter than sucrose
- Used in manufacture of baked goods and infant foods

Sweeteners

- Xylitol

- Polyhydric alcohol
- Non-carcinogenic property
- Used in chewing gums

- Occurs naturally
- Intermediary product of carbohydrate metabolism in humans and animals

Sweeteners

- Xylitol
- Produced commercially by hydrolysis of xylan to xylose
- Xylose hydrogenated to to produce xylitol
- Purified and crystallized

- Imparts a sweet taste and cooling effect
- Not metabolized by many microorganisms

Sweeteners

- Sorbitol

- Polyhydric alcohol
- Found in red seaweed and in fruits such as apples, cherries, peaches, pears

- First isolated from sorb berries

Sweeteners

- Sorbitol
- Used as an additive for its humectant property as well as its sweetening effect
- Used in
- Cough syrup
- Toothpaste
- Mouthwash

Sweeteners

- Sorbitol
- Produced industrially by electrochemical reduction or catalytic hydrogenation of glucose
- Not easily fermented by many microorganisms
- Largely transformed to fructose by liver enzymes

Sweeteners

- Mannitol

- Polyhydric alcohol
- Naturally occuring in many plants, algae and molds

- Can also be made by reduction of monosaccharides

Sweeteners

- Mannitol
- Occurs in sap on manna trees native to southern Italy
- Used in chewing gum and pharmaceuticals

Sweeteners

- Aspartame

- Common name for aspartyl-phenylalanine
- Combination of two amino acids

- 180 times sweeter than sucrose
- Leaves no bitter after taste

Sweeteners

- Aspartame

- Approved and later banned by FDA
- Reinstated as safe after exhausting evidence was presented

- About 200 times more expensive than sucrose
- Needed in much smaller quantities than sucrose

Sweeteners

- Saccharin
- Imide of *o*-benzosulfonic acid
- Used as sodium or calcium salt
- 300 times sweeter than sucrose
- Leaves bitter after taste

Food Biotechnology

Module # 114



Food Biotechnology

Starches

Starches

- Starch
- Although differs from each other somewhat
- Depending on the plant from which they are extracted

Starches

- Starch
- Usually sufficiently chemically similar
- Two basic starch polymers are
- Amylose
- Amylopectin

Starches

- Starch
- Used as a source of carbohydrate
- Relatively inexpensive
- Often used as an extender

Starches

- Starch
- Properties of starch make it useful as a thickening agent
- Major source is starch from corn
- Also produced from sorghum, potatoes and wheat

Food Biotechnology

Module # 115



Food Biotechnology

Gums

Gums

- Class of complex polysaccharides
- Defined as materials that are dispersible in water and make the water more viscous
- Many gums occur naturally

Gums

- Examples of gum are:

- Gum Arabic
- Agar

- Vinyl polymer gums are synthetic
- Cellulose derivate gums are modified or semi synthetic

Gums

- Used to:

- Stabilize ice-cream and desserts
- Thicken beverages and preserves
- Stabilize foam in beer
- Emulsify salad dressing
- Form protective coatings from meat and fish

Gums

- Gums add “body” and prevent settling of suspended particles in chocolate milk, ice cream and desserts
- May also prevent formation of large ice crystals in frozen desserts

Gums

- Use of gum also in production of certain low-calories food
- Oils in salad dressing can be replaced with gums
- Replacing oil with gum allows product to keep normal appearance, texture and taste while losing calorie count

Food Biotechnology

Module # 116



Food Biotechnology

Polyhydric Alcohols

Polyhydric Alcohols

- Also called polyols
- Used to improve texture and moisture retention
- Present in foods naturally
- Glycerine (glycerol)

Polyhydric Alcohols

- Only four allowed polyols

- Glycerine
- Sorbitol
- Mannitol
- Propylene glycol

Polyhydric Alcohols

- Propylene glycol has an undesirable bitter after taste
- Acceptable in small amounts
- Sorbitol imparts a cool sensation
- Glycerine imparts a hot sensation

Polyhydric Alcohols

- Polyols are used in the production of dietetic products

- Beverages
- Candies
- Gum
- Ice cream

Polyhydric Alcohols

- These compounds have less adverse effect on teeth than sugar
- Not fermented as easily as sugar
- Usually washed away before they can be consumed

END

Food Biotechnology

Module # 117



Food Biotechnology

Surface Active Agents

Surface Active Agents

- Affect the physical force at the interface of surfaces
- Commonly called surfactants
- Present in all natural foods
- Play a role in growth process

Surface Active Agents

- Organic compounds that affect the surface activities

- Act as:

- Wetting agent
- Lubricant
- Dispersing agent
- Detergent
- Emulsifies

Surface Active Agents

- Wetting agents
- Reduce surface tension of materials to permit absorption of water by the material
- Powdered chocolate mixes

Surface Active Agents

- Used in processing of food to prevent sticking in untreated peanut butter
- Used in cleaning detergents used on food equipment
- Stabilize or break down foams

Surface Active Agents

- Emulsifiers such as lecithin
- May be added to bakery products
- Improve volume and texture of final product
- Also used in cake mixes, frozen desserts, ice cream

END

Food Biotechnology

Module # 118



Food Biotechnology

Leavening Agents

Leavening Agents

- Used to enhance the rising of dough in the manufacture of baked products

- Chemical reagents that react to form carbon dioxide

Leavening Agents

- Inorganic salts mostly
- Ammonium salts
- Phosphate salts
- Favor growth of yeast that produces carbon dioxide
- Results in rise of dough

Leavening Agents

- Baking powder is a common household leavening agent
- Contains a mixture of chemical compounds
- React together to form carbon dioxide

Leavening Agents

- Baking soda can be of two types
- Single acting
- Double acting

Leavening Agents

- Single acting
- Sodium bicarbonate (base)
- Potassium acid tartrate (acid)
- Acid base reaction
- Carbon dioxide is liberated in one step

Leavening Agents

END

- Double acting

- Two acid reacting agents:
- Monocalcium phosphate monohydrate
- Sodium aluminium sulfate

- Portion of CO_2 released from sodium bicarbonate reaction with phosphate
- Remaining from sulfate

Food Biotechnology

Module # 119



Food Biotechnology

Chemical Preservatives

Chemical Preservatives

- Foods can be preserved altering them to make them incapable of supporting microbial growth
- Usually used in concentrations of 0.1% or less

Chemical Preservatives

- Sodium diacetate and sodium or calcium propionate
- Used in breads
- Prevents mold growth and development of bacteria that may produce a slimy material known as rope

Chemical Preservatives

- Sorbic acid and its salts
- Used in bakery products, cheeses, syrups, and pie fillings to prevent mold growth

- Sulfur dioxide
- Used to prevent browning in certain dried fruits
- Prevent wild yeast growth in wines used to make vinegar

Chemical Preservatives

- Benzoic acid
- Sodium benzoate

- Used to inhibit mold and bacterial growth in some fruit juices, oleomargarines, pickles, and condiments

- Benzoic acid is a natural component of cranberries

Chemical Preservatives

- Salt is an excellent inhibitor
- Suppresses water activity
- Effectiveness enhanced in conjugation with drying and smoking

Chemical Preservatives

- Sodium Chloride
- Makes water unavailable to microorganisms
- Lowers the water activity
- Most bacteria require water activity of 0.95
- Chloride ion inhibits growth

Chemical Preservatives

- Sodium Nitrite

- Inhibit microbial growth
- Enhance color
- Prevents oxidation of myoglobin in meat
- Tends to prevent growth of spores of botulinum
- Not to be used in fresh foods

Chemical Preservatives

- Sodium Nitrite
- Nitrite cured products cooked at high temperatures may develop nitrosamine
- Known to be extremely carcinogenic

Chemical Preservatives

- Oxidizing Agents

- Not ordinarily used in foods
- Used to sanitize food equipment
- Even walls and floors of food processing area
- Small residuals can get into food

Chemical Preservatives

- Oxidizing agents

- Chlorine
- Iodine

eND

- Hydrogen peroxide may be used to destroy natural flora of milk prior to inoculation with culture

Food Biotechnology

Module # 120



Food Biotechnology

Food Colorants

Food Colorants

- Color additives used to make food more desirable
- Three major types:
- Natural
- Nature-identical
- Synthetic

Food Colorants

- Natural colors are pigments obtained from animal, vegetable, or mineral sources

- Yellow from the annatto seed
- Green from chlorophyll
- Orange from carotene
- Brown from burnt sugar
- Red from beets, tomatoes

Food Colorants

- Nature-identical
- Synthetic counterparts of color and pigment derived from natural sources
- Pure carotenoids:
- Canthaxanthin (red)
- Apocarotenal (orange-red)
- Beta-carotene (yellow-orange)

Food Colorants

- Some colorants are derived from synthetic dyes
- Dyes and lakes

- Dyes are water soluble and available in powder, granules, liquids, blends, pastes
- Lakes are water soluble dyes on a substratum of aluminum hydrate or aluminum oxide

Food Colorants

- Lakes are useful in foods that have very little water such as coloring oils

- Used in:
- Fondant coatings
- Icing
- Cake
- Doughnut mixes
- Hard candy
- Gum products

Food Colorants

- Some compounds are not color additives but are added to produce a white color

- Oxidizing agents including benzoyl peroxide, chlorine dioxide, nitrosyl chloride, and chlorine are used at the end of the production cycle to whiten wheat flour which is pale yellow in color if untreated

Food Colorants

- Titanium dioxide
- Considered a color additive
- May be added to some foods such as artificial cream or coffee whiteners to add a white color

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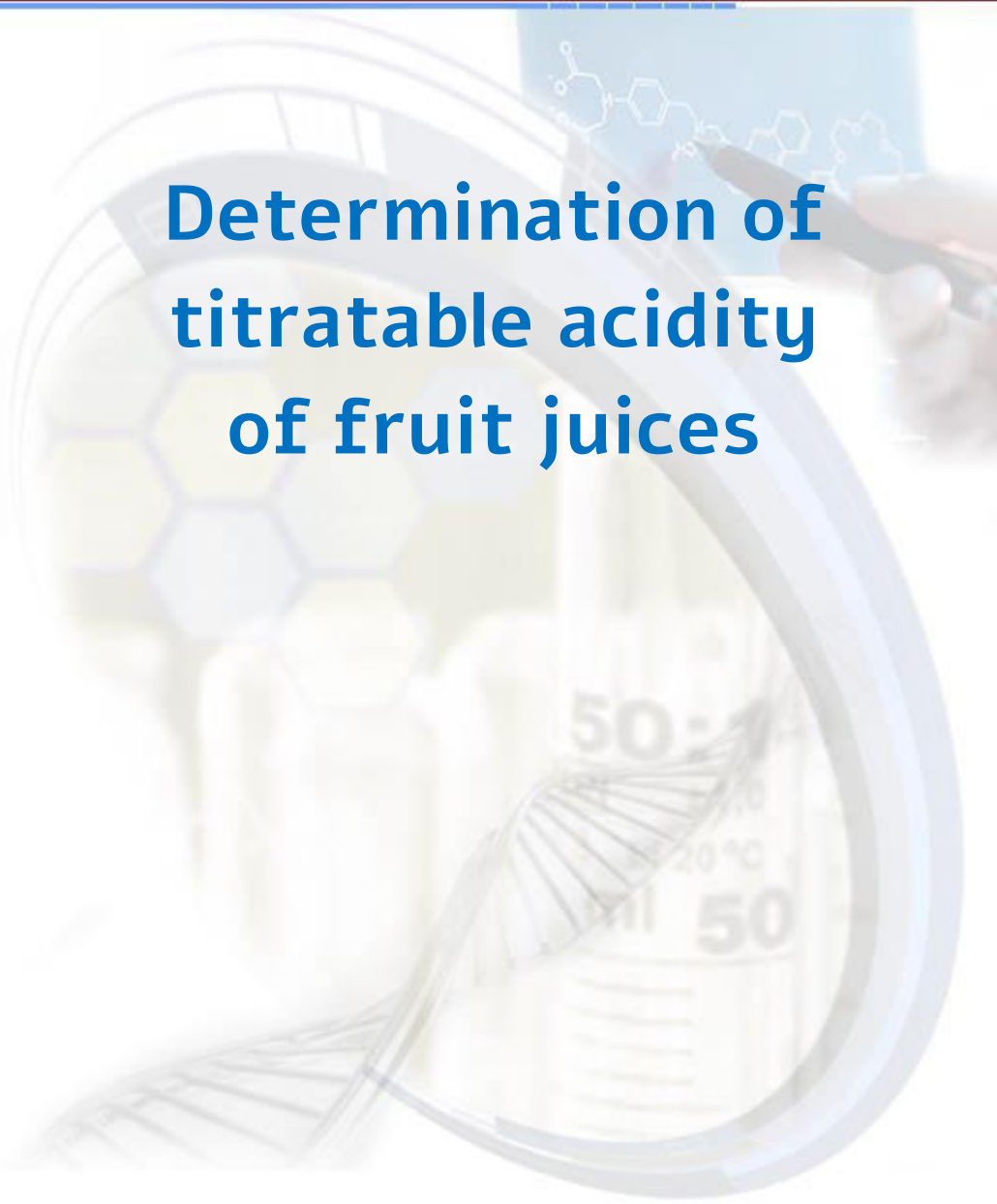
Food Biotechnology

Module # 121



Food Biotechnology

Determination of titratable acidity of fruit juices



Determination of titratable acidity of fruit juices

- The titratable acidity of fruits is used as an indicator of maturity
- To determine the titratable acidity, a known volume of food sample is titrated with a standard base to an endpoint

Determination of titratable acidity of fruit juices

- The volume of titrant used along with the normality of the base and the volume of the sample are used to calculate titratable acidity
- Expressed in terms of predominant organic acid

Determination of titratable acidity of fruit

- pH is used to determine the end point of an acid-base titration
- An indicator dye is used
- Phenolphthalein
- Bromothymol blue

Determination of titratable acidity of fruit

- All food acids are weak acids
- Buffering occurs in food whenever a weak acid and its salts are present in the same medium
- Naturally occurring food acids do not buffer in the region of phenolphthalein endpoint

Determination of titratable acidity of fruit

- Phenolphthalein
- Changes color from clear to red in pH range 8.0 to 9.6
- Bromothymol blue
- Changes color from yellow to blue is pH range 6.0 to 7.6

Determination of titratable acidity of fruit

- Deeply colored samples present a problem for end point indicators
- A potentiometric method is used for such samples
- All indicators are either weak acid or weak base
- Influence titration in excessive amounts
- 2-3 drops used typically

Determination of titratable acidity of fruit

- Sodium Hydroxide is the most commonly used base
- NaOH titrating solutions must be standardized against a standard acid
- Potassium acid phthalate is used (KHP)

Determination of titratable acidity of fruit

- An aliquot sample often of 10ml is titrated with a standard alkali solution (often 0.1N NaOH) to a phenolphthalein endpoint
- Titrant is dispensed at a slow uniform rate until the endpoint is reached
- Then added drop-wise until the endpoint fades

Determination of titratable acidity of fruit

END

- Calculation of titratable acidity
- % acid (wt/wt) =
- [N x V x Eq wt / W x 1000] x 100
- N= normality of titrant
- V= volume of titrant
- Eq wt = equivalent weight of predominant acid
- W = mass of sample
- 1000 = factor (mg/g)

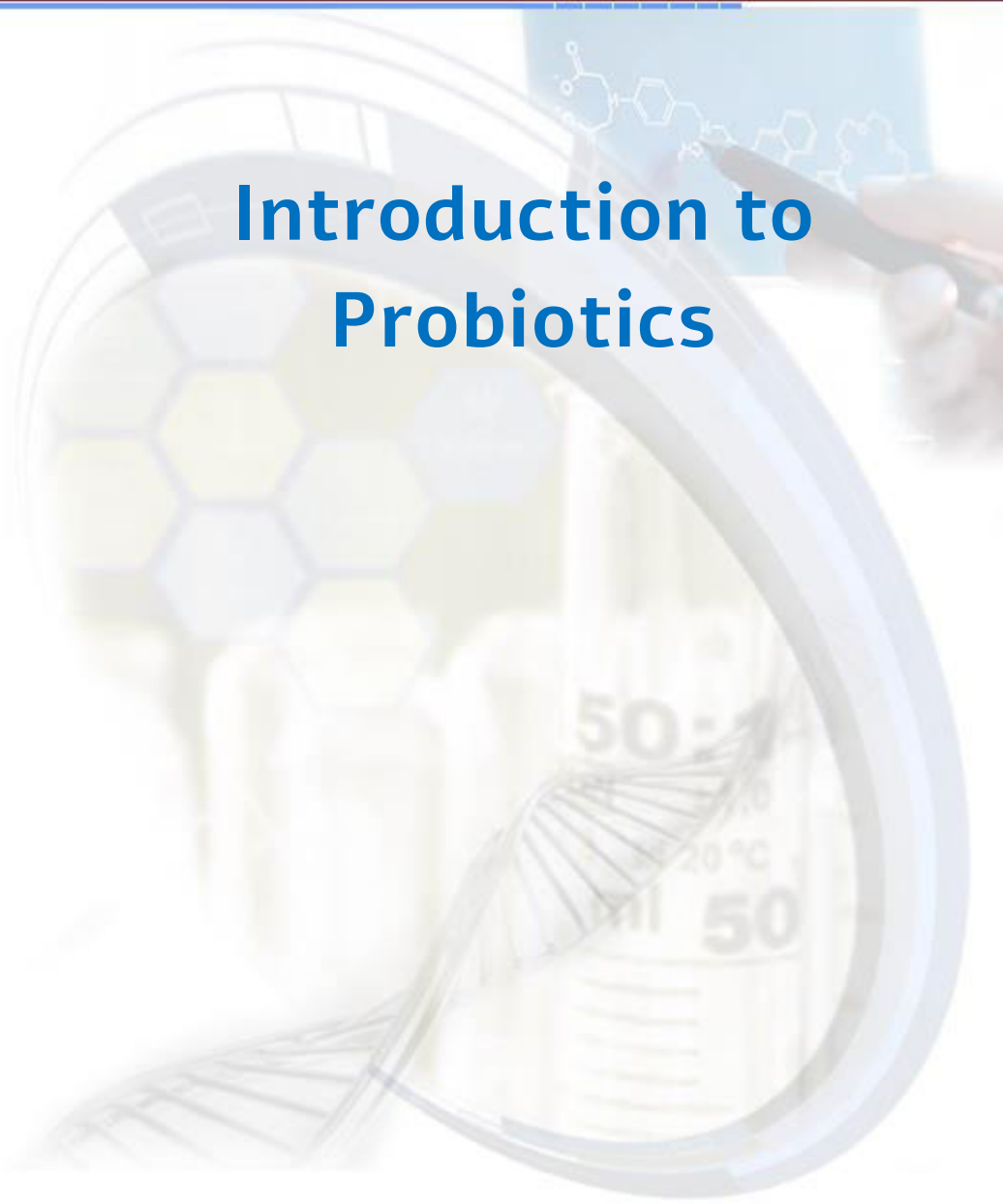
Food Biotechnology

Module # 122



Food Biotechnology

Introduction to Probiotics



Introduction to Probiotics

- Probiotics
- Best defined as live microorganisms which when reaching the intestines in sufficient numbers will exert positive effects
- Viable microorganisms that promote or support a beneficial balance of the microbial population of the GIT

Introduction to Probiotics

- Recent research has underlined the importance of a vital and “healthy” microbial population of the GIT

- The expression “probiotic” was probably first defined by Kollath in 1953

Introduction to Probiotics

- Although numerous definitions have been proposed
- None has been completely satisfactory
- Need for additional explanations, e.g., with regard to statements such as:
- “beneficial balance,” “normal population,” or “stabilization of the gut flora.”

Introduction to Probiotics

- A particular feature of probiotic cultures is that they regulate the balance of the gut bacterial population,
- By competition for epithelium contact sites and nutrients
- Also by modulation of the pH value

Introduction to Probiotics

END

- May not necessarily be constant inhabitants of the GIT
- Practically all organisms used in probiotic foods or food supplements are representatives of the genera Lactobacillus, Enterococcus or Bifidobacterium

Food Biotechnology

Module # 123



Food Biotechnology

Administration and Consumption of Probiotics

Administration and Consumption of Probiotics

- Probiotics may be administered in different forms

- Traditional fermented foods
- Food products
- Pharmaceuticals

Administration and Consumption of Probiotics

- Traditional fermented food
- Lactic acid fermented food of plant or animal origin
- Consumed in raw or uncooked state
- Presence of LAB strains with ability to survive passage of GIT

Administration and Consumption of Probiotics

- Food products
- Novel-type fermented food containing strains selected for functional properties
- Probiotic “yoghurts”
- Nutritional supplements as capsules, tablets
- Cereal based food containing live microorganism and prebiotics

Administration and Consumption of Probiotics

- Pharmaceuticals
- Viable microorganisms selected for their functional properties
- Administered in capsules
- Sometimes as micro-encapsulated lyophilised powders

Administration and Consumption of Probiotics

- A continued increase is observed among the dairy-type probiotic foods
- Also in the range of nondairy probiotic food products such as:
- Fermented meats
- Vegetable
- Fruit juices

Administration and Consumption of Probiotics

END

- Wide range of potential (fermentable) substrates
- Different conditions under which LAB strains may be challenged for “functional performance
- Expected developments toward new food-based probiotics

Food Biotechnology

Microbiota of GI Tract

Microbiota of GI Tract

- Increasing microbial populations are found throughout the GIT
- Many microbes isolated from the duodenum and jejunum are considered to be typical transients

Microbiota of GI Tract

- The estimated total population of 10^{14} viable bacteria in the adult human GIT represents about 10 times more than all body tissue cells

Microbiota of GI Tract

- Microbial population represents an immense metabolic potential
- Supports the digestion processes
- Interactive in detoxification and toxification processes
- Comprises the major part of the human immune system

Microbiota of GI Tract

- Bacteroides
- Eubacterium
- Bifidobacterium
- Predominate in the densely populate large intestine

- Role of the lactobacilli more important in small intestines where they comprise a higher proportion of the total population

Microbiota of GI Tract

- Considered as beneficial bacterial groups of major importance to the gut ecosystem, special attention was given to the Gram-positive genera Lactobacillus
- Bifidobacterium,
- And more recently, to Eubacterium

Microbiota of GI Tract

- LAB in the gut are mainly represented by the lactobacilli
- In contrast to their domination in the ileum, they only form a minor population in the colon
- Make up about 1% of the total bacterial population of human feces
- May be more numerous in the proximal colon

Microbiota of GI Tract

- In contrast to the oxygen-tolerant lactobacilli
- Study of anaerobic groups such as bifidobacteria and eubacteria was made possible by the development of anaerobic techniques developed in the early 1970s

Microbiota of GI Tract

- Compared to their domination in infants, bifidobacteria comprise only up to 3% of the total fecal bacteria of humans

- A reduced environment and special media are applied for the selective cultivation of bifidobacteria

Microbiota of GI Tract

- Functions of normal gut bacteria:
- Maintenance and restoration of barrier function
- Stimulation of the immune system
- Maintenance of mucosa nutrition and circulation

Microbiota of GI Tract

- Functions of normal gut bacteria:
- Improvement of bioavailability of nutrtients
- Stimulation of bowel motility and reduction of constipation

END

Food Biotechnology

Module # 125



Food Biotechnology

Probiotic Microorganisms

Probiotic Microorganisms

- Since at least 40 years in Japan and
- More than 30 years in Germany
- LAB cultures of human origin are applied in the manufacture of fermented milk products

Probiotic Microorganisms

- Viable probiotic strains with beneficial functional properties are at present found among a wide and diverse number of microbial species and genera

- Most strains currently in use as probiotics in food, nutrition, and in pharmaceutical preparations are members of the LAB

Probiotic Microorganisms

TABLE 1.5

Microorganisms Reported to Find Application as Probiotics Mainly for Humans

Lactobacillus Species	Bifidobacterium Species	Other LAB	"Non-lactics"[c]
L. acidophilus	*B. adolescentis*	*Ent. faecalis* [a]	*Bacillus cereus* ("toyoi")[a,c]
L. amylovorus	*B. animalis*	*Ent. faecium*	*Escherichia coli* (Nissle 1917)[c]
(L. casei)	*B. bifidum*	*Sporolactobacillus inulinus* [a]	*Propionibacterium freudenreichii*[a,c]
L. crispatus	*B. breve*		*Saccharomyces cerevisiae* ("boulardii")[c]
L. delbrueckii subsp. bulgaricus[c]	*B. infantis*		
L. gallinarum[a]	*B. lactis*[b]		
L. gasseri	*B. longum*		
L. johnsonii			
L. paracasei			
L. plantarum			
L. reuteri			
L. rhamnosus			

[a] Mainly for animals.
[b] Synonym of *B. animalis*.
[c] Mainly in pharmaceutical preparations.

Source: Modified from Holzapfel et al. (39).

Probiotic Microorganisms

- A number of “nonlactic” strains, e.g., *Bacillus cereus, Escherichia coli, and Saccharomyces cerevisiae*

- Available in the market mainly as pharmaceutic preparations and some also as animal feed supplements

Probiotic Microorganisms

- Three major requirements need to be met for approval of functional foods

1. Scientific evidence of the efficacy, including clinical testing
2. Safety for consumption
3. Analytical determination of the effective component

Probiotic Microorganisms

- Functional food products contain primarily strains of different bacteria

Probiotic Microorganisms

END

- Biotherapeutics for clinical applications are also based on selected probiotic strains
- Mainly LAB but may also include:
- *Escherichia coli* strains
- *Saccharomyces cerevisiae*
- *Enterococcus faecium*
- *E. fecalis*

Food Biotechnology

Module #126



Selection of appropriate strains

- The primary objective is to select microbial strains with functional properties.
- Requires efforts and resources for selection of an “ideal” strain.
- Desirable technical features and factors related to health promotion or sustaining health are important selection.

Selection of appropriate strains

Five key criteria for the selection of an appropriate functional strain:

1. General aspects:
2. Origin,
3. Identity, and
4. Resistance to mutations.
5. Technical aspects (growth properties *in vitro* and during processing, survival and viability during transport and storage)

Selection of appropriate strains

- **General aspects**
- resistance against environmental stress and to the antimicrobial factors prevailing in the upper GIT as encountered during the stomach-duodenum passage)

Selection of appropriate strains

- **Functional aspects and beneficial features:**
  - adhesion,
  - colonization potential of the mucosa,
  - competitiveness,
  - specific antimicrobial antagonism against pathogens,
  - stimulation of immune response,
  - selective stimulation of beneficial bacteria etc.

Selection of appropriate strains

- **Safety aspects:**
- No invasive potential
- No transferable resistance against therapeutic antibiotics
- No virulence factors.

Selection of appropriate strains

Major obstacles towards the large-scale production of functional foods containing probiotic strains like:

- Technical properties related to growth,
- Adaptation
- Persistence of some probiotic strains.
- The sensory properties of the resulting products.

Selection of appropriate strains

Use of *Lactobacillus* strains as probiotics.

- Association with traditional fermented foods, together with the high acceptability of lactic fermented foods.
- Their association with the human GIT, together with observations on their beneficial interactions in the gut ecosystem

Selection of appropriate strains

For the regulatory authorities,

- the safety and
- Non-pathogenicity of a new strain is considered of major importance.

END

Food Biotechnology

Module #126



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END

Food Biotechnology

Module #127



Health benefits of Probiotics

NUTRIONAL BENEFITS:

- Vitamin production, availability of minerals and trace elements
- Production of important digestive enzymes (e.g., β-galactosidase) for alleviation of lactose in-tolerance.

Health benefits of Probiotics

Barrier, Restoration, Antagonistic Effects Against:

- Infectious diarrhoea (traveller's diarrhoea, children's acute viral diarrhoea)
- Antibiotic-associated diarrhoea, irradiation-associated diarrhoea.

Health benefits of Probiotics

Cholesterol-lowering Effects By:

- Cholesterol assimilation.
- Modification of bile salt hydrolase activities.
- Antioxidative effect.

Health benefits of Probiotics

Stimulation and improvement of the immune system:

- Strengthening of nonspecific defence against infection.
- Increasing phagocytic activity of white blood cells.
- Increasing IgA production.
- Regulating many immune functions.

Health benefits of Probiotics

Enhancement of bowel motility,

(relief from constipation & reduction of inflammatory or allergic reactions) **by:**

- Restoration of the homeostasis of the immune system.
- Regulation of cytokine synthesis.

Health benefits of Probiotics

END

Adherence and colonization resistance, Anti carcinogenic effects effects in the colon by:

- Mutagen binding
- Inactivation of carcinogens, or prevention of their formation.
- Modulation of metabolic activities of colonic microbes.
- Immune response.

Food Biotechnology

Module #128



Health Trials of Probiotics

- Human trials (clinical or dietary intervention studies) are essential for proving health benefits of probiotic strains.

Health Trials of Probiotics

Different designs have been applied such as

- pre- and postintervention designs and
- placebo-controlled designs,
- parallel and
- crossover designs, and
- case-control studies.

The best evidence is probably coming from

- double-blind,
- randomized,
- placebo- controlled trials.

Health Trials of Probiotics

- Epidemiologic evidence relating probiotics or probiotic containing foods and disease incidence would be valuable but is hardly available.
- The studies would be difficult to control as fundamental parameters such as:
- specific strain and dose would be unknown for most probiotic-containing food products.

Health Trials of Probiotics

- In human studies certain markers are usually applied in order to observe the effect of pre- and probiotics or functional foods in general on the human body.

Health Trials of Probiotics

END

Markers can be classified into three categories:

a) Markers that relate to the exposure to the food component under study.

b) Markers that relate to the target function or biological response.

c) Markers that relate to an appropriate intermediate endpoint.

Food Biotechnology

Module #129



Technological application of probiotics in food products

Application of probiotic cultures in food products.

- Probiotic bacteria are applied in many different products worldwide.
- In addition to food products, probiotic cultures are also used in pharmaceuticals and animal feed.

Technological application of probiotics in food products

Dairy products

- applied in fermented dairy products for many years.
- In some cases fermented milk products are monocultures of probiotic bacteria.
- Usually support cultures are applied to speed up the acidification process and provide the desired texture and flavor.

Technological application of probiotics in food products

Dairy products

- Many *lactobacilli* and *bifidobacteria* survive in fermented milk products for 4 to 8 weeks.

Technological application of probiotics in food products

Several parameters influence the growth and survival :

- the starter culture,
- fermentation temperature,
- pH,
- sugar content,
- presence of oxygen,
- packaging material,
- fruit preparations, and other ingredients.

Technological application of probiotics in food products

DAIRY PRODUCTS

Probiotics may also be applied to unfermented milk products such as milk-based sweet or acidified drinks and ice cream.

Technological application of probiotics in food products

The applicability of probiotics in food products depends

- water activity,
- processing and
- storage temperature,
- shelf life,
- oxygen content,
- pH,
- mechanical stress,
- salt content, and
- content of other harmful or essential ingredients.

Technological application of probiotics in food products

Other food products

- Products with an unfavourable water activity are, for example, cereals, chocolate, marmalade, honey, and toffees.
- These products are too “dry” for applying live bacteria and too “wet” for the application of freeze-dried bacteria.
- Freeze-dried bacteria could be applied in these products.

Technological application of probiotics in food products

Other food products

- Lactic acid bacteria have been used for the fermentation of meat products,today some strains are also utilized as protective cultures.
- Freeze-dried probiotic bacteria are applied to infant nutrition powders and powdered milk drinks.

Technological application of probiotics in food products

END

Food supplements and over-the-counter products

- Most probiotic food supplements and over-the-counter (OTC) products are available as powders, tablets, and capsules.

Food Biotechnology

Module #130



Concept of food safety

- Food safety is the assurance that food will not cause harm to the consumer when it is prepared and eaten according to its intended use.
- All requirements relating to the safety characteristics of a food must be met; there must be no unacceptable health risk associated with a food.

Concept of food safety

The assurance that a food will not cause harm, injury, or illness is determined by:

1) whether all harmful substances present in the food have been eliminated, reduced to an established acceptable level, or prevented from exceeding the acceptable level;

Concept of food safety

2) the food has been prepared, handled, and stored under controlled and sanitary conditions in conformance with practices prescribed by government regulations.

Concept of food safety

END

The harmful substances in foods are food safety hazards and the prescribed conditions and practices for preparing, handling, and storing food are considered GMPs

Food Biotechnology

Module #131



Concept of food quality

- Food quality, as distinct from food safety, is the extent to which all the established requirements relating to the characteristics of a food are met.
- Common examples of quality characteristics of food, excluding the food safety characteristics, are:
- Identity of a food in relation to a standard (e.g., standardized food)

Concept of food quality

- Declared gross or net quantity (e.g., weight or volume) of a unit of the food or net fill of a food container
- Declared or claimed amount of one or more stated components of a food.
- Appearance (e.g., size, shape, color)
- Flavor
- Aroma
- Texture

Concept of food quality

- Viscosity
- Shelf-life stability
- Fitness for use as human food
- Wholesomeness
- Adulteration
- Packaging
- Labeling

Concept of food quality

- Some of these quality characteristics are covered in food laws and regulations.
- For instance, failure of a food to meet regulatory requirements relating to a standard of identity, the declared quantity, declared ingredients, or label claims, can be considered as misrepresentation, misbranding, or fraud.

Concept of food quality

- The spoilage, deterioration, or decomposition of foods with the absence of any resulting harmful substance that can lead to illness or injury, can be considered as failure to meet food quality requirements based on fitness for human use or wholesomeness criteria.

Concept of food quality

- In addition to the quality requirements established by government regulations, numerous requirements for food quality characteristics are also established by customers and consumers.

END

Food Biotechnology

Module #132



Responsibility of Food safety

The overall responsibility for food quality and food safety is shared by all segments of the food system, including

- the various food industry sectors,
- government regulatory agencies, and
- consumers in general.

Responsibility of Food safety

- The food industry has both the legal and moral responsibility for providing customers and consumers with foods that meet all established quality and safety requirements.
- Within a food company, overall responsibility for the implementation and effective use of these programs and systems rests with senior management.

Responsibility of Food safety

Governments worldwide have enacted food laws and regulations designed to ensure that foods are fit for human consumption. Such laws protect consumers from

- harm resulting from unsafe foods and
- from deception resulting from misrepresentation
- or fraud relating to certain established food quality characteristics

Responsibility of Food safety

Governments have also established various agencies that enforce these food laws and regulations; this legal framework is intended to provide consumers with confidence in the safety and quality of foods.

Responsibility of Food safety

Within the food supply chain, customers who purchase

- raw materials,
- ingredients and
- food contact packaging materials for manufacture of consumer foods,

must ensure that these materials are safe and fit for use.

Responsibility of Food safety

In particular, customers and consumers must pay attention to the instructions for handling, storage, preparation, and use of foods.

END

Food Biotechnology

Module #133



Factors of Food Quality and Systems

Food quality is the extent to which all the established requirements relating to the characteristics of a food are met.

Common examples of quality characteristics of food, excluding the food safety characteristics, are:

- Identity of a food in relation to a standard (e.g., standardized food)
- Declared gross or net quantity (e.g., weight or volume) of a unit of

Factors of Food Quality and Systems

- The food or net fill of a food container
- Declared or claimed amount of one or more stated components of a food
- Appearance (e.g., size, shape, color)
- Flavor
- Aroma
- Texture
- Viscosity
- Shelf-life stability
- Wholesomeness
- Fitness for use as human food

Factors of Food Quality and Systems

- Adulteration
- Packaging
- Labeling

Factors of Food Quality and Systems

Characteristics covered in food laws and regulations are:

- Failure of a food to meet regulatory requirements relating to a standard of identity,
- the declared quantity, declared ingredients,
- or label claims, can be considered as misrepresentation, misbranding, or fraud

Factors of Food Quality and Systems

Based on fitness for human use or wholesomeness criteria these can be considered as failure:

- The spoilage, deterioration, or decomposition of foods with the absence of any resulting harmful substance that can lead to illness or injury, can be considered as failure to meet food quality requirements based on fitness for human use or

Factors of Food Quality and Systems

Numerous requirements for food quality characteristics are also established by customers and consumers.

Factors of Food Quality and Systems

The food industry, used basic

- quality control programs,
- and more complex quality assurance programs and
- quality management systems,

in its efforts to achieve food quality; some food companies use the ISO 9000 Quality Management System Standard.

Factors of Food Quality and Systems

END

GMPs and the HACCP system can be integrated into a food industry,

- quality management system, or
- inspection and monitoring of materials, products, and
- processes for food safety hazards can be part of a quality control program.

Food Biotechnology

Module #134



Food Safety and Food Safety Systems

Systems and Programs for Food Safety

Food Safety and Food Safety Systems

- For decades, the food industry has depended on the use of quality programs based on inspection and testing of food products for hazards, and on GMPs for addressing food safety.
- Since the late 1980s, there has been widespread use of the HACCP system.
- HACCP system is designed to achieve food safety:
- Addresses food safety primarily on the basis of prevention or elimination of unacceptable hazard levels.

Food Safety and Food Safety Systems

- The GMPs, were used to address food safety requirements prior to the use of the HACCP system.
- GMP have also been incorporated into prerequisite programs for the HACCP system.
- A food company that does not operate with the HACCP system must continue to use the GMPs.

Food Safety and Food Safety Systems



Food Safety and Food Safety Systems

- **Food laws and regulations**
- The legal requirements protecting consumers and ensuring that foods are fit for human consumption
- scope varies from one country to another.
- The legal framework of food laws and regulations of a particular country depends on the overall government regulatory system of that country.

Food Safety and Food Safety Systems

- In the U.S. and Canada, the federal or national food laws are statements of government policies that cover both the general and specific aspects of adulteration and misbranding of foods, while the food regulations deal with the enforcement of government policies that are embodied in the food laws.

Food Safety and Food Safety Systems

Food laws and regulations apply to all foods produced

- domestically, as well as all
- foods imported into a country;
- foods cannot be imported if they do not conform to the food laws and regulations of the importing country.

Food Safety and Food Safety Systems

Examples of food laws are the:

- U.S. Federal Food, Drug, and Cosmetic Act (FDCA), which is the primary law governing the safety and quality of most foods in the U.S.
- Canada's Food and Drugs Act, which is the primary food law in Canada.

Food Safety and Food Safety Systems

The U.S. Code of Federal Regulations (CFR) Title 21 and Canada's Food and Drug Regulations are examples of food regulations that address food safety and food quality.

- .

Food Safety and Food Safety Systems

END

- Food laws protect consumers from illnesses and injury by prohibiting the presence of any poisonous or harmful substance in foods that are intended for human consumption.
- For example, in the U.S., adulterated food is regulated primarily under the FDCA, which covers all aspects of food safety and certain aspects of food quality.

Food Biotechnology

Module #135



Food Laws and Regulations

The responsibility for enforcing is assigned to regulatory agencies.

Enforcement activities fall into two categories:

First, they include

- inspection and audit of establishments that process, handle, and store food to ensure that the required sanitary and controlled conditions are followed;

Food Laws and Regulations

- audits are used by some regulatory agencies that enforce HACCP-based regulations.
- Second, they include inspection and analysis of foods for harmful substances to ensure that there is conformance to established limits and tolerances.

Food Laws and Regulations

- When a misbranded/ harmful substance or agent is detected in a food, it is determined that there is a likelihood for a harmful substance or agent to be present in a food, or an actual food-borne illness or injury occurs,
- food companies and government regulatory agencies take the necessary action to protect consumers.

Food Laws and Regulations

END

When such thing happen it often result in

- the food being recalled from the marketplace.

In addition, if it is determined that adulterated or misbranded food has been produced as a result of negligence on the part of a food company, legal action can be taken against the company.

Food Biotechnology

Module #136



Food Standards

- Food standards in addition to food laws and regulations also establish requirements for the safety and quality of foods.
- A food standard is part of food regulations (e.g., standard of identity in the U.S. CFR Title 21), it is not a legal requirement.
- The Codex Standards are the best examples of food standards.

Food Standards

- The Codex Alimentarius Commission has the mandate to implement the joint Food and Agricultural Organization (FAO)/ World Health Organization (WHO) Foods Standards Program.
- This has resulted in the Codex Alimentarius, a collection of standards for food quality, food suitability, and food safety.

Food Standards

- The food standards have been adopted by countries worldwide and are intended primarily to protect consumers and to facilitate international food trade.

Food Standards

END

- They include codes of practice such as:
- The Codex General Principles Of Food Hygiene, standards for maximum residual levels (MRL) for pesticides and for veterinary drugs in foods, and
- standards for specifications for food additives.

Food Biotechnology

Module #137



Food Safety Hazards and Health Risk

Food safety hazards and health risks

Food Safety Hazards and Health Risk

The safety of a food can be related directly to certain harmful substances that are present in the food; these substances are food safety hazards.

Any substance that is reasonably likely to cause harm, injury or illness, when present above an established acceptable level, is a food safety hazard.

Food Safety Hazards and Health Risk

There are three recognized categories of food safety hazards:

- biological hazards,
- chemical hazards, and
- physical hazards

Food Safety Hazards and Health Risk

The origin of these hazards in foods can be from:

- naturally occurring substances or
- agents in foods, from
- deterioration or decomposition of foods,
- or from contamination of the foods with the hazard at various stages of their production, harvesting, storing, processing, distribution, preparation, and utilization.

Food Safety Hazards and Health Risk

For many hazards, government regulatory agencies have established an acceptable level of the hazard in a food;

- the Codex Alimentarius has also established acceptable levels of certain hazards as part of its Food Standards Programme.

Food Safety Hazards and Health Risk

- For a known food safety hazard, the extent of the harmful effects of the hazard on the health of the consumer is established by risk analysis and by hazard analysis.
- Risk analysis is usually conducted by a national food or health regulatory agency and addresses a public health concern regarding a particular food safety hazard associated

Food Safety Hazards and Health Risk

A risk analysis is comprised of:

- risk assessment,
- risk management, and
- risk communication.

- A primary objective of risk analysis is to establish a national food safety objective for a hazard in a food.

Food Safety Hazards and Health Risk

- The food safety objective for a hazard is the maximum frequency and concentration of a hazard in a food at the time of consumption that provides the appropriate level of protection from the hazard.
- The food safety objective can be considered as the maximum acceptable level for the hazard in a food.

Food Safety Hazards and Health Risk

- Hazard analysis is the first of the seven HACCP principles, and is performed to determine the health risk associated with a hazard present in a food when it is produced, processed, handled, or stored, according to an established sequence of steps at a particular location.

Food Safety Hazards and Health Risk

END

- Once a food safety objective for a hazard has been established by risk analysis, it must be considered during the hazard analysis step of HACCP plan development

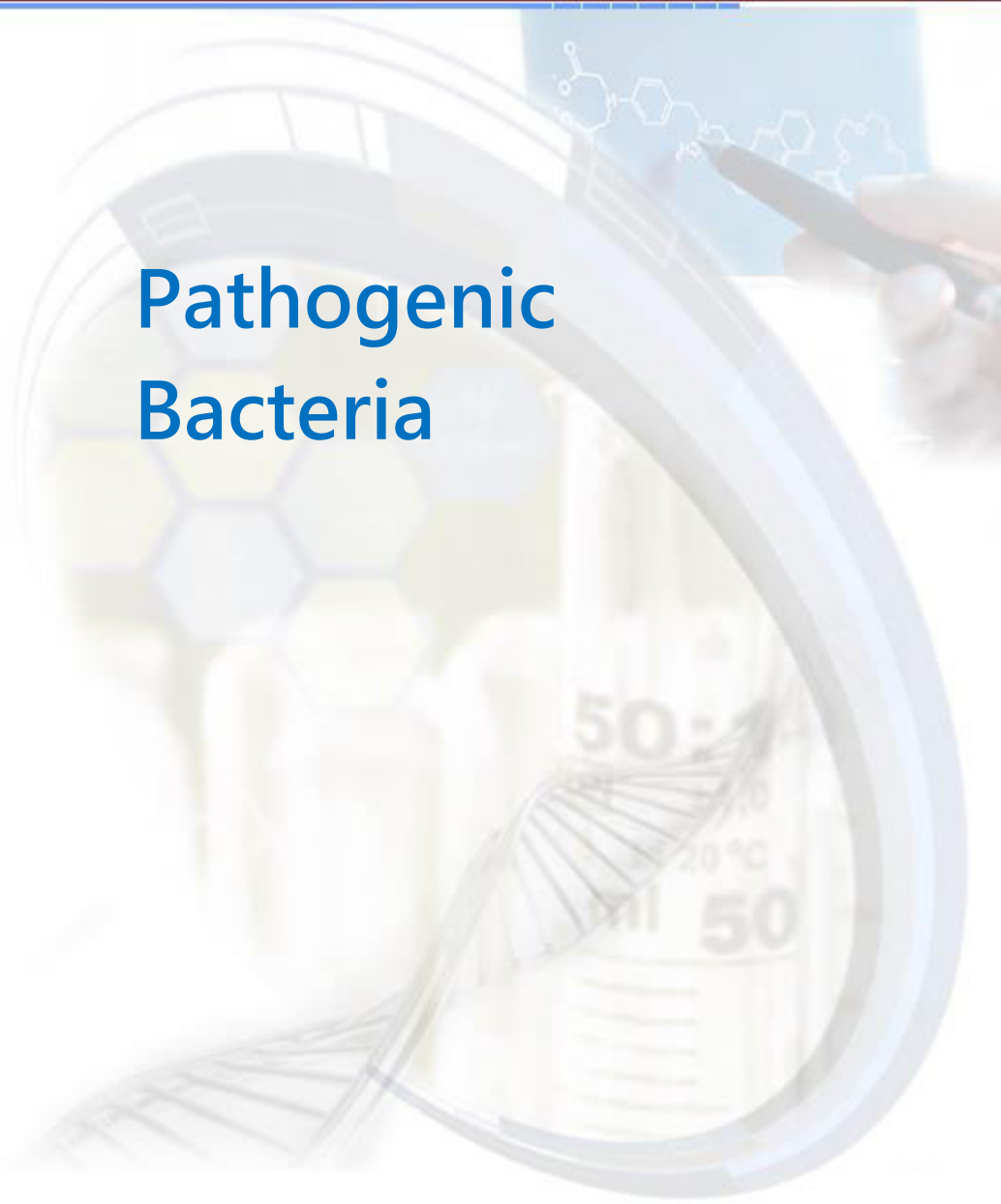
Food Biotechnology

Module #138



Biological Safety Hazards

Pathogenic Bacteria



Biological Safety Hazards

More than forty different pathogenic bacteria are known; however, following are most commonly associated pathogenic bacteria:

- *Salmonella* spp.,
- *Eschericha coli 0157:H7,*
- *Lysteria monocytogenes,*
- *Clostridium perfringens,*
- *Clostridium botulinum,*
- *Staphylococcus aureus,* and
- *Campylobacter jejeuni*.

Biological Safety Hazards

Food poisoning from these organisms occur frequently, with symptoms that include:

- headache,
- muscle pain,
- nausea,
- fatigue,
- chills or fever,
- stomach or abdominal pain,
- vomiting, and diarrhea.

Biological Safety Hazards

Numerous severe and fatal illnesses occur as a result of food poisoning from pathogenic bacteria; infants and the elderly are particularly vulnerable.

Biological Safety Hazards

The foods that are commonly involved in these food poisoning incidents include:

- meat and poultry and their products,
- seafood and seafood products,
- egg and egg products,
- milk and dairy products,
- fruits and vegetables and their products,
- low-acid canned foods, and water.

Biological Safety Hazards

Viruses:

- Foods can be the medium for transmission of certain viruses.
- Examples of viruses that are known to be food safety hazards are :
- the hepatitis A and E viruses,
- the Norwalk group of viruses, and
- rotavirus.

Biological Safety Hazards

END

Parasites:

- parasitic protozoan species (e.g.,
  - *Entamoeba histolytica,*
  - *Giardia lamblia,*
  - *Cryptosporidium parvum*), and
  - parasitic worms (*Ascaris lumbricoides,*
  - *Taenia solium, Trichinella spiralis*).

Food Biotechnology

Module #139



Food Biotechnology

Chemical Safety Hazards

Chemical Safety Hazards

Permitted food additives

- These substances are intended to impart some improved nutritional effect
- (e.g., vitamin fortification) or some specific technical function
- (e.g., preservative action, sensory attribute, stabilizing effect, etc.).

Chemical Safety Hazards

Naturally occurring harmful compounds

- Many foods contain inherent components that can be harmful if they are present in excess of certain levels; examples are:
  - oxalate in rhubarb,
  - alkaloids in potatoes,
  - toxins in mushrooms and in shellfish.

Chemical Safety Hazards

Unavoidable contaminats

- harmful substances that are not normal or inherent components of the foods.
- cannot be removed through processing or manufacturing practices; examples are
- aflatoxins from molds in peanuts and in some cereals.

Chemical Safety Hazards

Agricultural residues

- They include residues of permitted pesticides,
- herbicides,
- fungicides,
- drugs,
- hormones, and antibiotics.

Chemical Safety Hazards

Industrial contaminats

- Several harmful chemicals that enter the environment as a result of industrial activity have been shown to be present in foods.
- These substances include heavy metals,
- organo-chlorinated compounds such as polychlorinated biphenyls (PCBs).

Chemical Safety Hazards

Chemical residues

- These substances include chemical compounds used for cleaning and sanitizing food contact surfaces of processing, handling, and
- storage equipment, and for lubricating certain parts of food processing equipment.

Chemical Safety Hazards

PROHIBITED CHEMICALS

- No chemical substance is permitted for use in a food unless it meets all of the requirements that are covered in the applicable food laws and regulations.

Chemical Safety Hazards

END

Food allergens

- Most common food allergens:
  - peanuts,
  - soybeans,
  - milk, eggs,
  - fish,
  - crustacea,
  - tree nuts, and wheat.

Food Biotechnology

Module #140



Physical Safety Hazards

Broken glass

- In a food plant, the common potential sources of broken glass include light bulbs, glass containers, and gauges with glass covers.
- In addition, many foods are packaged, distributed and sold in glass containers. For these foods, the glass packaging itself can be a source of broken glass.

Physical Safety Hazards

Plastics:

- Both hard and soft plastic foreign objects are sometimes found in foods.
- In some food plants, some utensils and tools used for cleaning of equipment are made from hard plastic material.

Physical Safety Hazards

Metal pieces

- The most common sources of metal pieces in a food plant are food processing equipment, metallic cleaning tools, and equipment maintenance activities.
- In many food plants, magnets are used to eliminate some metals from foods, and metal detectors are used to detect the presence of metals in foods.

Physical Safety Hazards

Wood pieces

- The most common sources of wood pieces in a food plant are wood structures and wood pallets.
- The presence of these sources should be avoided whenever possible in food processing and production.

Physical Safety Hazards

Stones:

- Many plant foods and particularly field crops such as peas and beans can contain small stones that become incorporated with the foods during harvesting.
- In addition, in food processing plants, a common source of stones is concrete structures, particularly concrete floors.

Physical Safety Hazards

Personal Articles:

- Personal articles that have been found in foods include jewellery, pens or pencils or their parts, Band-Aids, and ear plugs.

Physical Safety Hazards

END

Other food safety concerns

- These include concerns relating to the safety of foods from biotechnology and particularly from genetically modified organisms, the safety of irradiated foods, and the safety of some herbal supplements and botanical products.

Food Biotechnology

Module 141



Food Biotechnology

Good Manufacturing Practices (GMP)

Good Manufacturing Practices (GMP)

(GMPs) that have been mandated by government agencies on the basis of scientific knowledge relating to known health hazards in foods, and the need to prevent unacceptable levels of these hazards or to eliminate them from food

Good Manufacturing Practices (GMP)

→ Some GMPs Addresses Quality assurance

→Consequently, the traditional GMPs, along with some additional requirements, are now universally regarded as Prerequisite Programs for the implementation of an HACCP system

Good Manufacturing Practices (GMP)

GMP programs or they should cover the basic controls required for raw materials, ingredients, packaging materials and products, and for the plant's facilities, employees, equipment, operations, and environment that influence the safety of a food.

Good Manufacturing Practices (GMP)

Components of GMP

1) GHP (Good Hygiene Practices)

2) GPP (Good Personnel Practices)

3) GVP (Good Vetrinary Practices)

→ Healthy, Fit , Clean animal

4) Good Sanitary Practices

Good Manufacturing Practices (GMP)

5) Good raw material practices

→ No cross contamination
→ Proper storage
→ Air, humidity ventilation

END

6) Good environment practices

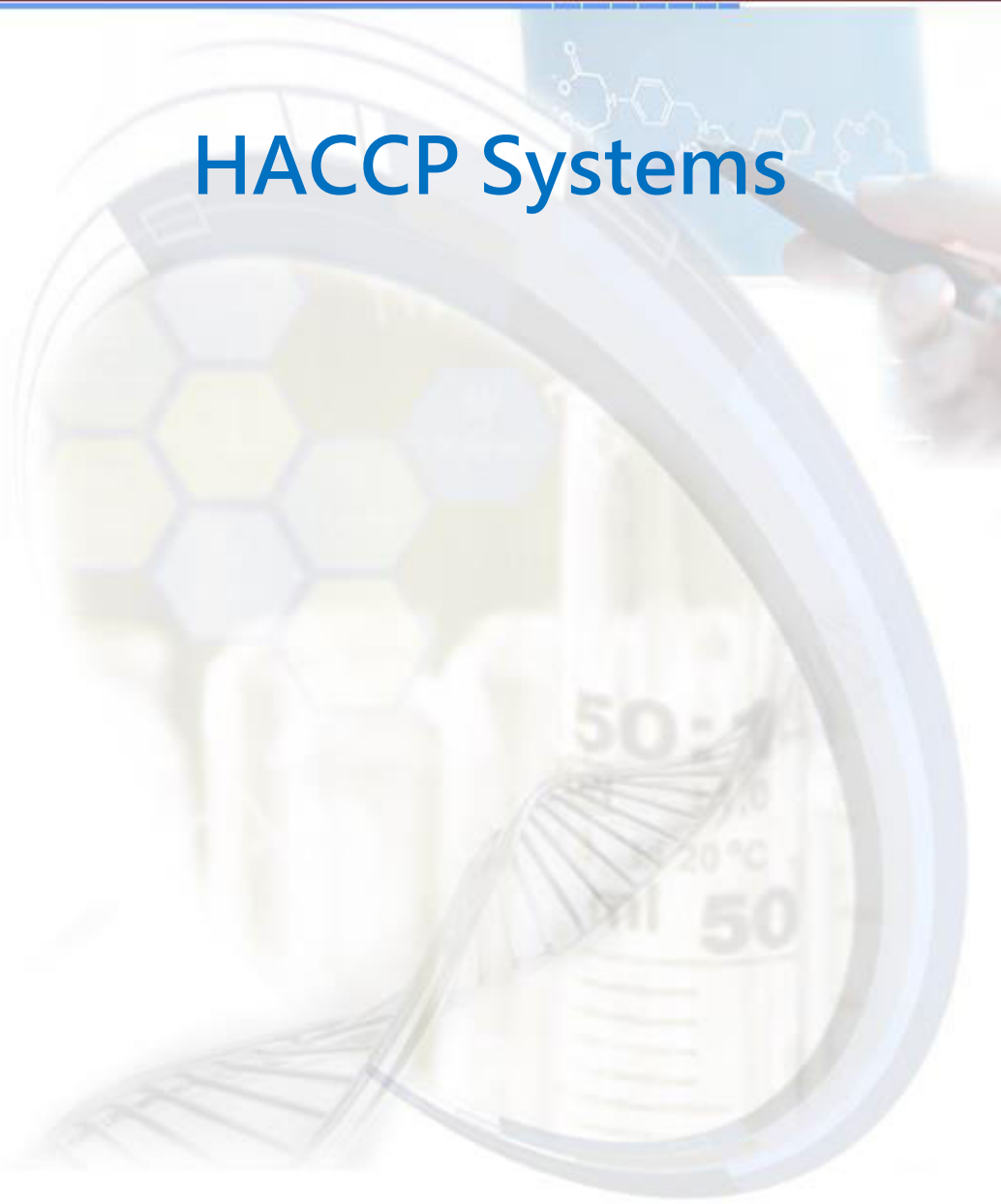
Food Biotechnology

Module 142



Food Biotechnology

HACCP Systems



HACCP Systems

HACCP is a systematic preventive approach to food safety from biological, chemical and physical hazards in production processes that can cause finished product to be unsafe and designs measures to reduce these risks to a safe level

HACCP Systems

Pre-requisite of HACCP :-

→ Different for different countries and foods

In Canada, Agriculture and Agri-Foods Canada Food Safety Enhancement Program (FSEP), HACCP prerequisite programs are covered by the following: Premises, Transportation and Storage, Sanitation, Equipment, Personnel Training, Recall Program

HACCP Systems

USDA have Food Safety and Inspection Services (FSIS) mandatory Pathogen Reduction: HACCP inspection program for the meat and poultry sector.

HACCP Systems

HACCP prerequisite programs are covered in sections on Establishment's grounds and facilities, Equipment and utensils, Sanitary operations, Employee hygiene, Tagging unsanitary equipment, utensils, rooms or compartments and mandatory Sanitation standard operating procedures (SSOPs).

HACCP Systems

Seven Principles of HACCP:-

1) Conduct a hazard Analysis

2) Identify Critical Control Points :-

Ccps are identified

For Milk Processing Ccps are

→ Pasteurization

→ Cooling Temperature

HACCP Systems

3) Establish Critical Limits :-

→ After the identification of Ccps Critical limits are established in order to control Ccps

e.g. For pasteurization Critical limits are 72ºC / 15 secs

HACCP Systems

4) Monitor Critical Limits :-

When Ccps are identified and Critical limits are set then Critical limits are monitored in order prevent safety hazards

HACCP Systems

5) Establish Corrective Actions :-

When Critical limits crossed then Corrective actions are taken in

→ Corrective actions are costly, time consuming so better monitoring is better than corrective actions

HACCP Systems

6) Verification :-

verification is done in order to see whether the corrective actions overcomes the problem or not

7) Record Keeping :-

→ Record of each and every step is compulsory in order to have better future plan

END

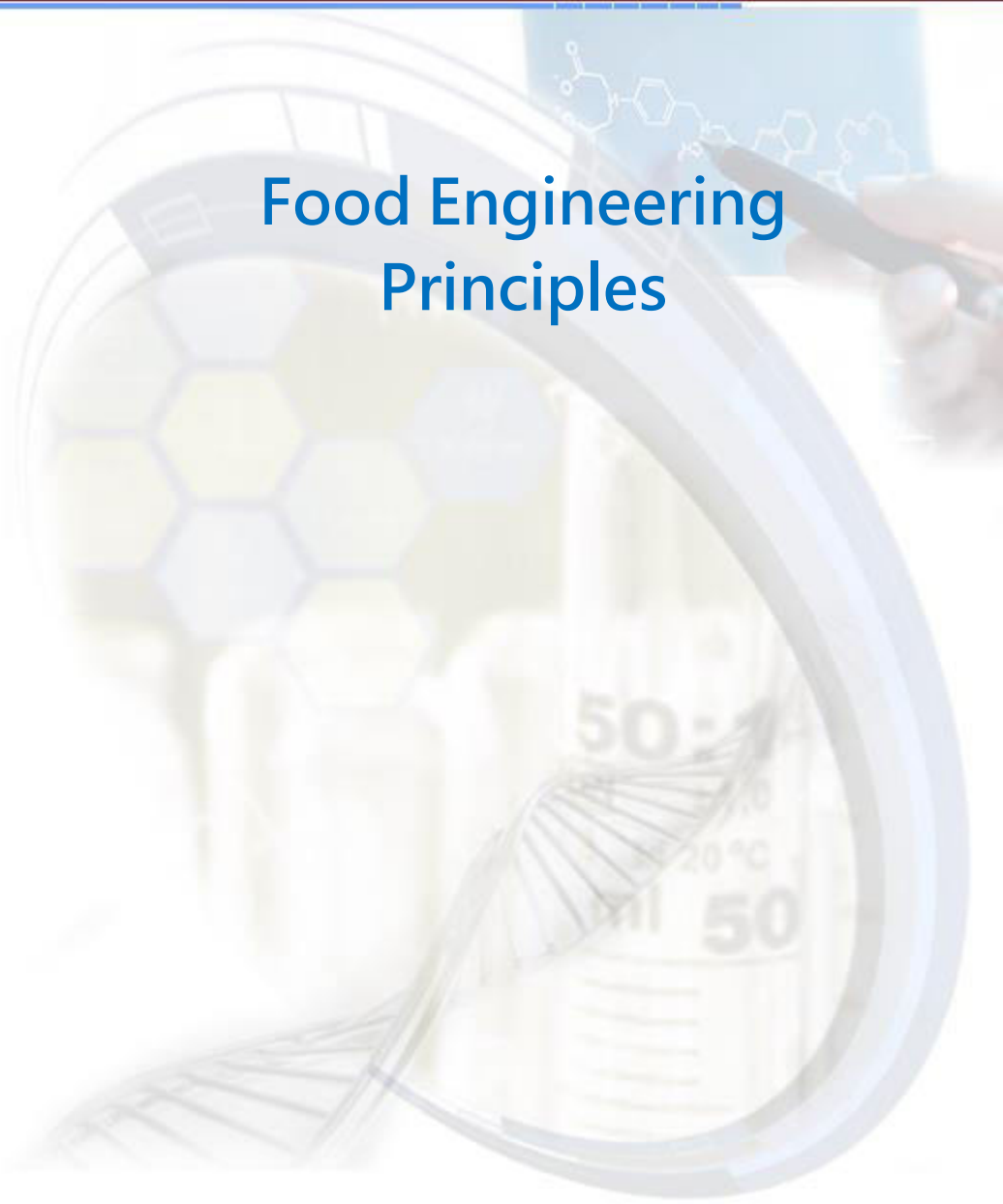
Food Biotechnology

Module 143



Food Biotechnology

Food Engineering Principles



Food Engineering Principles

Converting raw food into desired products involves equipment & processes. To design equipment, an engineer must consider numerous criteria that are inherent to a process. For example, in designing a heat exchanger, one must consider the heat transfer, fluid flow, and various physical, chemical, and biological aspects of food.

Food Engineering Principles

Overall engineering process involve energy balance and material balance.

Overall process can be broken down into series of operations. These are called unit operations.

Efforts are directed towards maximizing efficiency and reducing losses in every unit operations.

Food Engineering Principles

The essential concept is therefore to divide physical food processes into basic unit operations, each of which stands alone and depends on coherent physical principles. e.g. heat transfer is a unit operation and the fundamental physical principle underlying heat transfer principles.

Food Engineering Principles

Because of the dependence of the unit operation on a physical principle, or a small group of associated principles, quantitative relationships in the form of mathematical equations can be built to describe them. The equations can be used to follow what is happening in the process, and to control and modify the process if required.

Food Engineering Principles

Important unit operations include fluid flow, heat transfer, drying, evaporation, contact equilibrium processes (which include distillation, extraction, gas absorption, crystallization, and membrane processes), mechanical separations (which include filtration, centrifugation, sedimentation and sieving), size reduction and mixing.

Food Engineering Principles

The law of conservation of mass and energy is applied. Hence losses are calculated and discouraged in both materials and energy for best optimization of process.

END

Food Biotechnology

Module 144



Food Biotechnology

Importance of hygienic design and layout



Importance of hygienic design and layout

Hygiene is vital for the **production of a quality of a food product**.

The product quality depends not only on the **capability and safety of the process**

Preventive measures like **GMP, HACCP, maintenance**, etc.

But also on **the quality of the equipment** used.

Importance of hygienic design and layout

The efficiency of cleaning procedures is also **highly dependant** on **the design and installation** of equipment.

Therefore, when considering food safety, **hygienic design** and **appropriate cleaning** are, together with GMP, hygiene, HACCP, maintenance, etc., **essential preventive measures**.

Importance of hygienic design and layout

A prerequisite for high quality and safe products is **correct hygienic design** and **maintenance of the production systems**.

A correct hygienic design of process equipment has a tremendous impact on **diminishing the risks of contamination** of food during production.

Importance of hygienic design and layout

This also means that the shelf life of products is **lengthened**.

If the process equipment is of **poor hygienic design** residues (soil) may be retained in crevices and dead areas,

Which would allow micro-organisms **to persist, multiply** and then **cross-contaminate** subsequent batches of product.

Importance of hygienic design and layout

Poor hygienic design can affect the **equipment manufacturer**, the **food processing client**, **government legislative bodies** and the **final consumer**.

Many cases of food poisonings are caused by **incorrect equipment design**.

Importance of hygienic design and layout

For example the presence of salmonella in milk powder with the evidence of 76 cases of illness and 1 event of death were caused in 1985 in UK by contaminated water.

The source of contamination was traced to the fact that **birds had access to water** as a result of an **inadequate closure** of a washing water tank.

Importance of hygienic design and layout

Good hygienic building design **should minimize** the possibility of infestation by insects,

- birds,
- animals,
- micro-organisms,
- any accumulation of dust, surface water or condensed water or product.

These events are all **risk of contamination**.

END

Food Biotechnology

Module 145



Food Biotechnology

Food process equipment design



Food process equipment design

Risks due **to poor hygienic design** are mainly caused by **incorrect positioning of equipment and utility installations** by **horizontal surfaces**, **hollow bodies**, **dead spaces**, **bad drainage**, **insufficient cleanability / accessibility**, **non-resistant materials** etc.

Food process equipment design

The main issues of hygienic design are:

1. to **avoid contamination by foreign organisms** and materials like animals, insects, birds, micro-organism, migrations, construction/ equipment parts, lubricants

Food process equipment design

2. residues of other products and cleaning agents, etc.

3. to **avoid conditions which enhance the growth of micro-organism**, like accumulation of dust, water, moisture (condensation) and product

4. to **improve cleanability**

Food process equipment design

Product contact surfaces must be **inert** to the product as well as to

- detergents,
- disinfectants
- or sanitizers under the intended conditions of use.

Food process equipment design

They must **not migrate to, or be absorbed by, the food** and have to be

- corrosion resistant,
- non-toxic,
- mechanically stable,
- and their surface finish must not be adversely affected under conditions of use.

Food process equipment design

To ensure clean food processing equipment, the product contact surface has an important role. If the surface is rough and/or porous it will most likely allow food particles to build up and it will be more difficult to clean when compared with a smooth, polished surface.

Food process equipment design

The contact surface must not chemically interact with the food and its ingredients. It should be free of corrosion and it must be inert to the chemicals used during cleaning

Food process equipment design

→If the contact surface is hidden during visual inspection, it would be impossible to know whether it is properly cleaned.Therefore,all contact surfaces must be accessible during inspection and cleaning.

Food process equipment design

3)Incorporate design features that prevent harboring microbial accumulation and their growth

→ Porous surface → Microbial accumulation

→ Non accessible surfaces can also accumulate micro organisms and their growth

Food process equipment design

→ Equipment's that accumulate dust , water , oil etc. should be kept away from food to avoid cross contamination

Food process equipment design

Control measures in order to have control over cross contamination form

→ Equipment to material
→ Minimize microbial hazards
→ For proper inspection
→ To fulfill GHP
→ Better inspection
→ Minimize hazards

Food process equipment design

→ Motors used to power equipment should be placed where any lubricant used in the motor does not contaminate the product. Direct drive systems are generally preferable

Food process equipment design

→ Another source of contamination in drive systems is bearings. Use of food grade material such as nylon, sealed or self-lubricating bearings are preferable

→ Seals should be nontoxic and nonabsorbent. It should be easy to remove seals for inspection and sanitation purposes

Food process equipment design

→ For processing liquid foods, kettles should be of the self-draining type. For kettles equipped with mixers, the mixer lubricant should not enter the product.

END

→If steam is to be directly injected, then any additives used in the boiling water must be food grade

Food Biotechnology

Module 146



Food Biotechnology

Cleaning and sanitizing in food industry

Cleaning and sanitizing in food premises

Introduction :-

Cleaning and sanitizing in a food processing plant involves a number of steps.

1) Removal of gross soil
2) Chemical agent used for removal of visual soil
3) Rinse of cleaning agent
4) Sanitizer (For micro-organism removal, killing)
5) Final rinse

Cleaning and sanitizing in food premises

Cleaning is influenced by

→Temperature

→Time of cleaning,

→Concentration of chemicals,

→ Mechanical action used in cleaning

Cleaning and sanitizing in food premises

→Use of higher temperatures during cleaning of fat and grease is beneficial; however, it should not be excessively high to cause protein adhesion to a surface.

Cleaning and sanitizing in food premises

→ Many properties of soil are important in determining the ease or difficulty in removing it, for example, particle size,viscosity,surface tension,wettability,solubility of a liquid soil in a solid soil, the chemical reactivity with the substrate, the attachment of soil to a surface or entrapped in voids and attachment.

Cleaning and sanitizing in food premises

The selection of a cleaning compound depends upon:
→the type of soil on the surface; →the type of surface to be cleaned; -
→the amount of soil on the surface; →the method of cleaning (such as soaking, use of a foam, or clean-in-place);
→the type of cleaning agent – liquid or powder;
→ the quality of the water;

Cleaning and sanitizing in food premises

Cleaning involves first the separation of the soil from the surface, followed by dispersion of the soil in the detergent medium. It is important that the soil must not redeposit on the surface. The effectiveness of a detergent is evaluated based on its:

Cleaning and sanitizing in food premises

→penetration and wetting ability; →control of water hardness; →efficient removal of soil

→ease of rinsing;

→non corrosiveness to the surface.

END

→The desired properties of detergents are obtained by proper mixing of selected chemicals

Food Biotechnology

Module 147



Food Biotechnology

Process Control



Process control

Introduction :-

→Process control may be defined as manipulation of process variables so that the desired product attributes are obtained

→ The variables employed in a process have a marked influence on the final attributes of the product

Process control

→ By controlling process variables, the desired consistency of operation is achieved along with reduced costs of production and improvements in safety

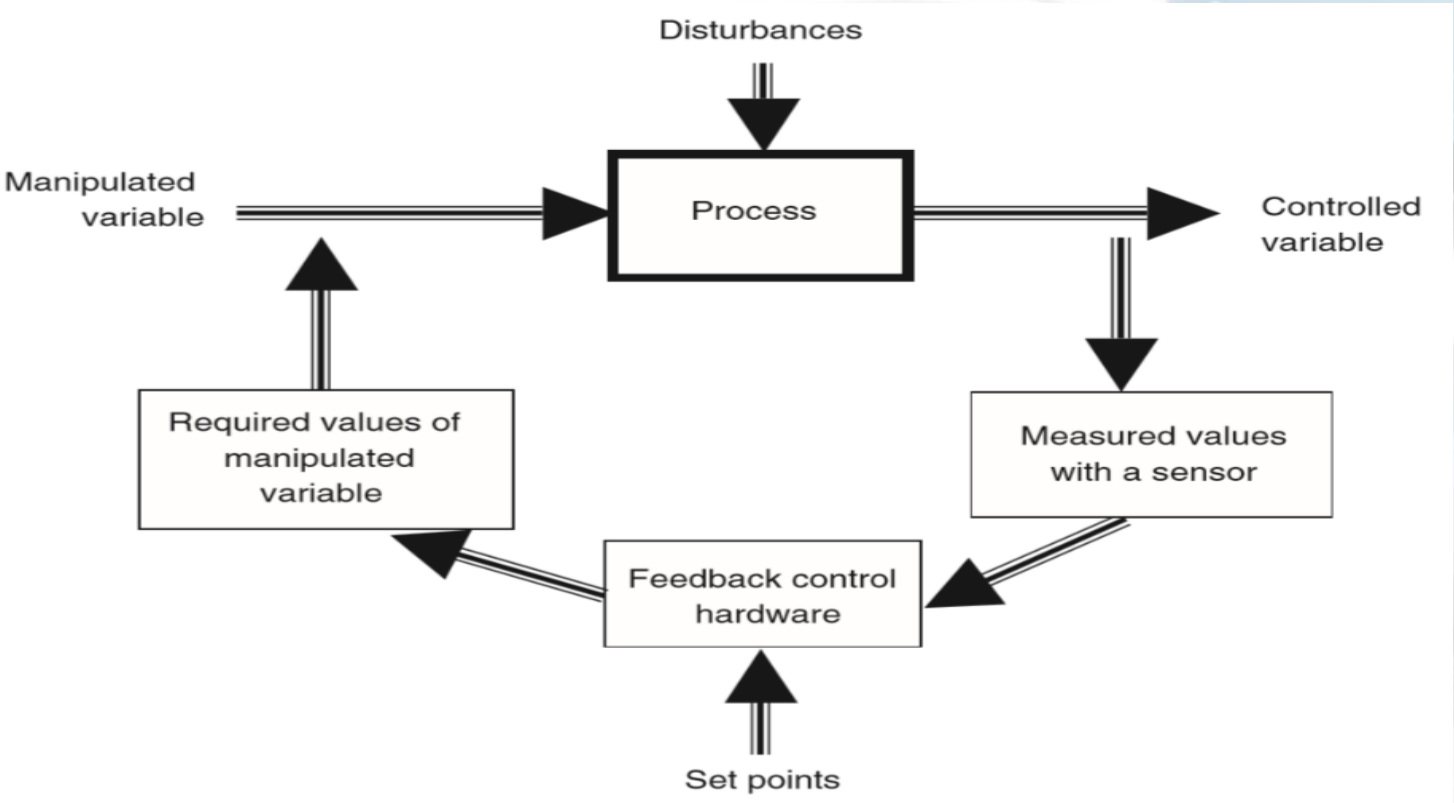
→ When equipment's deviate from what it is designed for, automatic controls provide a higher level of consistency than if human intervention is allowed that often leads to higher levels of variability.

Process control

→ The production productivity is enhanced as less out-of-specification product is produced

→ Automatic controls provide a higher level of screening of unsafe conditions to improve overall safety of the equipment

Process control



In this feed back process control diagram when a disturbance / change in a process or product specifications occurs then a control variable measure deflection or change in process with different sensors and sensor sends feedback to operator then operator make changes and adjustments to have constant and desired production

Process control

→Different sensors are employed in a process to detects undesirable changes during a process

→ Different valves are employed in order to have adjustments

END

Food Biotechnology

Module 148



Food Biotechnology

Handling of solid foods in processing plant

Handling of solid foods in processing plant

→The design and layout of the plant and how materials are handled between various processing equipment have a major impact on the production efficiency

→ In the case of transporting solid foods in a food processing plant, it is typical to consider the movement of product in any direction, horizontal or vertical

Handling of solid foods in processing plant

A variety of conveyors and elevators are used for this purpose. These include

→ belt,
→ chain,
→ screw,
→gravity,
→ pneumatic conveyors,
→ bucket elevators

In some cases fork-lifts and cranes are employed

Handling of solid foods in processing plant

1) Belt Conveyers :-

→ An endless belt operating between two or more pulleys is one of the most ubiquitous conveyors

→ used in transporting solid foods in a processing plant

Handling of solid foods in processing plant

2) Screw Conveyors :-

Screw conveyors are suited for handling

→ powders,
→ sticky
→ viscous products such as peanut butter, and granular materials

Handling of solid foods in processing plant

3) Bucket Elevators :-

→The bucket elevator consists of an endless belt with buckets attached to it

→ Bucket elevators are highly efficient as there is absence of frictional loss between the product and the housing material

Handling of solid foods in processing plant

4) Pneumatic Conveyors :-

In a pneumatic conveyor, particulate food is conveyed in a closed duct by a high-velocity stream of air

Handling of solid foods in processing plant

Belt , screw , pneumatic Conveyors and bucket elevators

increase the efficiency of →material handling
→reduce cost ,
→Minimizes losses and
→ most importantly it is time effective

END

Food Biotechnology

Module 149



Food Biotechnology

Food transport (Screw conveyors)

Food transport (Screw conveyors)

Screw conveyors consists of a helix turning inside a circular or U-shaped trough

Food transport (Screw conveyors)

→ Screw conveyors are suited for handling powders, sticky and viscous products such as peanut butter, and granular materials

→ Screw conveyors are useful to empty silos of flour and powder materials.

→ Often they are used as metering devices

Food transport (Screw conveyors)

→The flights of screw conveyor are made of a variety of materials including stainless steel.

→Their operating power requirements are high,

→They are used for distances less than 25 m

Food transport (Screw conveyors)

→Screw conveyors are suitable for horizontal as well as incline up to20 degrees.

→For horizontal screw conveyors an oval trough is used

→For a steep incline a cylindrical trough is necessary

Food transport (Screw conveyors)

The power requirement of a screw conveyor depends upon a number of factors including:

→the length of the conveyor

→elevation;

→pitch;

→speed;

→type of flights

→weight and properties of the material being conveyed;

→the coefficient of friction between the product and the material of the flights and housing.

Food transport (Screw conveyors)

The startup power requirement of a screw conveyor is generally higher than for continuous operation.

END

Food Biotechnology

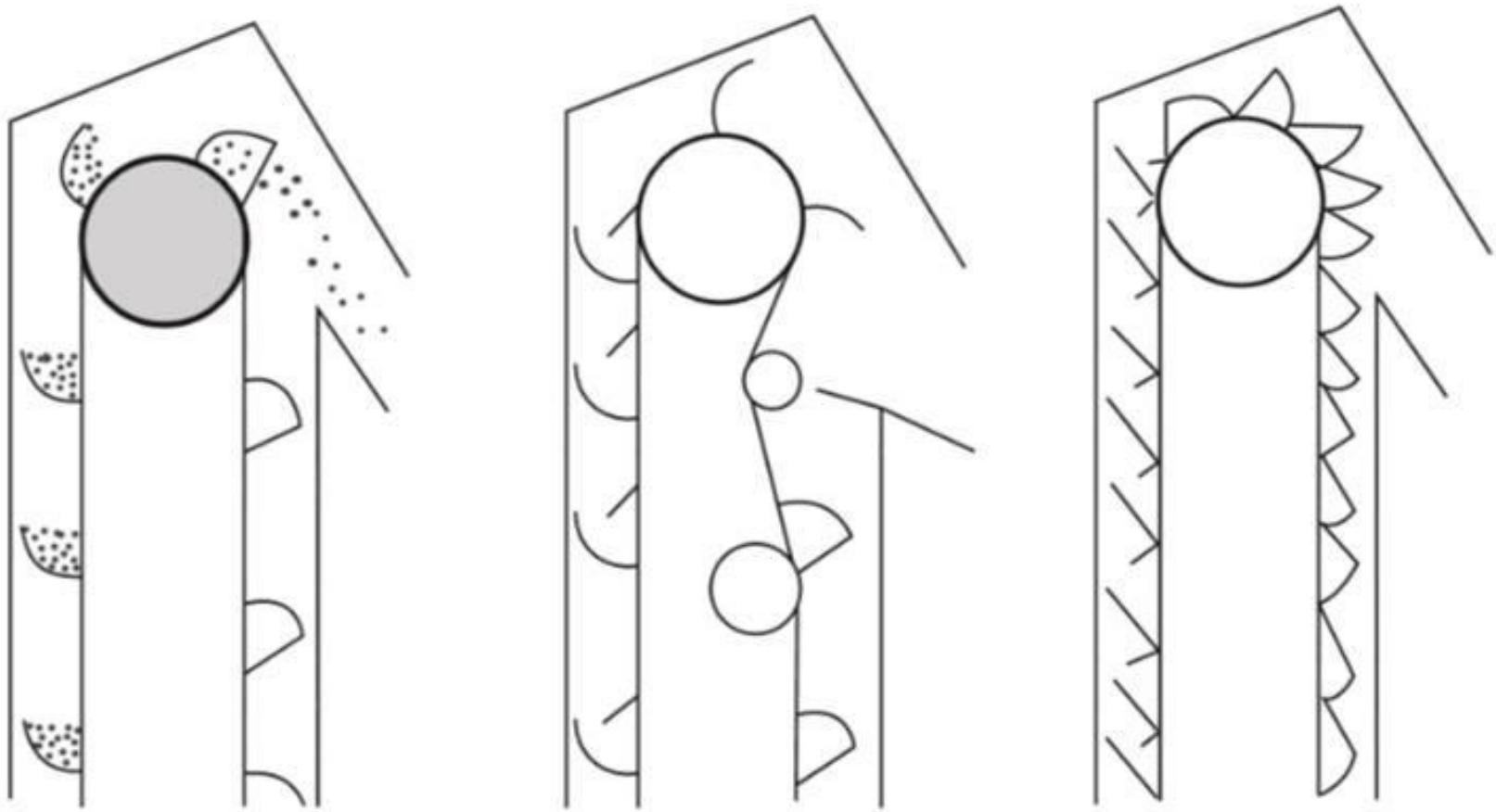
Module 150



Food transport (Bucket elevators)

The bucket elevator consists of an endless belt with buckets attached to it. The belt operates on two wheels; the top wheel is referred to as the head and the bottom is called the foot

Food transport (Bucket elevators)



Food transport (Bucket elevators)

→Bucket elevators are highly efficient as there is absence of frictional loss between the product and the housing materials

→The bucket elevators are enclosed in a single housing referred to as a leg; in certain cases, there turn is housed in a second leg

Food transport (Bucket elevators)

→ A chain or a belt is used to carry the buckets that are shaped with either rounded or sharp bottoms.

→ The belt or chain operates between two wheels – the head and foot. For longer lengths, idlers are installed to prevent belt whip

Food transport (Bucket elevators)

→The product carried by the buckets is discharged at the top when the bucket turns around the head wheel,

→Product is thrown out by the centrifugal force

Food transport (Bucket elevators)

→ The speed of the bucket as it goes around the head wheel must be maintained within limits to ensure that the product is discharged in a desired region

Food transport (Bucket elevators)

The conveying capacity of a bucket elevator depends on the

1)Product density,

2)Belt speed

3)Bucket size

4)Spacing of buckets on the belt

Food transport (Bucket elevators)

→Typical applications of bucket elevators are for Handling cereal grain, animal feed, and meal.

→The energy requirements of bucket conveyors for cereal grains range from 0.1 to 0.2 kWh/m3.

END

Food Biotechnology

Module 151



Food Biotechnology

Storage of Fruits and Vegetables

Storage of fruits and vegetables

→post harvest management, proper techniques of handling and storage of fruits and vegetables are essential to minimize losses.

→The range of post harvest losses varies anywhere from 5 to 50% or even higher

Storage of fruits and vegetables

In industrialized countries, major progress has been made in developing proper systems for the handling of fruits and vegetables.

Storage of fruits and vegetables

1) Respiration process and storage of fruits and vegetables :-

→Fruits and vegetables continue to undergo physiological changes after harvest. These changes are largely the result of the respiration process

→ Oxygen plays an important role in the respiration process

Storage of fruits and vegetables

→Aerobic Respiration (with oxygen)

→ Anaerobic Respiration (Without Oxygen)

Anaerobic respiration results in the production of ketones, aldehydes, and alcohol. These products are often toxic to the plant tissue and hasten its death and decay

Storage of fruits and vegetables

→ So if we limited the amount of O_2 then we can increase the shelf life of fruits and vegetables

2)Temperature Control

Heat is generated as a end product of aerobic respiration so by lowering temperature we can increase the shelf life of product

Storage of fruits and vegetables

3) Ethylene control:

→Climacteric (Produce ethylene and CO_2)

→ Non Climacteric (does not produce ethylene)

Ethylene control also prevent fruits decay

Storage of fruits and vegetables

Climacteric fruits		Nonclimacteric fruits	
Apple	Muskmelon	Blackberry	Olive
Apricot	Nectarine	Cacao	Orange
Avocado	Papaya	Cashew apple	Pepper
Banana	Passion fruit	Cherry	Pineapple
Blueberry	Peach	Cucumber	Pomegranate
Breadfruit	Pear	Eggplant	Raspberry
Cherimoya	Persimmon	Grape	Satsuma mandarin
Feijoa	Plantain	Grapefruit	Strawberry
Fig	Plum	Jujube	Summer squash
Guava	Sapote	Lemon	Tamarillo
Jackfruit	Soursop	Lime	Tangerine
Kiwifruit	Tomato	Loquat	
Mango	Watermelon	Lychee	

Storage of fruits and vegetables

Proper use of low temperature prevents

→ Chilling injuries
→Freezing injuries

Storage of fruits and vegetables

Fruits and vegetables must be stored under

→ Low oxygen
→ High CO_2
→ Proper storage temperature

END

Storage of fruits and vegetables

Fruits and vegetables must be stored under

→ Low oxygen
→ High CO_2
→ Proper storage temperature

END

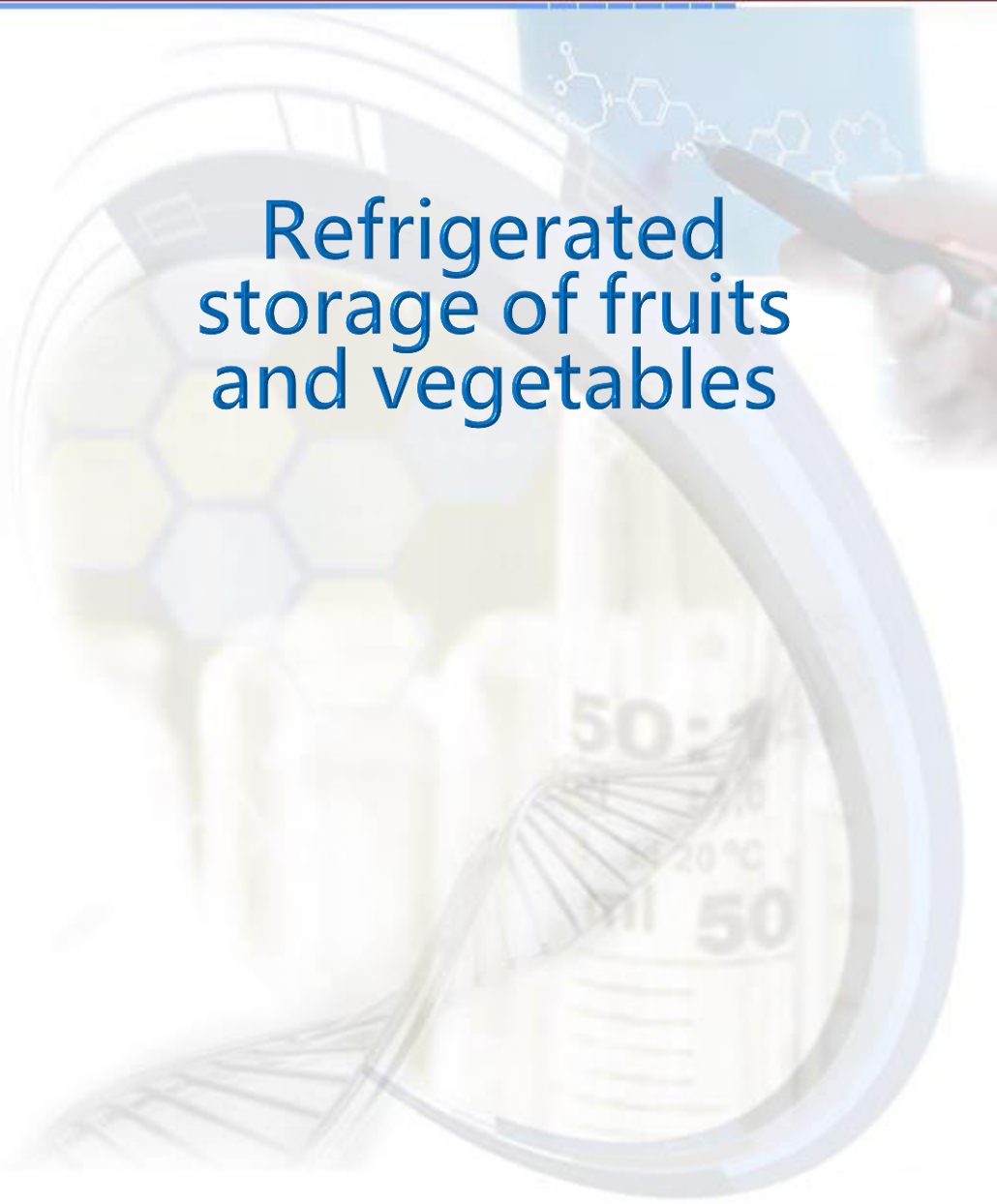
Food Biotechnology

Module 152



Food Biotechnology

Refrigerated storage of fruits and vegetables

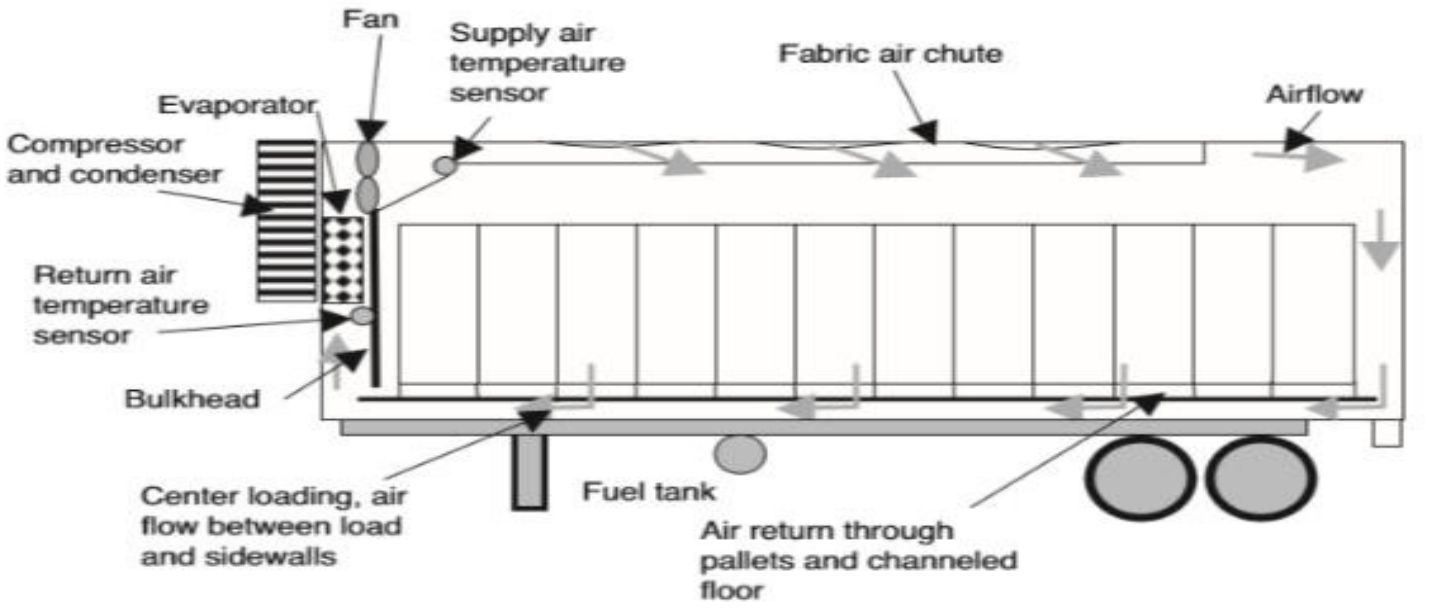


Refrigerated transport of fruits and vegetables

→One of the most common ways to transport perishable foods such as fruits and vegetables is by using a refrigerated trailer that may be either pulled via a truck on a highway or placed in a ship for trans-ocean shipment

→ Any perishable product stored in the trailer must be kept refrigerated for the entire duration

Refrigerated transport of fruits and vegetables



A typical refrigerated trailer, shown includes a refrigeration system to cool air and an air handling system to distribute air within the trailer. It is vital that there is uniform air distribution within the trailer; otherwise regions with no air circulation can lead to product heating and spoilage

Refrigerated transport of fruits and vegetables

→Temperature control of the air circulating inside the trailer is vital to the shipment of perishable foods

→ Modern trailers are equipped with temperature sensors and controllers that automatically control the refrigeration unit based on the temperature of air exiting the refrigeration unit

Refrigerated transport of fruits and vegetables

→ The temperature of the thermostat in these systems is set within 0.5◦C of the long-term storage temperature

→ The trailer used for frozen products should be set at −18◦C or colder.

→ Temperature of frozen product during loading should be less than -12 C

Refrigerated transport of fruits and vegetables

→ When a trailer load contains more than one type of produce, care must be taken that these are compatible in terms of their storage temperature and ethylene sensitivity

→ Ethylene-sensitive vegetables should not be mixed with ethylene-producing fruits

Refrigerated transport of fruits and vegetables

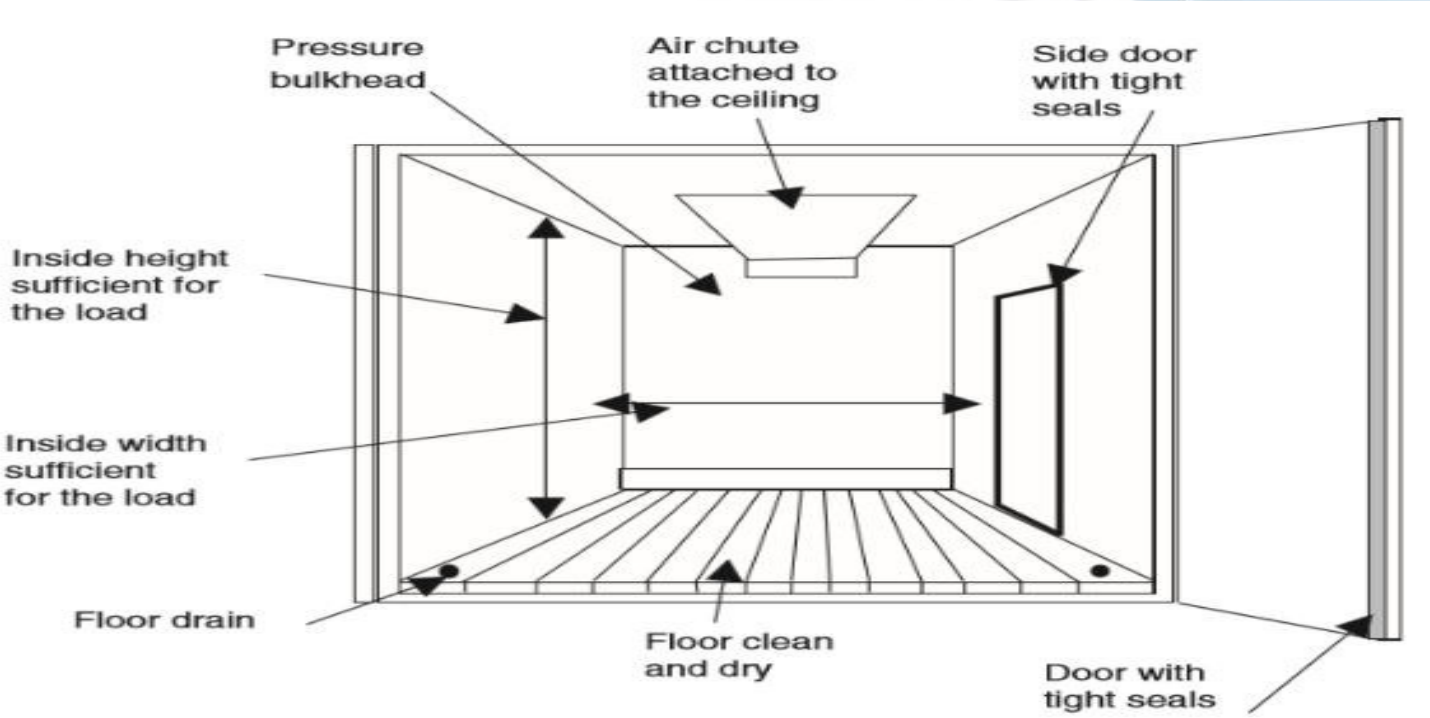


Figure shows a check list for trailer conditions before loading food

Refrigerated transport of fruits and vegetables

→ In long-distance shipping, vibration damage to the produce can be a significant factor.

→ Vibration damage to product is more severe for trailers using steel spring-suspended axles.

Refrigerated transport of fruits and vegetables

→ To prevent vibration loading of material should be done in such a way that there would be no or minimum inter material spaces

END

Food Biotechnology

Module 153



Food Biotechnology

Water quality and waste water treatment in food processing

Water quality & waste water treatment in food processing

→Water use in food processing has increased with the widespread mechanization of harvesting operations; raw agricultural products arriving at a food processing plant require large quantities of water for cleaning

Water quality & waste water treatment in food processing

→Inside the plant, water is used in a variety of processing and handling operations such as product conveying, peeling, blanching, cooling, generating steam, and washing equipment and floors

Water quality & waste water treatment in food processing

→ The quality of water used in food processing depends on the function of water in the manufacturing process. For example, water quality for initial cleaning of raw produce has a different requirement than water required for the formulation of carbonated beverages, beer, and bottled drinking water.

Water quality & waste water treatment in food processing

1)Treatment of hard water :-

If the water has high level of hardness due to the presence of dissolved solids, then processes such as

→ precipitation,

→ion exchange,

→distillation,

→reverse osmosis may be used.

Water quality & waste water treatment in food processing

2)Turbid water Treatment:

Turbid water often pretreated using

→coagulation

→flocculation

→sedimentation

→filtration

Water quality & waste water treatment in food processing

The presence of dissolved organics in water causes

Off-flavor
0ff-odor
Off-color

That is often removed by

→activated carbon adsorption.

Water quality & waste water treatment in food processing

Waste Water Treatment :-

→Primary treatment is often a physico-chemical process involving sedimentation;

→secondary treatment comprises biological treatment with sedimentation

→tertiary treatment involves removal of residual and non biodegradable materials.

Water quality & waste water treatment in food processing

→Dissolved organic matter in the wastewater is removed using biological treatment and adsorption

→Dissolved inorganic matter requires the use of ion exchange, reverse osmosis, evaporation, and/or distillation.

→Any suspended organic matter is typically removed using physico-chemical and biological treatment methods.

Water quality & waste water treatment in food processing

→→Other suspended inorganic or organic content is removed by screening, sedimentation, filtration, and coagulation

END

Food Biotechnology

Module 154



Food Biotechnology

Main techniques used for production of food ingredients by microbes

Main techniques used for production of food ingredients by microbes

→Microorganisms play an important role in the generation of natural compounds, particularly in the field of food ingredients

→ Microbial processes seem to be the most promising methods.

Main techniques used for production of food ingredients by microbes

→ Many microorganisms are capable of synthesizing useful compounds when growing on a culture media

→ They are also an economical source of enzymes, which can be utilized to enhance or alter the flavors of many food products

Main techniques used for production of food ingredients by microbes

→Biotechnological processes involved in the production of flavor compounds can be divided into two groups:

1)Microbiological Methods

→Biosynthesis
→ Biotransformation

2) Enzymatic Methods

Main techniques used for production of food ingredients by microbes

1)Microbial Methods (Biosynthesis and Biotransformation)

→The first is the production of chemical compounds by cells (fermentation or secondary metabolism).

→The second refers to the use of microbial cells in the specific modification of chemical structures

Main techniques used for production of food ingredients by microbes

→In fermentation, the production of flavors starts from cheap and simple sources such as sugars and amino acids.

→ The product is generated by the complex metabolism of the microorganism. When microorganisms are used in order to catalyze specific conversions of precursors and intermediates, the process is called biotransformation

Main techniques used for production of food ingredients by microbes

→Fermentation requires C and N sources, a specific substrate is necessary for microbial transformation. The enzymatic catalysis precedes a simple and specific transformation of the substrate molecule

Main techniques used for production of food ingredients by microbes

END

Flavors obtained from Fermentation :-

→ Yoghurt Flavor
→ Alcohol flavor
→ Wine
→ Beer
→ Pickles etc.

Acidic and alcoholic flavor due to fermentation

Food Biotechnology

Module :- 155



Food Biotechnology

Selection of microorganisms and development



Selection of microorganisms and development

Micro-organisms are

Prokaryotes
Eukaryotes
Autotrophs
Heterotrophs
Gram +ve
Gram –ve
Mesophiles
Thermophiles
Physchrophiles

Selection of microorganisms and development

Selective Enrichment:

Microorganisms of public health significance associated with foods are usually present as a small minority of the total microbial population. For both detection and quantitation of such microorganisms, selective enrichment cultivation is usually undertaken.

Selection of microorganisms and development

Selective enrichment involves conditions of cultivation that favor the development of the target organism over that of the usual majority of extraneous microorganism

Selection of microorganisms and development

For example most culture media for selective enrichment of *E. coli* make use of the fact that the organism is a common inhabitant of the intestinal tract of mammals and hence is normally in contact with surface-active agents such as bile salts, which are frequently used as selective agents for the organism.

Selection of microorganisms and development

Temperature Control for Selective Enrichment :-

→Incubation temperature has also been used successfully with selective chemical agents for the selective cultivation and enumeration of certain bacteria.

Selection of microorganisms and development

→The use of 45.5°C for the most probable number (MPN) enumeration of *E. coli* using the culture broth known as EC functions selectively in conjunction with the content of 0.15% bile salts .

Selection of microorganisms and development

Use of Selective incubation temperature for isolation :-

→ 55-65 ºC (Thermophiles)

→ 20 ºC (Physchrophiles)

Selection of microorganisms and development

(The Use of Mineral Salts Media for the Isolation of Unique Carbon Sources Utilizing Microorganisms)

The isolation of microorganisms capable of utilizing unique carbon sources is greatly facilitated if the organism being sought is capable of growth in a strictly glucose–mineral salts medium.

Selection of microorganisms and development

Unique carbon and energy sources such as methanol, cholesterol, and naphthalene, can then be added as sole carbon sources with the assurance that only organisms capable of attacking these unique substrates will grow.

Selection of microorganisms and development

→ Selective is for selective organisms by providing them suitable conditions

→ By knowing the temperature , pH , Salt etc. requirements of micro-organisms we can selectively grow them

END

Food Biotechnology

Module #156



Food Biotechnology

Culture Media and Upstream compounds



Culture media and upstream compounds

- The ideal culture medium will use inexpensive nutrient requirements.
- These ingredients are crop, animal, marine or yeast based components.
- The culture medium alone can represent 30 to 70% of the fermentation production costs.

Continue

- Slight changes in medium micronutrients can have a major impact on the fermentation.
- Thus, the food industry demands a consistent product from suppliers of these complex components.
- Failure to provide a consistent product will eliminate the commercial use.
- What decides the culture medium makeup?

Culture media and upstream compounds

- Essentially, it is the nutritional requirements of the microorganism of choice and its ability to biosynthesize essential elements such as amino acids, vitamins, lipids, and carbohydrates.
- For example, bacteria and yeast are high in protein (40–50%), whereas molds are not (10–25%).
- Yeast can generally grow in a minimal medium, whereas lactic acid bacteria require a host of micronutrients to grow.

Culture media and upstream compounds

- Each microorganism has different micronutrient requirements.
- Typically the exact micronutrient which benefits the fermentation that is being supplied by the complex nutrient employed is unknown.
- Generally it involves specific amino acids, vitamins, trace elements, and lipids.
- Furthermore, the concentration ratio of carbohydrate to nitrogen and phosphorous has a dramatic impact on microbial growth.

Culture media and upstream compounds

- Carbon–nitrogen imbalances can result in the production of other byproducts such as extracellular polysaccharide, and fermentation end products such as ethanol.
- In the case of ethanol production in yeast, excess glucose in the presence of oxygen will direct yeast to produce ethanol.
- This is called the Crabtree effect.
- Yeast cells typically consist of 48% carbon and 8% nitrogen on a dry weight basis.

Culture media and upstream compounds

- When the C:N ratio is 10:1, yeasts grow aerobically consuming little substrate while producing maximum cell mass, CO2 and H2O, but when the C:N ratio is 50:1 yeasts grow anaerobically, consuming much substrate while producing little cell mass, and much CO2 and ethanol.
- This difference in yeast growth is also called the Pasteur effect.

Culture media and upstream compounds

- Why the increase in industrial microbiology fermentation processes over the past 20 years? It is the result of the corn syrup sweetener industry and computer technology.
- Corn is 70% starch, and when dried to 19%, it can be stored for more than two years.
- glucose syrups from corn is consistent, low cost supply of substrate.

Culture media and upstream compounds

- For example, corn syrups containing 19% dextrose, 14% maltose, 12% maltotriose, and 55% higher saccharides are used to control microbial growth rates and biological heat production in many fermentations.
- Thus, glucose is the platform chemical used for the microbial production of organic acids, amino acids, vitamins, and more.
- In fact some food grade fermentation facilities have located adjacent to a corn sweetener facility to permit glucose syrup to be piped directly to their fermentors.
- Glucose corn syrup is also shipped via truck or rail cars as liquid or dried product.

Culture media and upstream compounds

END

- Furthermore, the advent of computer process controls of industrial scale fermenters has removed many of the fears associated with commercial scale fermentation by providing reliable and easy to operate dissolved oxygen, pH, foam, temperature, and sterilization controls of the process.

Food Biotechnology

Module #157



Food Biotechnology

Bioreactor monitoring systems and design

Bioreactor monitoring systems and design

- The stirred tank bioreactor design is the most common fermentor and consists of agitator, baffles, aeration sparger for aerobic culture growth, sterilizable monitoring probes for pH, dissolved oxygen, temperature, and antifoam, filling and draining ports, and often culture medium sterilization capabilities in the reactor tank.
- Reactor agitation is essential for temperature control, pH adjustments, oxygen absorption into the liquid medium, overall culture health, and mixing of any required additions of substrate, and nutrients.

Bioreactor monitoring systems and design

- Fermentation health requires real time monitoring system.
- Microbial growth can be monitored via several methods.
- The most common method is indirect measurement of biomass by absorbance of the fermentation broth at 620 nm by using a spectrophotometer.

Bioreactor monitoring systems and design

- The measured absorbance values can then be used to estimate biomass concentrations by using a standard curve.
- Standard curves are developed by collecting fresh log phase cells via centrifugation, washing cells with water or 0.1 M ammonium acetate pH 7.0 buffers, then serially diluting the pellet.
- Absorbance for each dilution is then determined spectrophotometrically at 620 nm.

Bioreactor monitoring systems and design

- The actual dry weight biomass (g/l) is determined for each dilution via direct biomass measurement after oven drying of each dilution in pre weighed boats at 70°C for 24–36 hr.
- This needs to be performed in at least replicates of three. By washing biomass with water, any influence of culture medium on dry weights can be eliminated or minimized.
- Finally, a standard curve can be constructed by plotting absorbance versus actual dry biomass weight (g/l).
- This method allows for quick, reliable, and easy conversion of absorbance to dry biomass (g/L).

Bioreactor monitoring systems and design

- For determining microbial health, viable cell counts can be rapidly performed by flow cytometer.
- A two color histogram is collected with gating on the bacteria only population from the two parameter light scatter distribution and is used for the analysis of green only (live bacteria), red only (dead bacteria), and both colors (stressed bacteria).

Bioreactor monitoring systems and design

- Substrate consumption and product formation rates can be monitored by high pressure liquid chromatography (HPLC) or by membrane bound enzymes biosensors, which requires 20 and 1 min to run, respectively.

Bioreactor monitoring systems and design

- Some compounds currently measurable are glucose, ethanol, maltose, lactic acid, and lysine. Sample preparation is simply filtration (0.45 :m) and dilution with water if the value falls outside the instrument window.
- Organic acid production can be continuously monitored via alkali addition rates for pH control.

Bioreactor monitoring systems and design

- Alkali consumption can be easily monitored by feeding alkali solution from a sterile burette.
- Microbial respiration for aerobic cultures can be continuously monitored via dissolved oxygen probe or the concentration of CO2 in the exit gas, which can be monitored via off gas analyzers or simply via alkali (4 N NaOH) traps followed by pH titration.
- However, more technologies are needed to acquire real time measurements of microbial growth, ensuring optimal fermentation time, and product formation in the shortest time possible.

Bioreactor monitoring systems and design

- For rapid analysis of biomass, substrate and product concentrations without any sample preparation, Fourier transform mid infrared (FT-MIR) spectroscopy has been successfully utilized for lactic acid and ethanol fermentation.
- This method can be used for online, real time analysis for monitoring and process control purposes.

ENd

Food Biotechnology

Module #158



Fermentation Types Employed:

The Actual Production Process

Fermentation types employed: The actual production process

- batch and fed-batch fermentation are most common.
- sequence starts with medium sterilization, reactor inoculation, incubation for complete microbial growth cycle with lag, log, and stationary phases, fermentation termination, draining the reactor for product recovery down stream, cleanup of the reactor, and starting over.

Fermentation types employed: The actual production process

- Percentage yield is calculated by taking the concentration of the product formed (g/l) divided by the concentration of substrate consumed (g/l) times 100, whereas productivity is a measure of the product formation rates. It is calculated by dividing the product concentration (g/l) by the fermentation time (hours); thus, it is presented in g/l/hr.

Fermentation types employed: The actual production process

- In fed-batch fermentation, additional carbohydrate is supplied to the batch fermentation during the run.
- High carbohydrate concentrations in the initial culture medium are toxic to many microorganisms.
- Thus, an optimal carbohydrate concentration is employed initially, which permits maximum culture growth to late log phase.

Fermentation types employed: The actual production process

- When the carbohydrate concentration is reduced to almost zero, additional sterile carbohydrate is injected into the bioreactor to bring the carbohydrate concentration back to the starting level.
- When this is consumed, the process is repeated until end product inhibition forces the whole bioconversion to stop. Ideally, at the end of the fermentation you will have a product concentration which is three to four times

Fermentation types employed: The actual production process

- This will generate the highest yield possible. Also, due to increased end product accumulation with each carbohydrate addition, culture production rates will decrease.
- Thus, the decision as to how many fed-batch fermentations to perform before harvest is based on the desired final product concentration and the optimal fermentor use time.

Fermentation types employed: The actual production process

- Continuous fermentations are open fermentations, whereby fresh medium is continually added to the bioreactor, while spent culture medium, cells, and product are continually leaving.
- It is desirable because it is 10 to 100 times smaller than batch fermentations, a steady stream of fermentation product is produced which will optimize downstream processes, bioconversion rates are always at maximum, operation costs are less, and the system can be fully automated and computer controlled to the point where only two operators are needed to manage the fermentation each day.

Fermentation types employed: The actual production process

- However, it requires a continuous supply of sterilized or pasteurized culture medium, dilution rates are linked to microbial growth rates and the operational speed of downstream recovery process.
- Startup is slow, so any facility shut downs have an impact on production, and you are constantly fighting contamination. Thus, not all fermentations can be operated this way.

Fermentation types employed: The actual production process

- Specific growth rate equals dilution rate. Finally a holding tank at the end is used to ensure complete bioconversion of any residual sugars to ethanol.
- The CO2 is collected and concentrated, then sold as another valuable byproduct. Fermented beverages (i.e., wine and beer) are still performed in batch.

Fermentation types employed: The actual production process

END

- For some continuous fermentations, an increased concentration of biomass in the reactor is required.
- This can be achieved by cell recycling, immobilized cell, and biofilm fermentations. Cell recycling reactors employ a filtration unit that allows for the constant bleeding of culture supernatant while retaining biomass.
- However, filtration unit

Food Biotechnology

Module #159



Food Biotechnology

Bioreactors design

Bioreactor design

- Entrapment type is very common bioreactor, where high concentrations of cells are trapped in a polymer matrix.
- high cell density is continuously retained in the fermentor continuously converting substrate to product.
- This results in higher productivities and yields.

Bioreactor design

- Migration of substrate through the matrix to the cell and the migration of product out is disadvantage, similarly end product inhibition, cell leakage from the polymer matrix due to cell growth, and bead swelling and disintegration over time causing the whole fermentation to be stopped, cleaned, and restarted.

Bioreactor design

- Biofilms type bioreactors are a natural form of cell immobilization in which microorganisms are attached to a solid surface.
- In this bioreactor, cells are continually growing, and sloughing off. Thus, the reactor is a mixture of immobilized and suspended cells.
- Continuous biofilm fermentations are truly open immobilized cell bioreactors.

Bioreactor design

- This is similar to a suspended cell continuous fermentation with the added advantage of increase biomass concentrations in the bioreactor.
- Biofilms are typically resistant to harsh conditions.

Bioreactor design

- Filamentous microorganisms such as fungi and actinomycetes are natural biofilm formers.
- For nonfilamentous bacteria to form a biofilm, an extracellular polysaccharide needs to be generated by the bacterium.

Bioreactor design

- Some bacteria will form biofilms on any surface such as metal, plastic, and glass.
- However lactobacilli, require something to stimulate this biofilm development. Plastic composite support (PCS) stimulate biofilm development of *Lactobacillus casei*, *Zymomonas mobilis*, *Saccharomyces cerevisiae*, and *Actinobacillus succinogenes*.

Bioreactor design

- PCS is a high temperature extruded material consisting of at least 50% polypropylene, plus ground soybean hulls, bovine albumin and various culture micronutrients.
- Soybean hulls keep the extruded product porous due to the release of steam as the PCS leaves the extruder die.

Bioreactor design

- Bovine albumin performs as a natural plastizer which protects the temperature sensitive micronutrients.
- Micronutrients are selected based on the specific cultural requirements for amino acids, vitamins, and lipids.
- Monosaccharides are avoided
due to poor PCS production.

Bioreactor design

- For example, the PCS blend for lactobacilli contains 50% (w/w) polypropylene, 35% (w/w) ground dried soybean hulls, 5% (w/w) bovine albumin, 5% (w/w) yeast extract, 5% (w/w) soybean flour, and mineral salts.
- PCS have shown higher percentage yields and productivity rates than suspended cell lactic acid fermentations.

Bioreactor design

- It will continue to grow as a biofilm and have good longevity.
- biofilm reactor washed with concentrated ammonium hydroxide, rinsed with mineral salts solution, and then reinoculated with a fresh culture and medium will reestablish itself overnight.
- quick vinegar process is the most common biofilm application.

Bioreactor design

- Solid substrate fermentation is when a substrate such as soybeans is ground, inoculated with *Aspergillus oryzae*, then incubated for three days for the production of soy sauce.
- It is a simple fermentation process and commonly used for aerobic fermentation due to its large surface area.

Bioreactor design

- Thus, oxygen concentration is high without using any mechanical forced air systems.
- Solid substrate fermentations require large areas or incubation space.
- A temperature controlled environment, intermittent monitoring for contamination and quality of starting material is essential for success.

END

Food Biotechnology

Module #160



Food Biotechnology

Future research in food fermentation



Future research in food fermentation

- Research is still needed for isolation of new microbial strains with improved production efficiencies and higher yields.
- More real time measurements are needed for culture conditions and metabolite formation.
- Recovery will continue to be the key factor associated with final product purity and cost.

Future research in food fermentation

- As an industry we cannot rely solely on genetic engineering as our method of improving current fermentations.
- As we have illustrated, there are many other techniques which can be employed to improve productivity and yield, including new inexpensive medium ingredients, more continuous fermentation processes, and new exotic microbial reservoirs in nature and in the food industry.

Food Biotechnology

Module #161



Food Biotechnology

Genetic modification of plant starches for food applications

Genetic modification of plant starches for food applications

- Most of the maize starch mutants affecting starch structure were discovered before biotechnology understanding.
- names were based upon the phenotype or on the properties of the starch, rather than that of the gene or enzyme affected by the mutation.

Genetic modification of plant starches for food applications

- For example, waxy, amylose extender, dull, shrunken, sugary-2, and sugary) have been described extensively with regard to their effects on carbohydrate composition and response to genetic background, allelic dosage, or interaction with other mutations.

Genetic modification of plant starches for food applications

- Examination of maize kernels with differing starch phenotypes has been instrumental in determining which enzymes are required for starch synthesis in this storage organ.
- Mutations that are responsible for most of the abnormal starch phenotypes have been located in genes encoding starch synthetic enzymes.
- Related isoforms, for which there are as yet no mutants available, have also been identified and characterized.

Genetic modification of plant starches for food applications

- It is possible to modify or engineer starch synthesis enzymes using modern biotechnology.
- This in turn opens up oppurtunities to consider using tools of breeding, genetics and biotechnology to genetically control starch deposition in plants so as to enhance starch yield and starch quality.
- instead of crudely controlling enzyme activity, it is possible to more precisely reduce the expression or activity of certain enzymes (such as by using antisense or RNAi technology).

Genetic modification of plant starches for food applications

- Unlike breeding, enzymes may be overexpressed:
- Additionally, novel enzyme isoforms can be selected in order to act synergistically with the existing starch pathway enzymes.
- With full sequencing of many homologous genes from different plants, we can identify domains within enzymes, which impart particular catalytic properties.
- This is an important new opportunity to gain new understandings of the origins, mode of action and potential for genetic modification of different enzymes.

Genetic modification of plant starches for food applications

- The starch created from such work has the potential to be extremely novel and valuable.
- Despite limitless opportunities, constraints remain for understanding links between starch structural changes and functionality.
- Additionally limitations are regarding our understanding of what determines the specific catalytic properties of the enzymes.

Genetic modification of plant starches for food applications

Enzyme and Specific Isoform	Mutant (Maize)	Effect on Starch
Granule Bound Starch Synthase		
GBSSI	Waxy (*wx*)	Low amylose content.
Soluble Starch Synthase		
SSI	None known	Not known
SSII	Sugary2 (*su2*)	Lacks intermediate glucan chains in amylopectin
SSIII	Dull1 (*du*)	Lacks longer glucan chains in amylopectin
SSIV	None known	Not known
Branching Enzyme		
BEI	BEI (*be1*)	None/Minimal
BEIIb	Amylose extender (*ae*)	High amylose content.
Debranching Enzyme		
ISAI	Sugary1 (*su1*)	Forms compound granules and phytoglycogen
ISAII	None known	Not known
ISAIII	None known	Not known
PUI	Pullulanase1 (*pu1*)	None/Minimal

Food Biotechnology

Module #162



Food Biotechnology

Modification of starches

Modification of starches

- Modifications in GBSS (granule bound starch synthase) activity result in changes in amylose content or amylose structure.
- In maize, the GBSSI mutant phenotype was named waxy because of waxy phenotypic appearance.
- waxy mutants of cereals is the only enzyme determining the amylose contents in cereals, the amf mutants of potato, the lam mutants of pea, and GBSSI antisense lines of potato show either a reduction or elimination of GBSS activity and a specific reduction of amylose in starch from tubers.

Modification of starches

- Some studies have attempted to restore the production of amylose in amylose free potato plants by transforming the plants with genes for GBSSI enzymes produced by other plants.

Modification of starches

- 3.5%-13% amylose was restored to amylose free mutants of potato by transformation with the cassava GBSSI enzyme.
- Similarly, amylose free potatoes transformed with pea GBSSI isoforms resulted in potatoes with amylose contents 0.8% -1%.
- Like, the other low amylose potatoes and pea starch, heterogeneity in amylose and amylopectin content was observed within the granules: the granules stained with iodine revealed amylose in concentric rings or having blue staining granule cores.

Modification of starches

- Japanese scientists has identified wheat mutants lacking GBSSI activity.
- substitution of genes producing functional GBSSI enzyme with the null alleles resulted in starches with a 22 to 23% amylose content rather than the 25.5% normal control.
- Similarly, elimination of an active GBSSI enzyme at 2 of the 3 loci in wheat endosperm results in a wheat starch that has an amylose content of 16% to 21%.

Modification of starches

- Thus, the presence of a functional GBSSI enzyme from a single locus pair is sufficient to produce a starch with an amylose content of at least 16%.
- between 14.1 and 16.7% amylose content wheat can be created through ethyl methanesulphonate (EMS) mutagenesis of the seeds.
- Amylose free wheat starches were created using triple null combinations of GBSSI mutants and using mutagenesis of a double null wheat known as Ike to generate a non null wheat.

Modification of starches

- Two functional Wx alleles of rice exist: Wxa, which produces a large amount of amylose, and Wxb, which produces a smaller amount of amylose.
- Studies of the effects of the two alleles on the gene expression at the waxy locus in rice showed that the Wxb allele resulted in an ineffective production of GBSSI enzyme and amylose in japonica rice, while the Wxa allele produced larger quantities of GBSSI enzyme and amylose in indica rice.

Modification of starches

- Wxa allele is shown less effective in the production of amylose than the Wxb allele based on analysis of 40 rice varieties.
- Similarly, two wild-type rice alleles, Wxa and Wxb, Wxb had a tenfold lower GBSS expression than Wxa at the protein and mRNA levels.
- The reason for this effect was result of a point mutation within the genetic sequence for the normal rice enzyme (Wxa allele).

Modification of starches

- Modifications in starch synthase activity result in changes in amylopectin content or amylopectin structure.
- Starch synthase may have either a subtle or profound impact on the starch, Although SSI is a relatively minor isoform in potato, it is the predominant isoform of SS in the cereals.

Modification of starches

- Of the starch synthases, the effects of SSIIa and SSIII on starch structure and composition are the best elucidated, especially in maize.
- The first and only reported SSI mutant was in rice having only a minor effect on starch content and quality.
- In maize, SSIIa maps to a locus known as Sugary-2 which when mutated produces the sugary-2 mutant.

Modification of starches

- The structural gene for SSIII in maize is known to be the Dull1 locus which when mutated gives the dull1 mutant.
- Starch from the dull1 mutant is relatively lacking in longer amylopectin chains.
- In potato the simultaneous antisense inhibition of SSII and SSIII resulted in a grossly modified amylopectin, with yet further changes in structure if GBSSI as well as SSII and SSIII are inhibited.

Modification of starches

- The first naturally high amylose starch was reported 50 years ago a maize endosperm mutant termed *amylose extender* (*ae*).
- We now know that the structural gene for SBEIIb is the *Ae* locus, which when mutated produces *ae* mutant starch having an amylose content of up to 50%.

Modification of starches

- relatively clear demarcation between amylose and amylopectin existing for normal starch is blurred for *ae* starch, giving rise to a third type of starch termed intermediate starch.
- Additional evidence for this comes from material collected as amylopectin from *wxae* intermediate starch which has no GBSSI activity and hence produces no true amylose, and has reduced amylopectin branching.

END

Food Biotechnology

Module #163



Functionality of Starches

Functionality of starches

- Crystalline packing of double helices of starch chains from amylose fiber shows diffraction pattern through wide angle x-ray scattering.
- When starch granules are heated sufficiently in excess water, this pattern is lost and an endothermic transition is observed.
- Double helical order is additionally lost during this endothermic event, and it is this loss and not the loss of crystalline order that is believed to be responsible for the endothermic event

Functionality of starches

- This loss of granular, crystalline, and double helical order is called gelatinization, and it is irreversible with respect to the three dimensional granule organization.
- Subsequent to gelatinization, starch chains of both amylose and amylopectin may organize into new double helical and crystalline structures in a process called retrogradation.

Functionality of starches

- In its native granular form mixed with water, starch makes a high solids mixture, useful in making batters and doughs.
- During heating, starch granules swells 30-fold its dry weight.
- This process is reversible provided temperatures and pressures are lower than those that lead to disruption of the organized intramolecular glucan chain associations within the granule.

Functionality of starches

- Above these temperatures and pressures, the granules will undergo irreversible changes and glucan chains will dissociate, permitting further swelling.
- additional heating and shear forces, swollen hydrated granules will eventually collapse to form an unorganized paste of starch molecules.
- results in a thickening effect, imparting texture and structure.
- In functional terms, this process of starch granule swelling and dissociation is known as gelatinization.

Functionality of starches

- The glucan polymers of a gelatinized starch can associate with each other or other components of food to impart stickiness, tackiness or a rubbery texture.
- amylose also readily leaches out of the granule during gelatinization and interacts with the food components.
- In mild cooking conditions, amylopectin does not so readily leach and hence is enriched inside the gelatinized granule.
- extreme cooking conditions, the amylopectin is much more dispersed, resulting in products with a tacky or sticky character.

Functionality of starches

- Once a cooked starch is allowed to cool, the glucan chains begin to reassociate in a process resulting in a change in the functional characteristics, including a decrease in paste clarity and gelation of the paste. In functional terms, this process of reorganization is known as retrogradation.
- Normal starches are generally recognized for their ability to retrograde within hours and usually form opaque gels.
- Retrogradation of amylopectin gels occurs during days or weeks of storage.

Functionality of starches

- As a consequence of their different molecular weights and chain length profiles, the rates of retrogradation of amylose and amylopectin are not the same.
- The rapid setting of the structure of breads is believed to be due to rapid (within seconds or minutes) amylose retrogradation to form a network structure.
- Starch functionality is therefore a consequence of the degree of gelatinization and is influenced by retrogradation, time, temperature, concentration, etc.

END

Food Biotechnology

Module #164



Food Biotechnology

Application of Starches

Application of starches

- Role of starch is in improving processing, shelf life, consistency and appearance by providing texture and thickening capability for suspending solids.
- Extensive applications of starches in foods and these uses have been extended due to application work by food technologists using physical and chemical modifications.
- These include oxidation, monostarch phosphorylation , distarch phosphorylation , acetylation, hydroxypropylation, and octenylsuccination

Application of starches

- **Amylose free starch:**
- Obtained from waxy maize and rice and in all cases are considered non-GM natural variants of normal maize and rice.
- Staining starch with iodine readily identifies amylose free types: normal starch will stain blue or purple whereas the starch produced from amylose free plants will stain red or brown or brownish red in color.
- Amylose free maize starches are generally considered to contain zero or almost zero amylose.

Application of starches

- Amylose free rice starches have been shown to contain between 0 and 3% amylose, though collectively these starches are referred to as amylose free or waxy rice starches.
- With these amylose free rice starches, it has been assumed that the differing cooking and paste properties are due to differences in the structure of the amylopectin of the starch rather than variations in the low levels of amylose of the starch.

Application of starches

- Amylose free starches are considered useful as water binders, viscosity builders, and texturizers in food as well as in industrial applications.
- However, these starches are less resistant to shear, acid, and high temperatures than are normal starches, and extended cooking results in stringy, cohesive pastes.
- Amylose free starches are generally recognized for their improved transparency and have better freeze and thaw stability compared to normal starches once cooked.

Application of starches

- They are also recognized for their improved long term storage capability as they require weeks to gel if they could be considered to gel at all.
- To correct for some of the negative paste attributes of amylose free starch, such as poor stability to temperature, shear and acid and undesirable paste quality, most amylose free starches are chemically modified by substitution, cross linking, or both.

Application of starches

- Developing amylose free starches in other crops so as to take further advantage of any species specific qualities of the starch produced from that species is being sought.
- However, although mutants are readily found in some plant species (such as maize, rice, and barley) this is more difficult in other plant species where there are multiple copies of each gene (such as the polyploid species like potato, oats, and wheat).

Application of starches

- In the case of wheat, the advances have been made by screening for mutants (non-GM), while in potato the waxy types were made using biotechnology (GM).
- Amylose free wheats are finding applications in foods such as noodles and baked goods including breads .

Application of starches

- Commercialized resistant starches include unprocessed starches (green banana, legumes, potato) or processed forms of high amylose starches.
- High amylose starches from other sources are also being advocated for their resistant starch properties. The most widely available high amylose starches originate from maize using the *amylose extender* mutation that results in a starch having 40–50% amylose.

Application of starches

- As a result of extensive breeding and selection work higher amylose contents (up to 90%) have been achieved.
- High amylose starches are also available in barley and rice.

END

Food Biotechnology

Module #165



Food Biotechnology

Novel Starches



Novel starches

- Starches obtained from different crops having varied functionalities due to differences in composition may now theoretically be readily transferred into crops from one or another different plant species.
- In general we can consider making changes in any aspect of amylose/amylopectin, phosphate, protein, phospholipids, crystallinity, gelation, and pasting characteristics, flavor and starch granule morphology.

Novel starches

- One interesting example of a cross species difference that may be exploited is phosphate content in potato.
- starch phosphate content may be attributed to a novel glucan water dikinase enzyme. This discovery opens up the possibility of creating novel genetically modified starches with variable phosphate contents.
- Application in food include changing starch digestibility or starch viscosity after cooking.

Novel starches

- Another avenue associated with the genes responsible for endosperm texture in some cereal crops.
- In this case the puroindoline genes, described as PinA and PinB are interesting candidates. Transforming the wheat genes into rice successfully created the softness trait in the seed.

Novel starches

- Although starch granule morphology varies extensively across species and starch from different species is highly valued for certain applications, there are few specific reports that granule morphology is the attribute desired for valuable food applications.
- Furthermore, although there have been some reports of progress in understanding the mechanisms that control and influence starch granule size, for granule morphology there is little that can be defined as clear enough to arouse interest in a genetic approach.

Novel starches

- Similarly, flavor can be considered to vary extensively across species and a bland flavor of (e.g., tapioca starch) is valued over certain cereal starches that impart a mealy character. However, as with granule morphology, our understanding of the genetic components determining this character is rather poor, making flavor a difficult target for genetic modification at the present time.

Novel starches

END

- Another novel concept has come from bifunctional domains involving cellulose and starch, which may enable developments with novel biomaterials.
- Another possibility is to explore incorporation of inulins with starch.
- Even more novel possibilities to consider include adding new functionality to starch using gene constructs containing fusion proteins.
- The fusion proteins can be proteins, which might affect starch properties or enzymes, or other bioactive proteins.

Food Biotechnology

Module #166



Biotechnology of food flavors production

Biotechnology of food flavors production

Biotechnology of food flavors production

- Plants were the sole sources of flavor compounds historically.
- However, active compounds are present in low concentrations , which makes their isolation difficult.
- flavors are dependent on factors such as the weather or plant diseases.
- The production of flavor by biotechnological methods is an alternative due to consumers' prefernatural ingredients. Microbial processes seem to be the most promising methods

Biotechnology of food flavors production

Main Technologies:

- micro organisms growing on a culture media.
- They have the ability to perform conversions that would require multiple chemical s steps.
- They catalyze specific steps.
- They are also an economical source of enzymes , which can be utilized to enhance or alter the flavors of many food products.
- production of flavor compounds can be divided into two groups : microbiological and enzymatic.

Biotechnology of food flavors production

- Microbial methods are of two types.
- **Biosynthesis** is production of chemical compounds by cells (fermentation or secondary metabolism).
  Biotransformation is use of microbial cells in the specific modification of chemical structures.
- In fermentation, the production of flavors starts from cheap and simple sources such as sugars and amino acids. When micro organisms are used in order to catalyze specific conversions of precursors and intermediates, the process is called biotransformation.

Biotechnology of food flavors production

- Although fermentation requires C and N sources, a specific substrate is necessary for microbial transformation.
- The enzymatic catalysis precedes a simple and specific transformation of the substrate molecule.
- “non-natural” substrates can also be biotransformed.
- It is important to distinguish research with the purpose of obtaining complex products with natural characteristics from those that try to obtain isolated molecules.

Biotechnology of food flavors production

- The first consists in the experience of nature imitation and in developing a process with one or more microorganisms and enzymes .
- The second tries to obtain a higher yield of the characteristic components.
- The choice between them determines the methodology, which will be employed in vivo or in vitro, through biosynthesis or bioconversions .

Biotechnology of food flavors production

How to obtain flavors

- **Flavor from Fermented Foods:**
- The organoleptic properties of fermented foods are superior to unfermented substrate and are depend ant upon the biochemical activities of the associated microorganisms.
- Fermentation is technology for the production of organic acids, flavor compounds , and other biologically important chemicals.
- New aroma and flavors includes acids,alcohols, carbonyl com pounds, esters, and pyrazines.

Biotechnology of food flavors production

- Fermented products are preserved, modified, and flavored using micro organ isms.
- Lactic acid and alcoholic fermentations are the two important processes responsible for fermented food flavors .
- The creation of new fermented products can result in the development of novel flavors and textures.

Biotechnology of food flavors production

How to obtain flavors- Dairy Products:

- Cheese , yogurt and buttermilk flavors find application in snacks, sauces, baked goods, and several other products.
- The cheese flavor results from the action of microorganisms and enzymes on milk's proteins, fats, and carbohydrates.
- Numerous breakdown products are formed, among them, short-chain fatty acids, acetic and lactic acids, alcohols, aldehydes, ketones, esters, sulfur and nitrogen compounds .

Biotechnology of food flavors production

- The starter cultures used in dairy technology are mainly prokaryotes like *Lactococci, Lactobacilli, Leuconostocs, Bifidobacteria, Propionibacteria, Streptococci, and Brevibacterium linens.*
- Yeasts, such as *Kluyveromyces, Debaromyces, Candida* or *Trichosporon* are present in many manufactured milk products.
- These microorganisms modify the sensory characteristics of the products by synthesizing or assimilating volatile nitrogen and sulfur compounds

Biotechnology of food flavors production

- Several chemical reactions take place in the surface of ripened cheeses such as Camembert and Brie during maturation due to fungi growth.
- The Fungal mycelium of Penicillium roqueforti grows rapidly and the resulting products are used directly for flavoring foods with a blue cheese-type flavor.

Biotechnology of food flavors production

How to obtain flavors-Alcoholic beverages:

- Flavor compounds are produced as byproducts of yeast metabolism during alcoholic fermentation. Many flavor compounds have been identified in alcoholic beverages.
- The main compounds are: During alcoholic fermentation, yeasts transform sugars (glucose, fructose, and sucrose) into ethanol and carbon dioxide by the Embden–Meyrhof–Parmas pathway.

Biotechnology of food flavors production

- This is the main bio reaction, but several secondary byproducts are formed.
- Higher alcohols, organic acids, and esters are the main flavor compounds.
- Higher alcohols are major portion of the secondary products of yeast metabolism. They include n-propanol, isobutyl alcohol, 2-methyl butanol, amyl alcohol, isoamyl alcohol , and 2-phenyl ethanol.
- Isoamyl alcohol accounts for more than 50% of the total concentration of higher alcohols.

Biotechnology of food flavors production

- Esters at appropriate concentrations impart flowery and fruity flavors . They are formed by esterification of fatty acyl -CoA or of organic acid by alcohols.
- Diacetyl and pentane-2,3-dione (vicinal diketones) have a characteristic flavor described as “buttery”, “honey” or “toffee-like”. They have a very high off-flavor potential, dependent on the fermentation temperature.
- The threshold for diacetyl in lager-type beers is 0.10–0.14 mg /L. At levels above 1 mg /L, it becomes increasingly “cheese-like” and sharp.

Biotechnology of food flavors production

- Volatile acids are usually present in beer at concentrations of 20–150 mg /L. Butyric and iso-butyric acids in concentrations of 6 mg /L cause a “butyric” or “rancid” flavor.
- Valeric and iso-valeric acids cause an “old-hop” and “cheesy” flavor. Fatty acids with 6
to 12 carbon fatty acids give the characteristic flavor of “cheesy”, “goaty”, or “sweaty” (Smogrovicova´ and Do¨me´ny 1999).

Biotechnology of food flavors production

END

How to obtain flavors-Bakery products:

- Although Candida yeast has occasionally been used for baking and some *Saccharomyces carlsbergensis* strains have been patented for use as baker's yeast, pure strains of *Saccharomyces cerevisae* are almost universally employed.

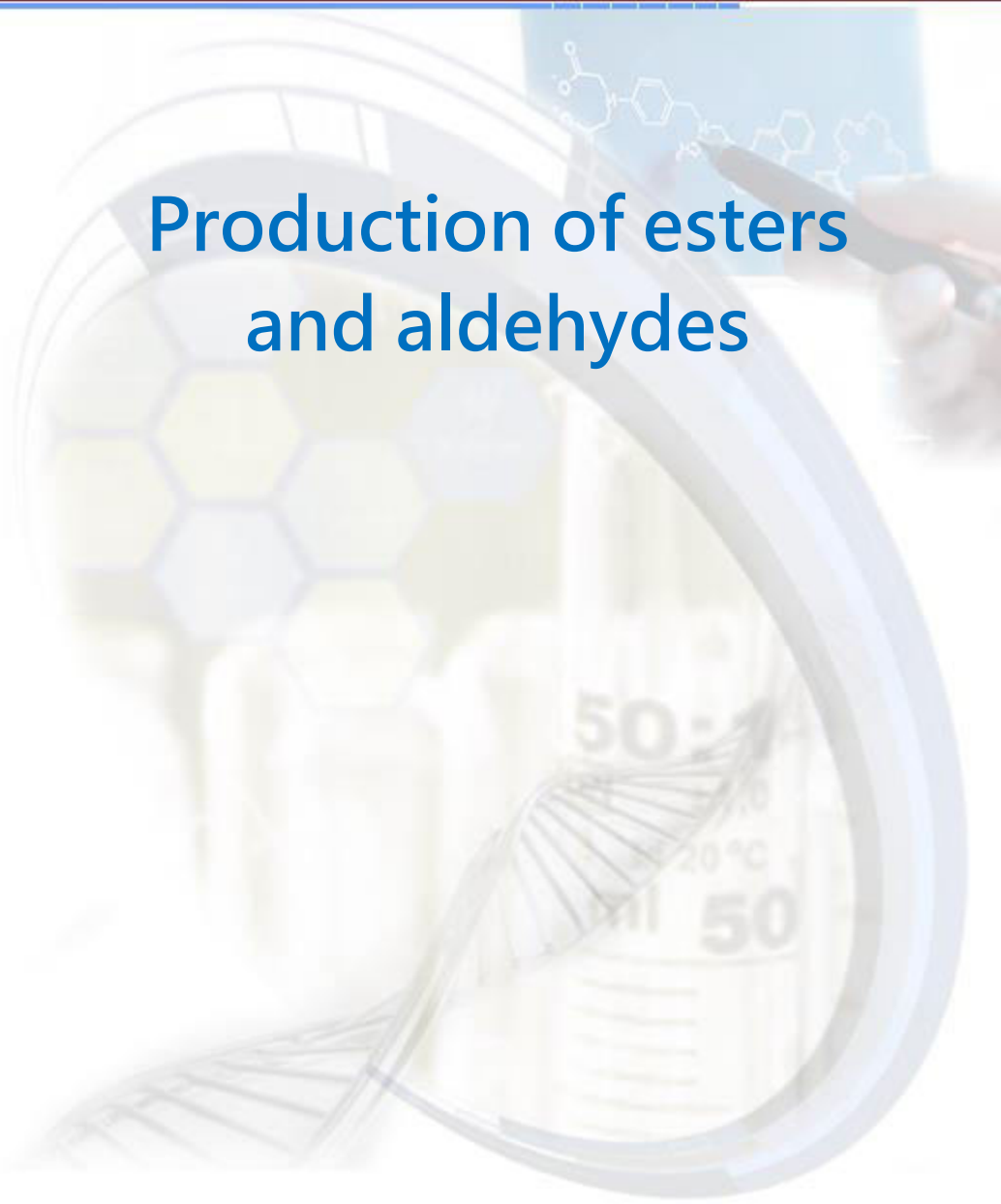
Food Biotechnology

Module #167



Food Biotechnology

Production of esters and aldehydes



Production of esters and aldehydes

Ester Production:

- Esters are flavor compounds of fresh fruits and fermented foods, which are found in concentrations 1-100 ppm
- The production of the ester ethyl acetate by the yeast *Candida utilis* from glucose is observed when the yeast grows on a medium containing a specific initial ethanol concentration.

Production of esters and aldehydes

- Esters of low molecular weight are responsible for fruity odors and consist of acids and their derived compounds such as acetates, propionates, and butyrates.
- Some examples are ethyl butyrate and isoamyl acetate, which are found in strawberry and banana .

Production of esters and aldehydes

- The presence of esters such as ethyl acetate and butyric acetate in the culture medium can eventually describe a detoxification mechanism by which the microorganism avoids the accumulation of toxic compounds.
- The production of acetates occur s in order to detoxify the medium by converting acetic acid and high alcohols .
- Two metabolic pathways can be followed in the format ion of esters : alcoholysis of acyl-CoA com pounds and the direct esterification of an organic acid.

Production of esters and aldehydes

- Yeasts follow predominantly the first path way, and filamentous fungi and bacteria prefer the second .
- Some ramified amino acids are important precursors of flavor compounds and are related to fruit maturation.
- The initial reaction is called the enzymatic Strecker degradation.
- Several micro organ isms including yeasts and bacteria such as *Streptococcus lactis* can modify the majority of the amino acid structures.

Production of esters and aldehydes

- Ethyl acetate comprises, with other compounds, banana flavor.
- 2-Methyl-ethyl-butyrate has a great impact on characteristic apple flavor .
- fruity banana flavor produced by the yeast *Hansenula mrakii* and by the fungus *Geotrichum penicilliatum* in submerged fermentation using a synthetic medium.
- Seventeen compounds were identified in concentrations greater than 50 mL /L, including ethanol, ethyl acetate, isobutanol, ethyl propionate, isobutyl acetate, and isopenthyl acetate.

Production of esters and aldehydes

- Alcohols were formed in the exponential growth phase, but the esters were formed in the stationary phase.
- Ethyl acetate was the main product. In both studies, some precursors of fruity esters were added, such as vanillin, leucine, isoleucine, and phenylalanine.

  Thirty-three compounds were identified in concentrations greater than 50 mL /L: ethanol, ethyl acetate, ethyl propionate, and others.
- Ethyl acetate was produced in the highest concentrations (9924.1 mL /L).

Production of esters and aldehydes

- Strains of Ceratocystis were also identified as ester producers. Lanza and others (1976) studied the production of acetates with different carbon (glucose, galactose, and glycerol) and nitrogen (urea and leucine) sources.
- They concluded that the type of flavor compounds produced depended on both sources (carbon and nitrogen), which is different for other microorganisms such as Trichoderma viride.
- Collins and Morgan (1961) identified esters synthesized by different species of Ceratocystis (C. moliniformis, C. major, C. coerulescens,

Production of esters and aldehydes

- A strong banana flavor was detected when using dextrose and urea due to the presence of isoamyl acetate, which was also identified in leucine- or isoleucinebased media.
- For the combination galactose–urea, the main flavor was citric due to the formation of terpenes.
- The yeast Kluyveromyces marxianus produced some compounds with characteristic fruity flavor, with ethyl acetate found in higher concentrations in solid-state fermentation of different agro-industrial residue s such as cassava bagasse (Medeiros and others 2000, 2001).

Production of esters and aldehydes

- Banana flavor has also been identified in some plants when micro organisms (Erwinia caratovora ssp atroseptica) infected them.
- Spinnler and Dijian (1991) identified the volatiles formed in a synthetic medium, similar to the ones in the infected plant.
- From 13 isolated microorganisms with capacity to produce esters and alcohols, several compounds were identified; including ethyl acetate, 2-methyl-1-propanol, and propyl acetate.
- In this study, different media were tested with different nitrogen and carbon sources.

Production of esters and aldehydes

- The microbiological population was identified and quantified by Park and others (1982), with counts between 6 *105 and 1.9 *106 CFU, as being predominantly Aspergillus niger and Pecylomyces sp.
- Yoshizawa and others (1988) identified some volatiles produced by a strain of Neurospora, isolated by Park and others (1982), in submerged fermentation.
- These include ethanol, ethyl acetate, isoamyl acetate, ethyl hexanoate, and acetoine.

Production of esters and aldehydes

Aldehyde production:

- Aliphatic, aromatic, and terpenoid aldehydes are important contributors to the flavor of fermented dairy products.
- They are synthesized by microorganisms as intermediates in the formation of alcohols from keto acids.
- An example is the bioconversion of ethanol to acetaldehyde by Candida utilis.

Production of esters and aldehydes

END

- Flavor production using immobilized lipase from the yeast Candida cylindracea in a nonaqueous system has been studied for producing a broad range of esters including ethyl butyrate, isoamyl acetate, and isobutyl acetate.
- Ethyl butyrate has a pineapple–banana flavor, which has a large market demand, and sells at a price of US$150 /kg upwards.
- This process has shown a great stability of the enzyme (more than a month) if kept hydrated intermittently (Singhal and Kulkarni 1999).

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Module #168



Food Biotechnology

Production of alcohol and terpenes

Production of alcohol and terpenes

Alcohol production:

- Alcohols do not contribute as a flavor component unless present in high concentration.
- They are formed as a primary metabolite from microorganisms' activity or due to the reduction of a carbonyl.
- Fusel alcohols can be formed from either carbohydrate or amino-acid metabolism and are the predominant volatiles of all fermented beverages, in addition to ethanol.

Production of alcohol and terpenes

- Different alcohols can be found in the culture of yeasts such as ethanol, propanol, isobutanol, and phenyl ethyl alcohol.
- In filamentous fungi it is possible to find methyl-3-buthanol, butanol, isobutanol, pentanol, hexanol, octanol-3, and phenyl ethanol from the metabolism of amino acids such as leucine, valine, isoleucine, and phenylalanine (Welsh and others 1989).

Production of alcohol and terpenes

Terpens production:

- Terpenes are the most important natural components of essential oils to be used as flavors.
- Microorganisms are able not only to synthesize but also to degrade or transform terpenes.
- Fungi are the main microorganisms responsible for terpene production, but bacteria are capable of synthesizing a few volatile terpenoids, such as geosmin and cadin-4-ene-1-ol (Berger 1995).

Production of alcohol and terpenes

- The synthesis of monoterpenes by *Ceratocystis variospora* is reported.
- Numerous other microorganisms are able to synthesize monoterpenes, among them *Ceratocystis moniliformis, Kluyveromyces lactis, Sporobolomyces odorus,* *Trametes odorata* and *Trichoderma viridae*.
- Microbial bioconversion of terpenes has been extensively studied.

Production of alcohol and terpenes

- Monoterpenoid compounds like citronellal, citral, limonene, and menthol (acetates) can be biotransformed in citronellol, geranic acid, carveol and 1-menthol, respectively.
- Valencene is a sesquiterpenoid available from orange oil and has little commercial use.Some bacteria are capable of transforming valencene to nootkatone, a main flavoring component of grapefruit.

Production of alcohol and terpenes

- From the economic point of view, the development of biotechnological processes for the production of terpenes is not viable due to the low yields obtained and the abundance of vegetable sources available.
- The real meaning of these studies is the understanding of the steps of the catabolism of terpenes.

END

Food Biotechnology

Module #169



Food Biotechnology

Determination of specific gravity of sugar

Determination of specific gravity of sugar

- Density, the mass per unit volume at a specified temperature, can be used to identify foods.
- A related parameter, specific gravity, is often used.
- Specific gravity is the ratio of the density of a substance at a specified temperature to that of water at 4°C. The specific gravity of water at 4°C is 1.0.
- The density of water at 20°C is 0.99823 g/cm3, which is close enough to unity that the temperature of the water will not need to be at 4°C for most uses.

Determination of specific gravity of sugar

- Specific gravity helps determine adulteration, quality, and composition of food such as the water or butterfat content of milk, the syrup concentration of canned fruits or juices, and the alcohol content of beverages.
- Specific gravity of liquids can be measured with a hydrometer.
- The function of the hydrometer is based on Archimedes' principle that a solid suspended in a liquid will be buoyed up by a force equal to the weight of the liquid displaced.

Determination of specific gravity of sugar

- The lower the density of the sample, the lower the hydrometer will sink.
- Saccharometers are hydrometers graduated to indicate percentage of sucrose by weight (degrees Brix).
- The refractometer provides an alternative means of measuring the soluble solids of sugar syrups and fruit products and is based on the index of refraction of the solution.

Determination of specific gravity of sugar

- Specific gravity of solids:
- Measures:
- Specific gravity by weight of known volume.
- Uses:
- Measure amount of air incorporated in products such as whipped cream, egg white foams, creamed shortening, and cake batters.

Determination of specific gravity of sugar

Proceedure:

- Weigh a dry container to the nearest gram.
- Fill container with cooled, boiled deionized water at room temperature. Complete fill on balance; judge at eye level. Weigh to nearest gram.
- Fill dry cup with test material. Do not pack. Remove excess with spatula. Wipe outside of container. Weigh to nearest gram.

Determination of specific gravity of sugar

- Calculate the specific gravity as follows:

Specific G= wt. of filled container- wt. of container/vol. of container.

- where volume container = (weight container + water) – weight container.
- Since specific gravity is the density of a substance relative to water it has no units.

END

Food Biotechnology

Module # 170



Food Biotechnology

Biotechnological approaches to improve nutritional quality and shelf life of fruits and vegetables

Biotech approaches for nutrition and shelf life of fruits and vegetables

- Fruits and vegetables consumed as fresh or processed
- Important ingredients of a healthy diet
- They are valuable sources of :
- Vitamins
- Minerals
- Antioxidants
- Fiber

Biotech approaches for nutrition and shelf life of fruits and vegetables

- Important quality factors of fruits and vegetables are:
- Color
- Flavor
- Texture
- Nutritive value

Biotech approaches for nutrition and shelf life of fruits and vegetables

- Quality is a concept involving measurable attributes :
- Degree of purity
- Firmness
- Flavor
- Color and size
- Maturity condition,
- Any other distinctive attributes or characteristics of the product

Biotech approaches for nutrition and shelf life of fruits and vegetables

- Qualities of the produce are influenced by :
- cultivar
- environmental conditions affecting growth
- Cultural practices
- pests and diseases
- Time of harvest
- Post harvest and storage conditions

Biotech approaches for nutrition and shelf life of fruits and vegetables

- With advancement of technology in several areas, the only factor on which the grower has no control of is the environment of the field conditions

- Heredity plays a major role in determining the quality of fruits and vegetables

Biotech approaches for nutrition and shelf life of fruits and vegetables

- Traditional crop breeding is still being used as one means of crop improvement
- Continuing advances in knowledge and technology have dramatically expanded the biotechnological tools available for genetic improvement and production of vegetables and fruits

Biotech approaches for nutrition and shelf life of fruits and vegetables

- The term biotechnology is broad, encompassing a wide range of disciplines in science focus will be on plant transformation:

- Genes are modified or transferred by molecular means, and the resulting improvements in fruit and vegetable quality

Biotech approaches for nutrition and shelf life of fruits and vegetables

- Parameters and mechanisms that affect nutritional quality and shelf life
- Other than pests and diseases
- Which can be improved or modified by genetic engineering

END

Food Biotechnology

Module # 171



Food Biotechnology

Factors effecting reducing sugars in potatoes

Factors effecting reducing sugars in

- Potato is an important agri commodity.
- Non enzymatic browning of the product occurs at high temperature, when reducing sugars combine with the alpha-amino groups of amino acids Resulting in dark and more bitter products that are unacceptable to the consumer
- Known as Maillard reaction

Factors effecting reducing sugars in

reducing sugars accumulate in potatoes by:

1. Low temperature sweetening (LTS) when stored below 10° C in storage.

2. Metabolism of starch in tuber.

- To avoid browning potatoes are generally stored at around 10° C but it leads to sprouting.
- chemical sprout inhibitors are used

Factors effecting reducing sugars in

The only solution to avoid this problem is using cultivars that are resistant to LTS. Benefits include

- Control of sprout growth and senescent sweetening
- Reduction of physiological weight loss due to decreased respiration and losses associated with bacterial and fungal pathogens
- Extended marketability

Factors effecting reducing sugars in

Disadvantages:

- Associated with it is the accumulation of reducing sugars and the resulting browning of processed products such as chips and fries
- The mechanism is complex and may involve the interaction of several pathways of carbohydrate metabolism and the genes that regulate these pathways

Factors effecting reducing sugars in

Metabolism of starch in tuber

- Potato tubers contain 60–80% starch Of which sugars up to 3% on a dry weight basis
- The principal event in LTS is the cold induced synthesis of sucrose
- The carbon needed for the synthesis of sucrose and reducing sugars for LTS is generally, but not always, provided by a net breakdown of starch

Factors effecting reducing sugars in

- Increase in potato tuber sugar content occurs early during cold storage;
- Over two to three months at storage temperatures of 1–3° C tubers can convert as much as 30% of their starch content.
- sweetening response to low temperatures is fairly consistent
- Is influenced by cultivar, locality, and conditions prior to cold storage

Factors effecting reducing sugars in

- After cold storage, potatoes are reconditioned at 18° C where sugar content decreases and starch content increases
- Response to reconditioning is neither consistent nor completely restorative
- Tends to be cultivar dependent

Factors effecting reducing sugars in

- Preconditioning treatment has been used to lessen chilling injury in chilling sensitive plants

- Storage at 10° C prior to cold storage can acclimatize tubers and lessen the LTS effect

END

Food Biotechnology

Module # 172



Food Biotechnology

Starch Synthesis

Starch Synthesis

- Starch is synthesized in plastids (amyloplasts) upon tuber initiation
- number of starch grains and grain size increase during tuber growth.
- Potato starch is comprised of 21–25% amylose and 75–79% amylopectin

Starch Synthesis

- Starch is synthesized from ADPglucose by the concerted action of:
- ADPglucose pyrophosphorylase (ADPGPase),
- Starch synthase
- Starch branching enzymes

Starch Synthesis

- It starts with conversion of sucrose into hexose phosphates in the cytosol:
- Then Glucose-6-phosphate is transported into the amyloplast where it is converted into Glucose-1-phosphate.
- Subsequently converted to ADP glucose by the action of ADPGPase.

Starch Synthesis

Starch synthases:

- Catalyze the polymerization of the glucose monomers into alpha-1,4-glucans
- Using ADP glucose as a substrate

Starch branching enzyme catalyzes the formation of the alpha-1,6-linkages of amylopectin

Starch Synthesis

- ADPGPase is a rate-limiting step in starch synthesis subjected to allosteric activation by 3-P-glycerate and inhibition by inorganic phosphate
- Strategies to alter the starch metabolism in tubers by genetic manipulation of ADPGPase may be helpful in reducing the accumulation of reducing sugars during LTS

END

Food Biotechnology

Module # 173



Food Biotechnology

Starch Degradation

Starch Degradation

- The differential response of potato cultivars to LTS is result of starch properties that affects enzymes ability to degrade it.
- Chemical modifications of glucose units by attachment of covalently bound phosphate esters attached to C3 or C6 glucosyl residues of amylopectin in inter branch chains but absent around branching points. This affects the cleavage sites and degradation pattern.

Starch Degradation

- Similarly negative surface charge from surface phosphate, lipid, or protein can affect the properties of enzymes and other soluble compounds.

Starch Degradation

- The degradation of starch is believed to occur in the amyloplast
- The widespread distribution of alpha-glucan phosphorylase, alpha-amylase, beta-amylase, and maltase suggests that starch breakdown could be phosphoryltic, hydrolytic, or both.

Starch Degradation

upon transfer of tubers to chilling temperatures sucrose is the first sugar to accumulate

Amylase is quite inactive at such cold temperature to catalyze the required rate of starch degradation

Starch breakdown in cold stored potato tubers is mainly conducted by starch phosphorylase

Starch Degradation

Two types of potato phosphorylases are recognized based on:

- Glucan specificity
- Monomer size
- Intracellular location.

They are non-interconvertible proteins with different primary structures and different immunological properties

Starch Degradation

- Type 1 isozyme also known as type H localized in the cytoplasm has low affinity for maltodextrins and high affinity for branched polyglucans,
- Cross reacts with type H phosphorylase from potato leaves

Starch Degradation

- Type 2 isozyme also known as type L located in the amyloplast has a high affinity for maltodextrins, low affinity for branched polyglucans
- Cross reacts with type L enzyme from the leaf

Starch Degradation

Effect of Inorganic Phosphates:

- Intracellular compartmentalization of Pi influence carbon partitioning in non-photosynthetic potato tubers
- Increased inorganic phosphate in the amyloplast shifted metabolic activities toward starch degradation rather than accumulation

Starch Degradation

- Increased Pi concentration inhibits ADPGPase and enhances starch breakdown by alpha-glucan phosphorylase

- Another source of Pi in plant cells is the vacuole. major portion of Pi is stored in potato tuber vacuoles where it is compartmentalized away from the cytoplasm

END

Food Biotechnology

Module # 174



Food Biotechnology

Starch-Sugar Balance

Starch–Sugar Balance

Relationship between starch and sugar levels when potatoes are cooled from 10 to 2° C and then re-warmed from 2 to 10° C after an interval, has given rise to a misleading concept of a starch to sugar balance in which the overall change between the two compounds is seen as being reversible

Starch–Sugar Balance

- Sucrose is formed from starch by an irreversible pathway

- Starch is formed from sucrose by separate, but likewise irreversible, routes

Starch–Sugar Balance

The very close relationship between starch and sucrose in stored potatoes may be due to the fact that starch is the only possible source of carbon for sugar synthesis in the cold

Strong evidence to indicate that G-6-P is transported into the amyloplast of potato tubers to support starch synthesis

Starch–Sugar Balance

Pathways of starch and sugar biosynthesis compete for the same pool of precursors

The coexistence of the pathways of sucrose synthesis and starch breakdown in stored tubers may be regulated by fine control mechanisms

Starch-Sugar Balance

Genotypic variation in the capacity to maintain an active starch synthesizing system may help in alleviating the rate of sucrose accumulation

END

Food Biotechnology

Module # 175



Food Biotechnology

Concept of Edible Vaccines

Concept of Edible Vaccines

Involves the process of incorporating the selected desired genes into plants and then enabling these altered plants to produce the encoded proteins

The process is known as transformation

Altered plants are known as transgenic plants

Concept of Edible Vaccines

Edible vaccines like traditional subunit vaccines consist of antigenic proteins and are devoid of pathogenic genes.

Concept of Edible Vaccines

Traditional subunit vaccines are:

- Unaffordable
- Technology-intensive
- Require purification, refrigeration
- Produce poor mucosal response

Concept of Edible Vaccines

Edible vaccines

- Posses good genetics
- Heat stability
- Do not need cold-chain maintenance, can be stored at the site.
- Syringes and needles are also not required, thus reduces the incidence of various infections

Concept of Edible Vaccines

- Act by stimulating mucosal and systemic immunity
- Dual action of edible vaccines provide first-line defense against pathogens attacking via mucosa
- Mycobacterium tuberculosis and carriers causing diarrhea, pneumonia, STDs, HIV etc.

Concept of Edible Vaccines

At present edible vaccines are produced for various human and animal diseases such as measles, Cholera, FMD, Hep B, C and E.

Also be used to prevent exceptional diseases like:

- Dengue, Hookworm, Rabies by combining with other vaccination programs enabling multiple antigen delivery.

Concept of Edible Vaccines

Methods:

Selected gene obtained from the microbes encoding specific antigen can be handled in two different ways

1. Suitable plant virus is engineered to produce desired peptides/proteins is incorporated into the plant, which enables it to produce a huge number of new plants from which chimeric virions are isolated and purified and applied.

Concept of Edible Vaccines

END

2. In another method, the desirable gene is incorporated with plant vector by transformation

   Many other approaches have been utilized which can be categorized into following groups:

- Agrobacterium mediated gene transfer
- Biolistic method
- Electroporation

Food Biotechnology

Module # 176



Food Biotechnology

Examples of Edible Vaccines

Examples of Edible Vaccines

Transgenic potatoes for diarrhea:

- First successful human trial was conducted in 1997
- Transgenic potatoes, which possessed the b-subunit of the *E. coli* heat-labile toxin, responsible for diarrhea
- 4-fold increase in serum antibodies was manifested in ten out of the 11 volunteers

Examples of Edible Vaccines

Next clinical trial took place at the Boyce Thompson Institute at Cornell University, USA, in which 20 volunteers ate the potatoes containing the Norwalk virus (responsible for vomiting and diarrhea), out of which19 showed an immune response.

Major drawback is cooking potato denatures protein and makes it ineffective

Examples of Edible Vaccines

Transgenic tomatoes against diarrhea:

- Transgenic tomatoes were produced against the Norwalk virus
- Responsible agent for severe diarrhea.
- The transgenic tomatoes are capable to produce surface protein specific to the virus.
- mice fed with transgenic tomatoes showed immune response towards the virus

Examples of Edible Vaccines

Other transgenic plants:

Presently, banana is being exploited as a good source for edible vaccine production

- It does not require cooking
- Locally grown plant

Examples of Edible Vaccines

- The protein expression in transgenic banana is tissue specific promoter dependent.
- Several other examples involve rabies glycoprotein expressed by viral vectors in spinach
- Hepatitis B surface antigen in case of lettuce and potato

Examples of Edible Vaccines

- Applications of edible vaccines include:

- Cancer therapy
- Birth control
- Chloroplast transformation
- Role in autoimmune disease
- Recombinant drugs/proteins

END

Food Biotechnology

Module # 178



Food Biotechnology

Food Marketing Principles

Food Marketing Principles

Marketing activities are generally carried out in all types of industries

However, the nature of the industry does lead to some differences in the activities

Food Marketing Principles

Food companies contribute to the development of marketing theory as objects of case study

Nestle, Unilever, Coca-Cola, McDonald's

Popular cases in recent years

Food Marketing Principles

Methodology for the entire food industry are confronted with one difficulty

‘Food’ is a general term for edible products that human beings ingest in daily life

Different companies produce different kinds of food

Food Marketing Principles

Nestle is famous for dairy products, chocolate, coffee

Unilever make margarine

Coca-Cola is a soft drink company

Mc-Donald's is a hamburger chain

Food Marketing Principles

The food industry comprises plural sub-industries

There is a need to consider many types of marketing applications that correspond to the different products in the food industry

Food Marketing Principles

Marketing activities implemented by companies and organizations in the food industry are described as ‘food marketing’

END

Food Marketing Principles

- The marketing concept
- The marketing process
- The marketing research
- The formulation of a marketing strategy
- The devising of a marketing plan

Food Biotechnology

Module # 179



Food Biotechnology

Marketing Principles

Marketing Principles

Marketing is different than selling and promotion only, now a days it is consider much broader, such as:

"the total of activities involved in the transfer of goods from the producer or seller to the consumer or buyer, including advertising, shipping, storing, and selling”

Marketing Principles

The basic principles of marketing include the 4 P's

- Product
- Price
- Place
- Promotion

Marketing Principles

Two parts

1. Create value for customers
2. To build strong customer relationships

Marketing Principles

END

The 4 P's have been updated to now 7 P's

- Product
- Price
- Place
- Promotion
- People
- Positioning
- Packaging

Food Biotechnology

Module # 180



Food Biotechnology

The Marketing Concept

The Marketing Concept

Marketing is defined as:

- ‘Marketing is the process by which companies create value for customers and build strong customer relationships in order to capture value from customers in return’

- Advertisements are a part of marketing

The Marketing Concept

- Sometimes, you are requested to sample new products such as a new flavor of cheese or a new taste of beverage, in a supermarket.
- Sales people explain to you how the new sample differs from existing products.
- This selling activity is also a part of marketing

The Marketing Concept

- Advertising and selling constitute only a part of marketing
- Marketing begins with understanding the customer's needs and wants
- Finding potential needs and wants that the customer is still unaware of also plays a key role

The Marketing Concept

- In order to create value for customers, it is necessary to realize customers' needs and wants

The Marketing Concept

- Knorr's soup stock
- Accepted by customers since it realized one such customer need:
- It not only reduced the soup to powder but also met the demand for quick and easy soup preparation

The Marketing Concept

- Building strong customer relationships is also integral to marketing

- Goal of marketing is to capture value from the customer
- Advertising, sales and promotions are marketing activities carried out to deliver the value created for customers

The Marketing Concept

- Once a customer buys a product, some measures must be taken to ensure that he/she makes repeat purchases

- Retaining customers is an important part of marketing

The Marketing Concept

- It is important to establish a brand name for the customer to recognize a product
- Branding helps in customer retention

END

Food Biotechnology

Module # 181



Food Biotechnology

Marketing Process

Marketing Process

As mentioned previously, the aim of marketing is to capture value from the customer.

To arrive at this value, marketers go through five steps

Collectively referred to as the ‘marketing process’

Marketing Process

1. Understand the marketplace and customer needs and wants

2. Design a customer-driven marketing strategy

3. Construct a marketing program that delivers superior value

Marketing Process

4. Build profitable relationships and create customer delight

5. Capture value from customers to create profits and customer equity

Marketing Process

understanding marketplace and customers

- Marketing research is one method
- Understand main influences on marketing activities
- Collect, analyze, and evaluate information and data

Marketing Process

Design of customer-driven marketing strategy

- Marketing strategy formulation depends on the targeted customers
- Forced to decide who customers are
- Carryout market segmentation
- Deliver goods to target customers

Marketing Process

Preparing a practical marketing program.

- Begin with planning the details of the marketing mix
- Marketing mix is a combination of the firm's marketing tools – product, price, promotion and place
- Draft a marketing program that will meet the wants of the market

Marketing Process

Customer and Partner relationship management:

- The former pertains to the manner in which firms build relationships with their customers
- The latter pertains to how firms build relationships with their trade partners.

Marketing Process

Capturing value from the customer

- Attempt to build customer loyalty by enhancing customer satisfaction
- Focus on increasing their market share
- Also on increasing customers' mind share

Marketing Process

- By building customer loyalty, firms are able to increase their profits substantially

END

Food Biotechnology

Module # 182



Food Biotechnology

Application of social, legal, ethical and environmental principles to marketing situations

Application of social, legal, ethical & environmental principles

Corporate Social Responsibility (CSR)

The idea of social responsibility supposes that the corporation has not only economic and legal obligations, but also certain responsibilities to society which extend beyond these obligations

Application of social, legal, ethical & environmental principles

Economic responsibilities demand that the corporation should produce goods and services that society wants and sell them at fair prices

Application of social, legal, ethical & environmental principles

Fair prices do not naturally imply low prices

A fair price is one that includes dividends to the investors and sufficient profits for the continuance of the business

Application of social, legal, ethical & environmental principles

In recent years, legal responsibilities have come to the force due to the keen market competition

Compliance with laws is an important aspect of a business responsibilities

Application of social, legal, ethical & environmental principles

A company's ethical responsibilities toward society go beyond its economic and legal obligations

Ethical responsibilities embrace those activities and practices that are expected or prohibited by societal members even though they are not codified into law

Application of social, legal, ethical & environmental principles

Application of CSR to marketing situations is linked to producing a positive image that the company contributes to society and that its existence is valuable to society

Positive image is not a tangible asset of the company but is stored in the minds of customers

Application of social, legal, ethical & environmental principles

- Neglecting CSR can lead to major problems
- Food scandals and reputation loss

Application of social, legal, ethical & environmental principles

CSR includes the important topic of environment protection

The significance of this topic came to light with the environmental pollution of the 1970s

In the 980s, there emerged the environmental concerns, such as global warming, depletion of the ozone layer, reckless deforestation and sea pollution.

Green marketing

Application of social, legal, ethical & environmental principles

- For companies to build their reputation in the market, it is important that they carry out green marketing

- Customers who are very sensitive to global environmental problems constitute an important market that firms cannot afford to ignore

END

Food Biotechnology

Module # 183



Food Biotechnology

Marketing Research

Marketing Research

For marketing research, we must first:

Understand the main influences on the marketing process within the macro and micro environments

Marketing Research

- Analyze and respond to the issues associated with buying behavior in both consumer and organizational markets
- Evaluate the need for information in marketing and understand the different methods of collecting and analyzing data
- Apply the marketing research process in selected food markets.

Marketing Research

- For effective marketing, marketers need to understand all relationships that surround the company
- These relationships are called the:marketing environment'. It is imperative that companies adapt to the changing environment

Marketing Research

- The marketing environment comprises a macro environment and a micro environment

- Macro environment is formed of some societal forces that influence its marketing activities

Demographic, Economic, Natural, Technological, Political, Cultural

Marketing Research

Demographic : study of the human population. Includes the study of the size, structure and distribution of populations and how populations change over time

- Baby boomers (born between 1946-1964),
- Most powerful market
- Developing products for this market has gained prominence

Marketing Research

Economic: level and distribution of the income effect and consumer purchasing power and purchasing patterns

In the case of a company that approaches a foreign market, the price of the imported food is higher than local food

Important to know the size of the market for the high-priced product

Marketing Research

Natural: concern for the global environment has been increasing steadily

Companies need to develop products that do not lead to global warming and air and water pollution

The production processes should be modified to suit this need

Marketing Research

Technological: new technologies create new markets and opportunities

New technology replaces older technology

New technology also creates new regulations.

For example, it is mandatory to check the safety of genetically modified organism (GMO) foods before they are delivered to the market.

Marketing Research

Political: laws, government agencies and lobby groups form the political environment

These factors influence businesses in various ways

The number of laws and regulations is increasing year by year

Marketing Research

Cultural: people in a society have common basic beliefs and values, namely, culture.

These core beliefs and values affect their perceptions, preferences and behaviors.

Eating habits are especially influenced by the food culture

Marketing Research

END

The micro environment consists of the actors close to the company that affect its ability to serve its customers

Following are the forces considered to constitute micro environment

Company, Suppliers

Marketing intermediaries

Customer markets, Competitors, Public.

Marketing Research

- Company: marketing plan that requires the coordination of many intra-firm functions – top management, finance, research and development (R&D), purchasing, operations and accounting – is formulated.
- These interrelated functions constitute the internal environment

Marketing Research

- Suppliers: companies need many kinds of resources to produce goods and services
- These resources are provided by suppliers. If there are shortages and late deliveries of these resources, marketing activities are affected.
- The prices of resources influence the prices of products and services.

Marketing Research

- Marketing intermediaries: in making products accessible to the final customer, marketing intermediaries, for example, resellers, physical distribution firms, marketing services agencies and financial intermediaries, play an important role
- Companies need to build good relationships with their marketing intermediaries

Marketing Research

- Customer markets: the three well-known types of customer markets are personal consumption markets, business markets and government markets
- Personal consumption markets are subdivided into individuals and households.
- Marketers today need to recognize both the international market and the domestic market

Marketing Research

- Competitors: companies produce goods in order to satisfy the needs of their target consumers

- These goods must realize greater customer satisfaction than those of their competitors

Marketing Research

- Public: public relations influence marketing activities
- A company's relationships with investors, the media, government, consumer and citizen groups and local communities play an extremely important role
- Workers, managers and the board of directors constitute the internal public

Marketing Research

- It is important to make employees feel good about their company

Food Biotechnology

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Food Biotechnology

Consumer Behavior

Consumer Behavior

As markets are classified; Buying behavior is also classified

- Consumer buyer behavior in the consumer market
- Business buyer behavior in the business market

Consumer Behavior

- Marketers have to consider two things:
- Factors influencing consumers
- The buyer decision process

Consumer Behavior

Factors influencing consumers, the following four are well known:

- Cultural
- Social
- Personal
- Psychological

These four factors are presented in a kind of progression, from factors that have a broad influence to those that have a personal influence

Consumer Behavior

Marketers need to understand the three components of the buyer decision process

Need to understand:

- The buying decision maker
- Types of buyer decisions
- The steps in the buyer decision process

Consumer Behavior

The buying decision maker has a role in the buying decision process

A man who purchases daily food items from a supermarket by referring to a list is the purchaser

His partner, who made the list, is the decision maker of these purchases

Marketers have to identify the buying decision maker and approach him/her

Consumer Behavior

The decision-making process of a purchaser depends on what he/she wants to buy

The types of buyer decisions are classified along two axes:

- The commitment level to the buying process
- The differences between brands

Consumer Behavior

This model depicts the five steps in the buyer decision process. The buyer:

1 recognizes a need

2 seeks information

3 evaluates alternatives

4 decides the purchase

5 makes a post-purchase assessment

Consumer Behavior

The buying process begins long before the actual purchase

Continues long after it is completed

Therefore, marketers need to focus on the entire buying process rather than on merely the purchase decision

Consumer Behavior

Business buyer behavior

- Consists of organizations that purchase other companies' products and services in order to produce their goods
- Characteristics of a business market differ from those of a consumer market

Consumer Behavior

- The business market has fewer buyers than a consumer market
- The buyers in a business market are bigger than those in a consumer market
- There exists a close relationship between the supplier and the customer

Consumer Behavior

Business buyer behavior is influenced by various factors:

- Environmental
- Organizational
- Interpersonal
- Individual factors

These factors include many factors that differ from those that influence the general public and those that are related to individual transactions.

Consumer Behavior

Marketers also need to understand the three components of the buyer decision process:

- Buying decision maker
- Types of buyer decisions
- Steps in the buyer decision process

Consumer Behavior

- First, the decision is made in the buying centre.
- The buying centre includes organization members who play five roles in the purchase decision process:
- ‘users’ use the product or service
- ‘influencers’ help to prepare the specifications

Consumer Behavior

- ‘buyers’ select the supplier for routine buying
- ‘deciders’ have the final authority to select suppliers
- ‘gatekeepers’ control the information flow into the buying centre

Consumer Behavior

- Second, there are three orientations in the case of the buyer decision
- ‘purchasing orientation’ is a market transaction; the buyer aims to buy goods that are cheaper
- ‘sourcing orientation’ builds more cooperative relationships with suppliers

Consumer Behavior

- By building close relationships, the buyer aims to simultaneously achieve quality improvement and lower costs.
- ‘supply management orientation’, the buying centre plays a broader role than in other orientations; the company aims to enhance value through the entire value chain that it formulates

Consumer Behavior

END

Following eight steps are suggested to constitute the buyer decision process:

1 recognize problems

2 describe the general needs

3 specify the products

4 seek suppliers

5 solicit proposals

6 select the supplier

7 order the desired specification

8 review the performance

Food Biotechnology

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Food Biotechnology

Market Research Process in Food Market

Market Research Process in Food Market

Making decisions needs exact information about market. This information comes through market research.

Market research has 4 steps.

1.Define problem & objective.

2. Develop research plan.

3. Implement research plan.

4. Interpret and report findings

Market Research Process in Food Market

Defining Problem & Objective:

Most difficult step to assign cause to some effect/market observation. e.g. decrease in demand for a product. Then we make objectives through 3 approaches:

1. Exploratory research
2. Descriptive research
3. Cause & effect research

Market Research Process in Food Market

2. Developing Research Plan:

It outlines the sources of existing data and spells out the specific research approaches, contact methods, sampling plans and instruments to gather new data. It comes in the form of research proposal.

Secondary and primary data collection.

Market Research Process in Food Market

3. implementing research plan.

collection, processing and analysis of information.

Collection through research methods

For processing and analysis of information, we need to isolate the important information and findings.

Market Research Process in Food Market

4. Interpret and report findings

Interpretation is vital and must be correct to understand new perspectives.

Sharing must be reported to internal and external experts and be ready to embrace results.

Market Research Process in Food Market

References:

1. **Food preservation techniques,** Edited by Peter Zeuthen and Leif Bùgh-Sùrensen (2003) CRC Press Woodhead Publishing Limited Cambridge England.
2. **Food Biodeterioration and Preservation,** Edited by Gary S. Tucker (2008) 2008 by Blackwell Publishing Ltd
3. **Food Processing Technology,** 2nd Edition P.J Fellows (2000) CRC Press Woodhead Publishing Limited Cambridge England
4. **Food Science and Technology,** Edited by Geoffrey Campbell-Platt 2009 Blackwell Publishing Ltd
5. **Food Science,** Revised 2nd edition, Sumati R. Mudambi *(*2006) New Age International (P) Limited, Publishers
6. **Fundamental Food Microbiology ,** 3rd edition Bibek Ray CRC Press London (2005)
7. **Handbook of Food Preservation .** Edited by Shafiur Rehman. 2007 by Taylor & Francis Group, LLC.

Market Research Process in Food Market

8. Handbook of Food Products Manufacturing, Principles, Bakery, Beverages, Cereals, Cheese, Confectionary, Fats, Fruits and Functional Foods. Edited by Y. H. Hui. 2007 by John Wiley & Sons, Inc.
9. Probiotics in Food Safety and Human Health . *edited by* Ipek Goktepe et al, 2006 by Taylor & Francis Group, LLC
10. Food Quality Assurance, Principles and Practices CRC PRESS Boca Raton London New York Washington, D.C. Inteaz Alli (2004) by CRC Press LLC.
11. Elementary Food Science, 4th Edition, ERNEST R. VIEIRA, 1996 Springer Science+Business Media Dordrecht
12. Food Biotechnology edited by Kalidas Shety. 2006 by Taylor & Francis Group, LLC
13. Naeema *et al*. (2016) An Overview on Edible Vaccines and Immunization. Austin Journal of Nutrition and Food Sciences. 2(4). pp 1-5.

Final Term Syllabus
Lecture 92 to 185

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